Central Bank Digital Currency Handbook for Africa

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Abbreviations

AML/CFT	Anti-money-laundering/countering the financing of terrorism
ECA	Economic Commission for Africa
fintech	financial technology
GDP	gross domestic product
IMF	International Monetary Fund
OECD	Organisation for Economic Co-operation and Development

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Executive summary

The *Central Bank Digital Currency Handbook for Africa* offers central bank executives and policymakers across Africa and beyond a tailored, policy-driven approach to exploring, designing and implementing digital currencies. Recognizing the diverse economic priorities, financial sector capabilities and technological readiness of Africa, the handbook is based on a distinctive archetype-based framework, enabling African central banks to develop digital currency strategies that are specifically tailored to their unique contexts.

Central to the handbook is a five-step strategic framework guiding central banks to systematically assess digital currency feasibility, prioritize policy objectives, evaluate readiness, design appropriate operating and governance models and outline clear implementation pathways. The framework ensures that each stage of digital currency development is aligned with national economic strategies and carefully calibrated to institutional capacities.

The main strategic recommendations highlighted in the handbook include:

1. Archetype-based customization

Categorizing countries on the basis of shared characteristics, such as economic size, mobile money penetration, financial vulnerability or regional integration, enables central banks to precisely evaluate their specific digital currency readiness, objectives and design considerations.

2. Alignment with national development goals

Digital currency exploration must integrate seamlessly with broader economic policies, digital economy blueprints and financial sector strategies. Clear linkages between digital currency policy objectives (financial inclusion, payment efficiency, monetary sovereignty) and national priorities maximize policy coherence and public value.

3. Framework for linking design choices to objectives and readiness

The handbook contains a structured approach for explicitly linking digital currency design choices, including technology stacks, operating models and governance frameworks, to articulated policy objectives and assessed levels of technical, regulatory and institutional readiness.

4. Iterative development and stakeholder feedback loops

Emphasis is placed on iterative and phased development, and regular engagement with stakeholders, including continuous feedback mechanisms, is stressed. Each phase, from exploratory research to pilot testing to full implementation, integrates ongoing evaluations of user experiences, stakeholder satisfaction and operational performance, fostering adaptive improvements and reinforcing public trust.

5. Tailored pathways for different readiness levels

Varying degrees of institutional and technological readiness are recognized, and customizable implementation pathways are outlined. This ensures that central banks can strategically adapt their digital currency development processes on the basis of their respective capabilities, infrastructure and policy priorities.

6. Comprehensive readiness assessment

A multidimensional readiness assessment, covering technical infrastructure, regulatory frameworks, financial sector maturity, social acceptance and institutional governance, enables central banks to identify and address critical gaps prior to moving forward with pilot phases or broader deployments.

7. Continuous monitoring, evaluation and adaptation

The establishment of clear, measurable success metrics and decision points at every stage, supported by structured monitoring frameworks, facilitates continuous evaluation and informed adjustments to digital currency strategies, ensuring alignment with evolving stakeholder needs and national development objectives.

By applying this structured, policy-driven methodology, African central banks can effectively leverage digital currencies as strategic policy instruments to promote economic resilience, financial innovation and inclusive growth. The diverse dynamics of African economies are set out in the handbook, which includes a detailed, practical guide for digital currency exploration, drawing upon best practices and lessons learned from both global and regional digital currency initiatives. Complemented by extensive references to authoritative resources, practical decision-making tools and detailed case studies from across the continent, the handbook contains actionable insights, tailored specifically to national contexts, for policymakers and central banks.

1 Introduction

The global financial landscape is evolving, and Africa is experiencing significant transformations with regard to its payment systems, financial services and monetary policies. With the widespread adoption of new forms of digital money, increasing integration in terms of cross-border trade and the expansion of alternative payment systems, many African economies are exploring how digital financial innovations, such as central bank digital currencies, can enhance economic resilience and financial inclusion. The exploration and potential adoption of a digital currency will, however, require a context-sensitive approach that takes into account the diverse financial, social and economic realities of the continent and its individual countries, including:

- Variations in financial infrastructure among large economies, small island economies and fragile economies
- Differing monetary policy frameworks, including fixed exchange rate regimes, highly dollarized economies and resource-dependent States
- The role of remittances, mobile payments and informal financial systems in shaping monetary flows

The *Central Bank Digital Currency Handbook for Africa* outlines a way for central banks to explore digital currencies in a manner that is tailored to their national realities, constraints and policy objectives. Whether a country is in the early exploration stages or moving towards full-scale implementation, the handbook contains contextual practical guidance, decision-making tools and implementation road maps to support the informed and strategic adoption of a digital currency.

1.1 Purpose and objectives

Purpose of the handbook

As Africa strives to strengthen and advance its financial architecture, the development of a tailored handbook on digital currencies is both timely and essential, given the challenges and opportunities for economic prosperity presented by this innovative financial market infrastructure and new form of central bank money. The *Central Bank Digital Currency Handbook for Africa* is designed to provide a structured, actionable framework for the exploration, assessment and implementation of digital currencies. Recognizing the diverse economic landscapes, monetary systems and policy objectives of African countries, the handbook contains tailored guidance using an archetype-based approach that reflects the realities of different country contexts. By addressing the unique challenges and opportunities presented by various financial structures, institutional capacities and economic priorities, the handbook serves as a comprehensive resource to inform and guide central bank decision-making.

The rise of digital financial services, the increasing use of mobile money, the persistence of informal economies and the importance of cross-border payments all make digital currencies a strategic consideration for central banks across Africa. The exploration and implementation of digital currencies, however, requires careful alignment with national policy objectives, ensuring that digital currency initiatives contribute meaningfully to financial inclusion, monetary sovereignty, payment efficiency and economic stability. The handbook contains clear methodologies, assessment tools and strategic frameworks that will allow policymakers to evaluate whether, how and when to proceed with the adoption of a digital currency.

Objectives of the handbook

The *Central Bank Digital Currency Handbook for Africa* is designed to achieve the following objectives:

1. **Provide a policy-driven approach to digital currency exploration:**

   - Help African central banks to define clear policy objectives for digital currency exploration, ensuring alignment with broader national economic, financial and social priorities
   - Offer a structured decision-making framework that links digital currency design and technology choices to specific policy drivers, such as financial inclusion, monetary policy effectiveness and payment system modernization

2. **Establish an archetype-based framework for tailored guidance:**

   - Categorize African countries into practical archetypes on the basis of economic size, monetary policy frameworks, financial sector maturity and structural dependencies
   - Provide customized recommendations and approaches for each archetype, allowing central banks to adapt digital currency strategies to their specific conditions and constraints

3. **Deliver a comprehensive readiness assessment framework:**

   - Outline important technical, regulatory, financial and social factors that influence the feasibility of adopting a digital currency
   - Offer self-assessment tools and checklists that central banks can use to measure their country's readiness for the implementation of a digital currency
   - Highlight capacity-building needs to ensure that the adoption of a digital currency is supported by institutional and operational preparedness

4. **Guide central banks through digital currency design and technology choices:**

   - Explore digital currency operating models, including retail, wholesale and general-purpose designs, with a focus on regional interoperability and cross-border functionality
   - Evaluate technology options, such as centralized versus distributed ledger systems, offline capabilities and privacy-enhancing features, linking them to the specific policy needs of different archetypes
   - Address governance considerations, such as public-private partnerships, regulatory frameworks and cybersecurity safeguards

5. **Assess the costs and benefits and implementation viability of digital currencies:**

   - Provide a structured methodology for analysing the costs and benefits of deploying digital currencies
   - Address financial sustainability, including infrastructure investment, operational costs and long-term adoption strategies
   - Guide central banks in balancing economic, financial and technological feasibility with national policy objectives

6. **Outline a phased implementation road map:**

   - Define a step-by-step approach for digital currency exploration, from feasibility studies to pilot programmes to full-scale deployment
   - Offer phased action plans that have "go/no-go" criteria, ensuring that central banks proceed with digital currency implementation only when certain readiness conditions are met
   - Address risk management strategies, including contingency planning and post-launch monitoring

7. **Provide actionable tools for decision-making and monitoring:**

   - Include decision frameworks, matrices and templates to facilitate structured analysis at each stage of the digital currency process
   - Offer monitoring and evaluation frameworks to track the adoption of digital currency, measure policy impact and refine strategies over time
   - Incorporate case studies from both global and regional digital currency initiatives to provide real-world insights and best practices

1.2 Role of archetypes

Introduction to the archetype-based approach

The diverse economic, financial and institutional landscapes across African countries mean that a one-size-fits-all approach to digital currency exploration and implementation is neither practical nor effective. Countries vary in terms of monetary frameworks, financial infrastructure, reliance on remittances, degree of dollarization, debt vulnerabilities and mobile money penetration, among other factors. These differences have a direct impact on the motivations, feasibility and design choices for a digital currency.

To ensure that digital currency strategies are relevant and actionable, an archetype-based approach is used. Rather than treating all countries as homogeneous, archetypes group countries that have similar characteristics, challenges and policy considerations. This approach enables central banks to:

- Receive tailored guidance that reflects their specific financial, technological and institutional constraints
- Navigate trade-offs and risks unique to their country's economic model

- Compare experiences and best practices with countries that share similar conditions
- Develop digital currency strategies that are aligned with national policy objectives and financial sector priorities

Each archetype represents a distinct set of economic conditions that shape the rationale, feasibility and design choices for digital currencies. Some countries may fall into multiple archetypes that reflect overlapping characteristics and require a hybridized approach to digital currency planning. The handbook includes flexible frameworks that accommodate these complexities, ensuring that central banks can adapt recommendations to their unique national contexts.

How archetypes align with policy objectives and readiness levels

The successful exploration and implementation of a digital currency will require clear alignment between policy objectives and a country's level of readiness. While some countries may pursue digital currencies to enhance financial inclusion, others may prioritize monetary sovereignty, payment system efficiency or cross-border integration. Similarly, countries at different stages of digital currency exploration, whether at the early research or pilot-testing stage or nearing full-scale deployment, will require distinct strategies tailored to their specific needs.

The archetype-based approach ensures that digital currency design and technology choices are linked to both policy goals and national readiness levels, allowing central banks to adopt a structured, strategic approach to digital currency development.

Archetypes and their policy objectives

Each archetype is associated with specific policy drivers that shape digital currency motivations. By categorizing countries on the basis of economic structure, monetary system and financial sector maturity, central banks are able to identify the main policy outcomes they should prioritize when considering a digital currency (see table 1).

Table 1
Policy objectives and relevant archetypes

Policy objective	Economic archetypes
Enhancing financial inclusion	High mobile money penetration; fragile; small island
Improving payment system efficiency	Large; high remittance; small island
Strengthening monetary sovereignty	Highly dollarized; fixed exchange rate and currency-pegged; high debt and financial vulnerability
Reducing reliance on foreign currencies	Highly dollarized; high remittance
Increasing cross-border payment integration	High remittance; small island; large
Enhancing fiscal transparency and efficiency	Resource-dependent; high debt and financial vulnerability

Source: Author's elaboration.

For example:

- A mobile-driven economy, such as Kenya, may focus on interoperability between digital currencies and mobile money platforms to strengthen financial inclusion.
- A highly dollarized economy, such as Zimbabwe, may prioritize monetary sovereignty by developing a digital currency as a mechanism to reduce reliance on foreign currencies.
- A small island economy, such as Cabo Verde, may be most concerned with cross-border remittance efficiency, ensuring that its digital currency integrates seamlessly with global payment networks.

By aligning digital currency features and design considerations with national policy objectives, central banks can ensure that digital currency development is not a technology-driven exercise but a policy-driven transformation.

Archetypes and their readiness levels

Digital currency readiness varies significantly across African countries, with some central banks still in the early assessment phase while others are actively testing pilot projects. The archetype-based framework integrates readiness levels, allowing policymakers to adopt digital currency strategies that match their capacity, economic structure and financial infrastructure (see table 2).

Table 2
Digital currency readiness, stages and relevant archetypes

Readiness level	Stage of digital currency exploration	Economic archetypes
Early exploration	Conduct feasibility assessments; engage stakeholders; evaluate legal frameworks	Fragile States; highly dollarized
Technical and regulatory assessment	Focus on interoperability; technology selection; policy alignment	High mobile money penetration; high remittance; small island
Pilot testing	Test retail or wholesale models; refine technical and operational features	Large; fixed exchange rate and currency-pegged
Implementation-ready	Establish governance; scale adoption; integrate with existing financial infrastructure	Large; resource-dependent

Source: Author's elaboration.

For example:

- A fragile State, such as South Sudan, which faces institutional and infrastructure challenges, may need to first assess its legal and technical readiness before considering any pilot programmes.
- A country with high mobile money penetration, such as Ghana, may already have some regulatory and technological foundations in place, allowing for a faster transition to pilot testing.

- A large economy, such as Nigeria, which has an existing digital currency implementation (eNaira), will focus on scaling adoption, improving integration with commercial banks and refining governance structures.

This alignment among archetypes, policy objectives and readiness levels ensures that digital currency decision-making is both structured and strategic, helping central banks to develop a tailored approach to the implementation of digital currency implementation on the basis of their national priorities and existing financial capabilities.

1.3 Central bank digital currencies in the African context

Growing relevance of central bank digital currencies for African economies

As digital transformation continues to reshape global financial systems, African economies are undergoing rapid shifts in their monetary and payment landscapes. The proliferation of mobile money, the increasing volume of cross-border remittances, the persistence of informal economies and the challenges of monetary sovereignty all shape the way central banks consider financial innovations. Within this evolving environment, digital currencies can present an opportunity to modernize financial systems, enhance policy effectiveness and address unique economic challenges.

Digital currencies are particularly relevant for African economies because they align with national policy objectives, such as financial inclusion, payment efficiency and monetary stability; however, the motivations and design considerations for digital currencies vary across the continent. While some countries may prioritize reducing reliance on foreign currencies, others may seek to enhance interoperability with mobile money platforms or improve cross-border trade payments.

The timing of digital currency exploration is also strategically significant. As the African Continental Free Trade Area moves towards deeper regional economic integration, and the Pan-African Payment and Settlement System expands to enable instant cross-border payments in local currencies, digital currencies represent a natural extension of the broader digital and monetary cooperation agenda of Africa. Exploring digital currencies now allows central banks to align national innovations with continent-wide priorities, ensuring that future financial infrastructures are interoperable, inclusive and supportive of regional trade and digital public goods.

In recognition of these variations, a structured, archetype-based framework is outlined in the handbook, allowing central banks to assess the relevance, feasibility and potential impact of a digital currency within their unique economic context.

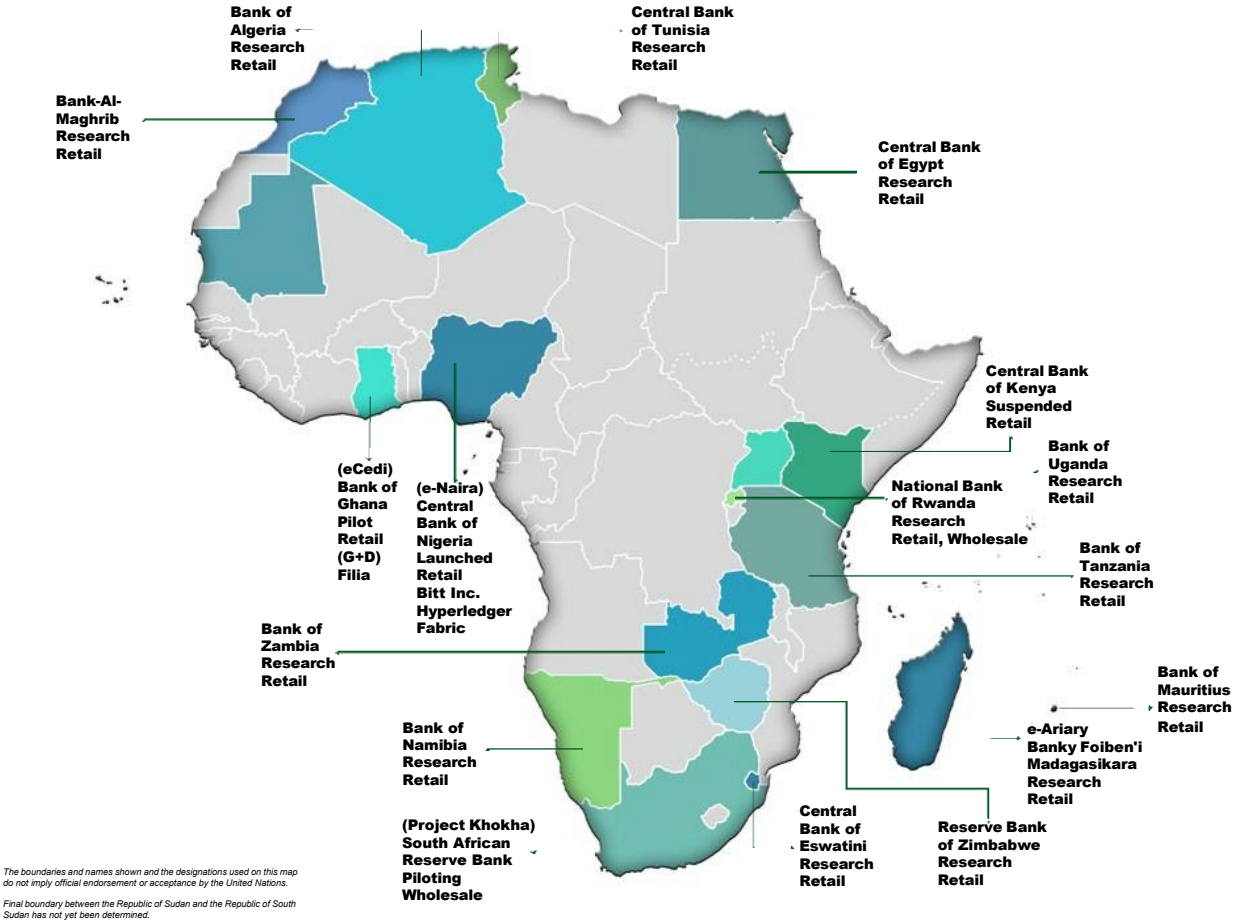
1.4 Central bank digital currency landscape in Africa: an evolving trajectory

The exploration of digital currencies globally and across Africa is gaining momentum,[1] with many countries initiating feasibility studies, engaging in public consultations and announcing pilot intentions. As illustrated in the digital currency readiness diagram by Agpaytech (2023) (see figure I), African countries vary widely in their approach and progress, ranging from live implementation to early stage research. The visual mapping helps to

[1] Alberto Di Iorio, Anneke Kosse and Ilaria Mattei, "Embracing diversity, advancing together – results of the 2023 BIS survey on Central Bank digital currencies and crypto", BIS Papers, No. 47 (Basel, Switzerland: Bank for International Settlements, 2024).

contextualize regional dynamics and is referenced throughout the present handbook to support comparative assessment.

Figure I
Central bank digital currency readiness and activity in Africa



Source: Adapted from Agpaytech, "The payment choice debate: single currency, CBDC, or interoperable mobile money in Africa?" Available at www.agpaytech.com.

Survey findings from ECA provide important insights into the policy motivations driving digital currency exploration across the continent. These results reflect how the unique economic structures, financial sector dynamics and developmental priorities of Africa are shaping national interests in digital currencies. Among the most cited drivers identified by African policymakers are:

- **Financial inclusion**. Many African countries continue to face persistent gaps in access to formal financial services, in particular in rural or underserved areas. Digital currencies are seen as a potential tool to bridge these gaps, offering a government-backed digital alternative that can reach populations currently excluded from the banking sector.
- **Payment system efficiency**. Fragmented and costly domestic payment systems hamper everyday transactions and contribute to economic informality. A digital currency could lower transaction costs, promote interoperability and support

innovation in digital financial services, in particular when integrated into existing mobile money systems.

- **Cross-border payment efficiency**. Intra-African trade and remittance flows are often burdened by high fees, long settlement times and reliance on foreign correspondent banks. Digital currencies, especially those designed with interoperability in mind, present a compelling opportunity to enhance the transparency, cost-effectiveness and speed of cross-border transactions, which are critical for deepening economic integration through such frameworks as the African Continental Free Trade Area.

Critical central bank digital currency considerations for African economies

1. Financial inclusion and digital payment expansion

Many African countries have made significant strides in expanding financial access through mobile currencies, yet a substantial proportion of the population remains unbanked or reliant on informal financial services. A well-designed digital currency could:

- Bridge the financial inclusion gap by providing a low-cost, accessible digital currency that integrates seamlessly with mobile money and digital wallets
- Extend financial services to remote and underserved populations, in particular in rural areas where banking infrastructure is weak
- Reduce transaction costs for small-value payments, making financial services more affordable for low-income households

However, digital currency adoption alone does not guarantee financial inclusion. It must be accompanied by:

- Infrastructure improvements (e.g. mobile connectivity and digital literacy programmes)
- Policy coordination to ensure interoperability with existing payment systems
- Public-private partnerships that involve banks, fintech firms and telecommunications operators

Significant considerations

- For economies that have high mobile money penetration (e.g. Ghana, Kenya, Uganda), a digital currency could enhance interoperability and reduce transaction costs across digital payment platforms.
- For fragile-State and rural economies (e.g. Chad, Malawi, South Sudan), offline-capable digital currencies may be essential in order to overcome infrastructure limitations.

2. Strengthening cross-border payments and regional trade integration

Africa has some of the highest cross-border remittance fees in the world, averaging 8 to 10 per cent per transaction. In addition, intra-African trade remains hampered by fragmented payment systems and increasing reliance on correspondent banks outside the continent. A digital currency designed for cross-border functionality could:

- Lower remittance costs and improve transparency for migrant workers sending money home
- Enhance regional trade efficiency by reducing settlement delays among African economies
- Strengthen financial sovereignty by reducing dependence on third-party intermediaries in international transactions

Significant considerations

- For small island economies and high-remittance economies (e.g. Cabo Verde, Comoros, Gambia, Lesotho), a cross-border interoperable digital currency could enhance economic resilience by lowering the cost of inbound remittances.
- For regional economic hubs (e.g. Egypt, Morocco, Nigeria, South Africa), digital currencies could facilitate the objectives of the Agreement Establishing the African Continental Free Trade Area by supporting seamless digital trade transactions.

Cross-border digital currency adoption does, however, require strong regulatory coordination and interoperability agreements, both regionally and internationally.

3. Enhancing monetary policy and reducing dollarization

Several African economies experience limited monetary policy effectiveness owing to such factors as:

- High levels of currency substitution (dollarization/"euroization")
- Fixed exchange rate regimes that constrain policy flexibility
- Volatile inflation and external shocks from commodity price fluctuations

A central bank digital currency can strengthen monetary policy by:

- Increasing the central bank's ability to influence liquidity conditions in digital financial markets
- Providing a trusted digital alternative to foreign currencies and reducing reliance on cash-based transactions in dollarized economies
- Enhancing financial transparency and improving data collection on money supply and economic activity

Significant considerations

- For highly dollarized economies (e.g. Democratic Republic of the Congo, Liberia, Zimbabwe), a central bank digital currency could help to rebuild trust in local currency if designed with robust usability and security features.
- For fixed exchange rate regimes (e.g. CFA Franc Zone, Morocco), a digital currency could enhance monetary control without disrupting currency stability mechanisms.

The success of a digital currency with regard to the strengthening of monetary policy, however, depends on adoption rates, regulatory enforcement and integration with broader economic reforms. More information can be found in box 1.

Box 1
Considerations for digital currencies in fixed exchange rate regimes

Fixed exchange rate regimes, in which a national currency is pegged to a foreign currency, often the euro or the United States dollar, present both opportunities and constraints with regard to digital currency exploration. Such countries as Morocco (pegged to a currency basket) and members of the CFA Franc Zone (pegged to the euro through a subregional monetary arrangement with the Treasury of France) must carefully assess the macroeconomic and operational implications of introducing a digital currency.

Significant considerations

- **Exchange rate stability**. Introducing a digital currency must not undermine the credibility of the peg. Careful liquidity management and convertibility rules are required to prevent speculative arbitrage or shifts in monetary aggregates that could strain the peg.
- **External reserve backing**. When issuing digital currency in a fixed regime, central banks should consider the adequacy of foreign exchange reserves. A retail digital currency that becomes widely adopted could affect reserve adequacy ratios and the central bank's ability to meet external liabilities if not properly calibrated.
- **Regional coordination**. In such cases as the CFA franc, where monetary sovereignty is pooled, the deployment of digital currency may require coordinated governance and oversight across multiple members and the subregional central bank (e.g. the Bank of Central African States), especially with regard to mandates related to its legal issuance and the harmonization of its design.
- **Capital flow management**. Some fixed regimes already implement capital account restrictions to preserve the peg. Digital currencies that have cross-border or programmable features could require enhanced controls or compliance layers to avoid exacerbating illicit flows or the substitution of unmonitored currency.
- **Policy leverage**. If designed appropriately, digital currencies in fixed regimes can enhance the transmission of monetary policy by improving transparency and increasing the velocity of official currency, especially in cash-heavy economies.

Design and roll-out implications

Central banks operating in fixed regimes may consider:

- Starting with domestic use cases to limit external exposure
- Imposing wallet limits, tiered know-your-customer policies and non-interest-bearing digital currency models to reduce substitution risks
- Embedding exchange control rules into smart contracts or back-end application programming interfaces, especially for cross-border pilot programmes

4. Increasing payment system efficiency and reducing transaction costs

Many African countries still rely heavily on cash transactions, which pose security risks and contribute to inefficiencies in tax collection and public finance management. A digital currency could:

- Reduce cash-handling costs for businesses and Governments
- Improve efficiency in social benefit payments, public sector wages and subsidy programmes
- Combat illicit financial flows by increasing transaction traceability

Significant considerations

- For large economies (e.g. Egypt, South Africa), a digital currency could improve the resilience of payment systems and modernize financial markets.
- For resource-dependent economies (e.g. Algeria, Angola, Zambia), a well-structured digital currency could streamline the collection of fiscal revenues from extractive industries.

The implementation of digital currency, however, must avoid excessive financial surveillance risks, ensuring privacy protection and compliance with data protection laws.

1.5 Global context

Growing momentum of digital currencies worldwide

As central banks worldwide explore digital currencies, the development of this type of currency has moved from theoretical discussions to active experimentation and implementation. More than 130 central banks globally are investigating digital currencies,[2] with projects ranging from pilot programmes to full-scale deployments. While motivations for developing digital currencies differ across economies, critical lessons from global experiences are particularly relevant for African central banks seeking to align digital currency design with regional monetary, financial and technological realities.

The present section provides an overview of global digital currency trends, emphasizing lessons that align with the archetypes of African economies. By examining successful implementations, ongoing pilot programmes and challenges encountered by central banks worldwide, African policymakers can gain valuable insights into the feasibility, risks and best practices for digital currency development.

Global central bank digital currency developments and their relevance to African archetypes

1. Retail versus wholesale digital currencies: differing use cases

As set out in table 3, globally, central banks are considering two primary digital currency models:

- Retail digital currencies designed for use by individuals and businesses as a digital alternative to cash
- Wholesale digital currencies used for interbank settlements, improving cross-border transactions and financial market infrastructure

[2] Alisha Chhangani, "Central bank digital currency evolution in 2023: from investigation to preparation", Atlantic Council, 6 November 2023.

Table 3
Digital currencies and lessons for African economies

Country	Type of digital currency	Lesson for African economies
The Bahamas (SandDollar)	Retail	Demonstrates how a digital currency can improve financial inclusion in small island economies that have weak banking penetration
China (e-CNY, also known as digital renminbi)	Retail	Provides a case study on State-backed digital payments and integration with existing fintech platforms
Sweden (e-krona)	Retail	Highlights the role of digital currencies in reducing reliance on physical cash in highly digitalized economies
France and Singapore (wholesale digital currencies)	Wholesale	Showcases how digital currencies can improve interbank settlements and cross-border trade payments
Ghana (eCedi) and Singapore	Retail	Showcases how digital currencies and supporting infrastructure can be used to support Ghanaian micro-, small and medium-sized enterprises in international cross-border trade
Switzerland	Wholesale	Explores the settling of tokenized securities with commercial banks

Source: Author's elaboration.

Significant considerations

- For small island economies (e.g. Cabo Verde, Seychelles), the SandDollar of The Bahamas provides a model for the implementation of digital currency in tourism-dependent economies that have limited banking access.
- For mobile-driven economies (e.g. Ghana, Kenya, Uganda), the e-CNY of China offers insights into the integration of digital currency with digital wallets and existing fintech platforms.
- For high-remittance economies (e.g. Gambia, Lesotho, Nigeria), wholesale digital currencies could reduce cross-border remittance fees and strengthen financial autonomy.

2. Financial inclusion and digital payment adoption

One of the strongest arguments for digital currencies in Africa is their potential to expand financial inclusion and enhance digital payments (see table 4). Global digital currency pilots have provided the following vital insights:

- Digital currencies should complement, not replace, existing financial solutions (e.g. mobile money).
- Low-cost digital infrastructure and offline functionality are critical for adoption in remote areas.
- Regulatory clarity is necessary to ensure smooth integration with commercial banking systems.

Table 4
Digital currency strategies for financial inclusion and lessons for African central banks

Country	Digital currency strategy for financial inclusion	Lesson for African central banks
India (e₹ digital rupee pilot)	Exploring digital currency for rural banking access and digital payments	Highlights the need for interoperability between digital currency and mobile payment systems
Jamaica (JAM-DEX)	Introducing digital currency for low-income households and unbanked populations	Demonstrates how Government-backed digital wallets can support digital currency distribution
Eastern Caribbean Currency Union (DCash pilot)	Exploring digital currency for underserved communities and broadening access to digital payments	Demonstrates how non-bank financial institutions can support digital currency distribution
Nigeria (eNaira)	Exploring digital currency for access to digital payments	Highlights the need for effective stakeholder engagement and Unstructured Supplementary Service Data, and mobile money integration

Source: Author's elaboration.

Significant considerations

- For economies with high mobile money penetration (e.g. Rwanda, United Republic of Tanzania, Zambia), the approach used by India with regard to mobile central bank digital currency integration could serve as a model for interoperability.
- For fragile-State economies (e.g. Central African Republic, South Sudan), the JAM-DEX shows how a government-issued digital currency can increase economic participation for low-income households.

3. Cross-border central bank digital currency initiatives: implications for African regional trade

African countries face high remittance costs and fragmented payment systems, making cross-border digital currencies a potential solution. Several global projects demonstrate how digital currencies can enhance regional payment efficiency and trade integration (see table 5).

Table 5
Global digital currency projects and lessons for Africa

Project	Participating countries	Lessons for Africa
Project mBridge	China; Hong Kong, China; Thailand; United Arab Emirates	Uses the interoperability of digital currencies to improve cross-border payments, reducing reliance on correspondent banks
Project Dunbar	Australia, Malaysia, Singapore, South Africa	Demonstrates how multiple central banks can issue and settle digital currencies on a shared digital ledger
Digital Economy Semi-Fungible Token	Ghana, Singapore	Demonstrates how innovation led by central banks can create trust and solve challenges for international trade

Source: Author's elaboration.

Significant considerations

- For regional economic hubs (e.g. Egypt, Morocco, Nigeria, South Africa), adopting wholesale digital currency models for trade settlement could enhance regional financial integration.
- For high-remittance economies (e.g. Liberia, Senegal, Somalia), multiple digital currency models could reduce remittance fees and enhance financial sovereignty.

4. Regulatory and monetary policy considerations

Several central banks have recognized the need for robust legal frameworks before rolling out digital currencies (see table 6). Global experiences highlight the importance of:

- Balancing financial privacy with compliance with AML/CFT policies
- Ensuring interoperability with commercial banks to avoid financial disintermediation
- Clarifying the legal tender status of digital currencies to support widespread adoption

Table 6
Regulatory approach and lessons for Africa

Country/area	Regulatory approach	Lessons for Africa
European Union (digital euro)	Prioritizes privacy protections and consumer safeguards	Highlights the need for clear legal frameworks before full implementation
Nigeria (eNaira)	Adopts a phased approach to implementation	Demonstrates the importance of public engagement and education to drive adoption

Source: Author's elaboration.

Significant considerations

- For highly dollarized economies (e.g. Democratic Republic of the Congo, Liberia, Zimbabwe), legal clarity will be essential to establish digital currency as a trusted alternative to foreign currency dependence.
- For resource-dependent economies (e.g. Algeria, Angola, Chad), regulations should consider the role of digital currencies in fiscal policy, taxation and revenue collection.

Critical takeaways for African central banks

Global digital currency experiences provide valuable insights for African policymakers, reinforcing that the design of digital currency must align with national economic priorities. The following lessons stand out:

1. Digital currencies must complement, not compete with, such existing financial solutions as mobile money.
2. Cross-border digital currencies can strengthen regional trade and remittance flows, but they require regulatory harmonization.
3. Legal and regulatory clarity is essential prior to implementation to ensure trust and adoption.
4. Stakeholder engagement is critical. Successful digital currency roll-outs require central banks to actively involve commercial banks, fintech companies and the public.
5. Pilot projects and phased roll-outs allow central banks to test digital currency models in a controlled environment, refine designs on the basis of feedback and progressively build public trust.

By applying these lessons within an archetype-based framework, African central banks can ensure that digital currency exploration is both context-sensitive and policy-driven, supporting long-term economic resilience and financial modernization.

Further reading

For a foundational perspective on digital currency design and implementation, central banks are encouraged to consult a publication by the Bank for International Settlements entitled "Central bank digital currencies: foundational principles and core features". This landmark publication contains core principles for central banks that take into consideration digital currencies, including their coexistence with cash, as well as promote innovation and efficiency and support broader policy objectives, offering essential context that complements the strategic orientation of the present handbook.

2 Central bank digital currency exploration and implementation framework

2.1 Overview

The successful exploration and implementation of a digital currency requires a structured, policy-driven approach that aligns with national economic priorities, financial sector capabilities and technological readiness. The handbook contains a five-step framework that

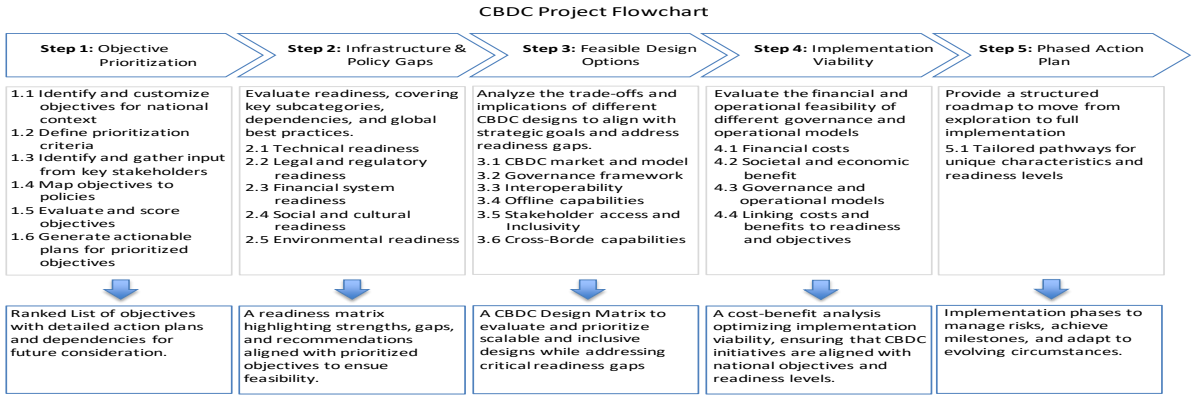
enables central banks to assess feasibility, design appropriate models and develop a strategic implementation pathway tailored to their specific contexts.

The framework is designed to be flexible and adaptable and to recognize the diverse economic landscapes across African countries. Whether a country is in the early exploratory phase or actively considering pilot testing and deployment, this structured methodology ensures informed decision-making, risk mitigation and alignment with broader policy goals.

2.2 Five-step approach to digital currency exploration and implementation

Figure II illustrates a flow chart that can be used to explore and implement digital currencies.

Figure II
Central bank digital currency project flow chart



Source: Author's elaboration.

Abbreviation: CBDC, central bank digital currency.

Step 1. Policy objectives and prioritization

The first step ensures that digital currency exploration is driven by clear national policy objectives. Central banks must define why they are considering a digital currency and how it aligns with macroeconomic goals, financial inclusion strategies, payment system modernization and monetary policy frameworks.

Relevant actions to be taken

- Identify the primary motivations for exploring a digital currency.
- Use an archetype-based policy alignment tool to match national priorities with digital currency objectives.
- Evaluate whether a digital currency is the most effective instrument to achieve the identified policy goals.

Output

A ranked list of policy objectives, ensuring that digital currency development is outcome-driven rather than technology-driven.

Step 2. Readiness assessment

Before moving forward, central banks must assess their readiness across essential dimensions, including technical infrastructure, regulatory preparedness, financial sector stability and societal acceptance. The success of a digital currency depends not only on policy alignment but also on institutional and technological capacity.

Relevant actions to be taken

- Conduct a multidimensional digital currency readiness assessment, covering:
  - Technical infrastructure (payment systems, digital identification frameworks, cybersecurity).
  - Regulatory environment (legal frameworks, AML/CFT policy compliance, data protection).
  - Financial system readiness (banking sector resilience, remittance flows, informal economy integration).
  - Social and cultural readiness (public trust, digital literacy, adoption potential).
- Identify strengths and gaps using a digital currency readiness scorecard.
- Compare findings against archetype-specific considerations to tailor the approach.

Output

A digital currency readiness report that contains gaps, opportunities and areas for capacity-building.

Step 3. Design choice evaluation

Once policy priorities and readiness levels are assessed, central banks must select the most suitable digital currency design and technology model on the basis of their specific needs.

Relevant actions to be taken

- Define the digital currency operating model (retail, wholesale or general-purpose).
- Select the technology framework (centralized database versus distributed ledger technology).
- Determine functional features, including:
  - Privacy and security considerations.
  - Offline capabilities (important for rural and remote areas).
  - Interoperability with banking and mobile money systems.

- Programmability features (e.g. smart contracts for conditional payments).

- Apply an archetype-based digital currency design matrix to ensure that the design aligns with national needs and policy drivers.

Output

A digital currency design blueprint detailing the proposed operating model, technological architecture and critical functionalities.

Step 4. Analysis of cost-benefit and implementation viability

Before committing to a digital currency initiative, central banks must assess the economic and financial feasibility of implementation. This step ensures that a digital currency is cost-effective, scalable and sustainable.

Relevant actions to be taken

- Conduct a cost-benefit analysis, estimating:
  - Infrastructure investment costs.
  - Operational and security expenses.
  - Potential efficiency gains and economic benefits.
- Measure expected benefits, such as:
  - Reduction in cash management costs.
  - Improvement in cross-border payment efficiency.
  - Enhanced monetary policy transmission.
- Evaluate implementation risks and establish a risk mitigation framework.

Output

A central bank digital currency feasibility report detailing economic impact, financial viability and strategic trade-offs.

Step 5. Phased action plan

A well-defined implementation road map ensures that digital currency deployment is incremental, manageable and aligned with national capacity. A phased approach allows central banks to test, refine and scale up digital currency initiatives while minimizing risks.

Relevant actions to be taken

- Define phases of implementation, including:
  - Phase 1. Feasibility study and stakeholder engagement.
  - Phase 2. Proof-of-concept and limited-scale pilot testing.
  - Phase 3. Gradual roll-out and full-scale integration.
- Set “go/no-go” criteria for each phase, ensuring readiness before advancing.

- Establish a monitoring and feedback mechanism to evaluate performance and user adoption.

Output

A digital currency implementation road map that outlines milestones, decision checkpoints and risk mitigation strategies.

Further reading

For additional insights on aligning digital currency policy objectives with macroeconomic conditions and institutional capacities, central banks may refer to the IMF publication entitled *Central Bank Digital Currency Virtual Handbook*. This resource contains a comprehensive framework for evaluating the motivations, risks and operational choices for digital currencies, including guidance that complements the objective prioritization and policy alignment discussions in the present section.

3 Policy objectives and drivers

Digital currencies are complex financial market instruments that offer several design choices for each layer of their construction. The success of a central bank digital currency initiative relies heavily on the clear definition of its objectives and alignment with national policies to identify critical priorities that enable the design of a digital currency that can address local challenges, deliver meaningful benefits and integrate seamlessly into the broader socioeconomic framework.

African countries exhibit diverse economic structures, financial systems and social contexts. Aligning digital currencies with local conditions ensures that their design and implementation effectively address such challenges as financial inclusion, payment efficiency and cross-border trade. The present chapter provides a structured, context-sensitive approach for African countries that seek to identify, prioritize and align their digital currency objectives with socioeconomic goals and policy frameworks.

3.1 Linking policy objectives to national development goals

Digital currency exploration should not be undertaken in isolation; rather, it must be strategically aligned with national development priorities and macroeconomic policy objectives. Whether the goal is to enhance financial inclusion, modernize payment infrastructure, strengthen monetary sovereignty or increase fiscal transparency, a digital currency should be designed as a policy tool that complements and reinforces broader national economic strategies (see table 7).

Table 7
Focus areas of national digital development plans

Country	Plan name	Focus areas
Kenya	Digital Economy Blueprint (2019)	Infrastructure, digital governance, skills, entrepreneurship
Nigeria	National Digital Economy Policy and Strategy (2020–2030)	Financial inclusion, digital identification, broadband, literacy
Rwanda	Smart Rwanda Master Plan	Sectoral information and communications technology integration, governance, innovation
Uganda	Digital Uganda Vision	Infrastructure, rural connectivity, government services
Ghana	Ghana Information and Communications Technology for Accelerated Development (ICT4AD) Policy	Financial inclusion, payment modernization
Senegal	Plan Sénégal Emergent – Plan d'Actions Prioritaires 3: 2024–2028	Digital modernization as a lever for development

Source: Author's elaboration.

In many African countries, national development plans now include digital economy blueprints or information and communications technology master plans that set out long-term visions for inclusive digital transformation. These blueprints typically prioritize such areas as digital identity, broadband expansion, fintech innovation and e-government services. A well-designed digital currency can directly support these agendas by enabling trusted digital payments for government-to-person transfers, digital tax collection, interoperable e-commerce systems and integrated digital identity verification. Linking digital currency exploration to these priorities ensures that the digital currency becomes a platform for innovation, not just a parallel payments instrument.

Similarly, financial sector development strategies, which are often developed by countries in partnership with regional development banks or international financial institutions, articulate national objectives aimed at financial inclusion, access to credit, payment modernization and regulatory transformation. Digital currencies can serve as enablers for these strategies, as they promote competition, reduce costs for underserved populations, facilitate real-time retail and wholesale payments and strengthen a central bank's capacity to supervise digital finance landscapes.

Moreover, as countries explore more extensive examples of the use of digital currency, alignment with sectoral development goals becomes even more critical. For example:

- Agricultural digitalization efforts may benefit from programmable digital currency disbursements to farmers.
- Green finance goals can be supported by traceable digital-currency-based subsidies or tax incentives for clean energy use.
- Regional trade facilitation under the African Continental Free Trade Area can be enhanced by digital currencies that support faster, lower-cost cross-border payments aligned with the objectives of the Pan-African Payment and Settlement System.

Ultimately, integrating digital currency design with these national and sectoral strategies enables central banks to ensure that the digital currency is not merely a technological experiment but a policy-aligned tool that has a measurable impact on development. This reinforces political buy-in, guides the prioritization of use cases and builds institutional momentum towards a coherent, whole-of-government approach to the implementation of digital currency.

For African central banks, aligning digital currency policy objectives with national development goals ensures that digital currency initiatives contribute meaningfully to socioeconomic progress, financial system resilience and regional economic integration. The present section explores the ways in which digital currency objectives can support critical national priorities, using an archetype-based framework to guide alignment with development strategies.

Common central bank digital currency objectives across Africa include:

- Financial inclusion: extending access to financial services for underserved populations
- Payment system efficiency: improving the speed, cost-effectiveness and transparency of transactions
- Monetary policy effectiveness: enhancing a central bank's ability to implement and transmit monetary policy
- Cross-border trade: reducing costs and delays in international payments and remittances
- Economic innovation: driving digital transformation and fostering fintech innovation

Governments and central banks across Africa have defined long-term economic development plans that often include such objectives as those set out below.

1. Promoting financial inclusion and economic empowerment

- Many African countries have large unbanked populations, which rely significantly on cash-based and informal financial systems.
- National financial inclusion strategies are often aimed at increasing digital financial participation, promoting mobile money adoption and expanding access to banking services.
- *Digital currency linkage*: a well-designed digital currency can support financial inclusion by providing a low-cost, accessible digital payment option, in particular in rural areas.
- *Example*: the financial inclusion strategy of Ghana prioritizes mobile-driven financial services, which could be complemented by a digital currency that is interoperable with mobile money platforms.

2. Modernizing payment systems and improving their efficiency

- Many African economies face high transaction costs, slow settlement times and fragmented payment systems.

- National policies often emphasize the need for secure, interoperable and cost-effective digital payment solutions.
- *Digital currency linkage*: a digital currency can serve as a modern digital payment instrument, reducing reliance on cash transactions and streamlining public sector disbursements.
- *Example*: the eNaira of Nigeria is aimed at improving government-to-person transfers, which reduces inefficiencies in social benefit payments.

3. Strengthening monetary policy and currency stability

- Some African economies face monetary instability, exchange rate volatility and weak currency control mechanisms.
- Many Governments intend to strengthen monetary policy tools and enhance domestic currency stability.
- *Digital currency linkage*: a digital currency can enhance the control of the central bank over monetary supply and its transmission, reducing reliance on foreign currencies.
- *Example*: Zimbabwe, a highly dollarized economy, could use a digital currency to promote trust in its local currency while reducing dependence on the United States dollar.

4. Enhancing fiscal policy and transparency

- Governments across Africa seek to improve public sector efficiency, enhance revenue collection and combat illicit financial flows.
- Many national fiscal policies prioritize digital tax systems, enhanced public finance management and anti-corruption efforts.
- *Digital currency linkage*: a digital currency that has programmable features could improve tax compliance, track public spending and enhance financial transparency.
- *Example*: a resource-dependent economy, such as Angola, could use digital-currency-based smart contracts to create transparent management of the revenue from its natural resources.

5. Increasing cross-border trade and regional integration

- The objectives of the Agreement Establishing the African Continental Free Trade Area are aimed at boosting intra-African trade, but cross-border payments remain expensive and inefficient.
- Many Governments support initiatives to enhance regional financial integration.
- *Digital currency linkage*: a cross-border interoperable digital currency could reduce foreign exchange dependence, lower remittance costs and strengthen trade efficiency.
- *Example*: the West African Economic and Monetary Union could explore a digital currency that enhances intraregional transactions while maintaining its currency peg.

6. Strengthening public sector delivery through programmable government payments

Digital currencies offer a transformative opportunity for Governments to align monetary innovation with national development goals, in particular the delivery of government-to-person and person-to-government payments. By enabling programmability, digital currencies can help to optimize the efficiency, transparency and accountability of public financial transfers, contributing directly to such strategic national outcomes as poverty alleviation, social protection, revenue collection and fiscal governance.

Government-to-person: social transfers and targeted disbursements

Digital currencies can be programmed to disburse welfare payments, pensions or subsidies that have built-in rules that govern usage (e.g. time limits, geographic restrictions or approved vendors). This enhances fiscal targeting while reducing the risk of leakages and fraud.

For example:

- The eCedi pilot of Ghana explored the use of a programmable digital currency for government-based social payments.[3] The initiative piloted the disbursement of funds to recipients in underserved areas through digital wallets programmed to be used within a specific time frame and limited to pre-approved merchants. This approach supported financial inclusion, minimized administrative costs and improved trust in the integrity of the disbursement process.
- The eNaira of Nigeria has been tested for conditional cash transfers to vulnerable populations; the programmable features of the digital currency enabled instant settlement, improved traceability and reduced manual reconciliation.[4] In doing so, the eNaira demonstrated the potential to modernize welfare systems and build public confidence in digital disbursement mechanisms.

These examples show how programmable digital currencies can directly advance national development strategies by enhancing the delivery and accountability of public services, a core objective in many African digital economy blueprints and social protection strategies.

Person-to-government: enhancing revenue collection and trust

On the revenue side, person-to-government payments, such as taxes, licence fees and utility bills, can benefit from automation and real-time confirmation enabled by digital currency infrastructure. Programmable digital currencies can include embedded compliance triggers, such as due dates or automatic reminders, thereby increasing the efficiency of collections and reducing transaction frictions.

This dual use of digital currencies in both government-to-person payments and person-to-government streams creates a feedback loop of transparency, improving public financial management and reinforcing citizen trust in the State, which is a critical enabler for inclusive digital transformation.

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[3] Bank of Ghana, "Join the Bank of Ghana eCedi hackathon in collaboration with EMTECH", October 2023. Available at www.bog.gov.gh/ecedihackathon/.

[4] Godwin Nwaobi, "An impact evaluation of digital cash transfers scheme on income poverty in Nigeria", MPRA Paper No. 118228 (9 August 2023).

Policy implications

Incorporating programmable government payments into digital currency design aligns directly with national development goals, such as:

- Expanding financial inclusion (especially in rural and informal sectors)
- Enhancing fiscal efficiency
- Strengthening the social contract between the Government and citizens

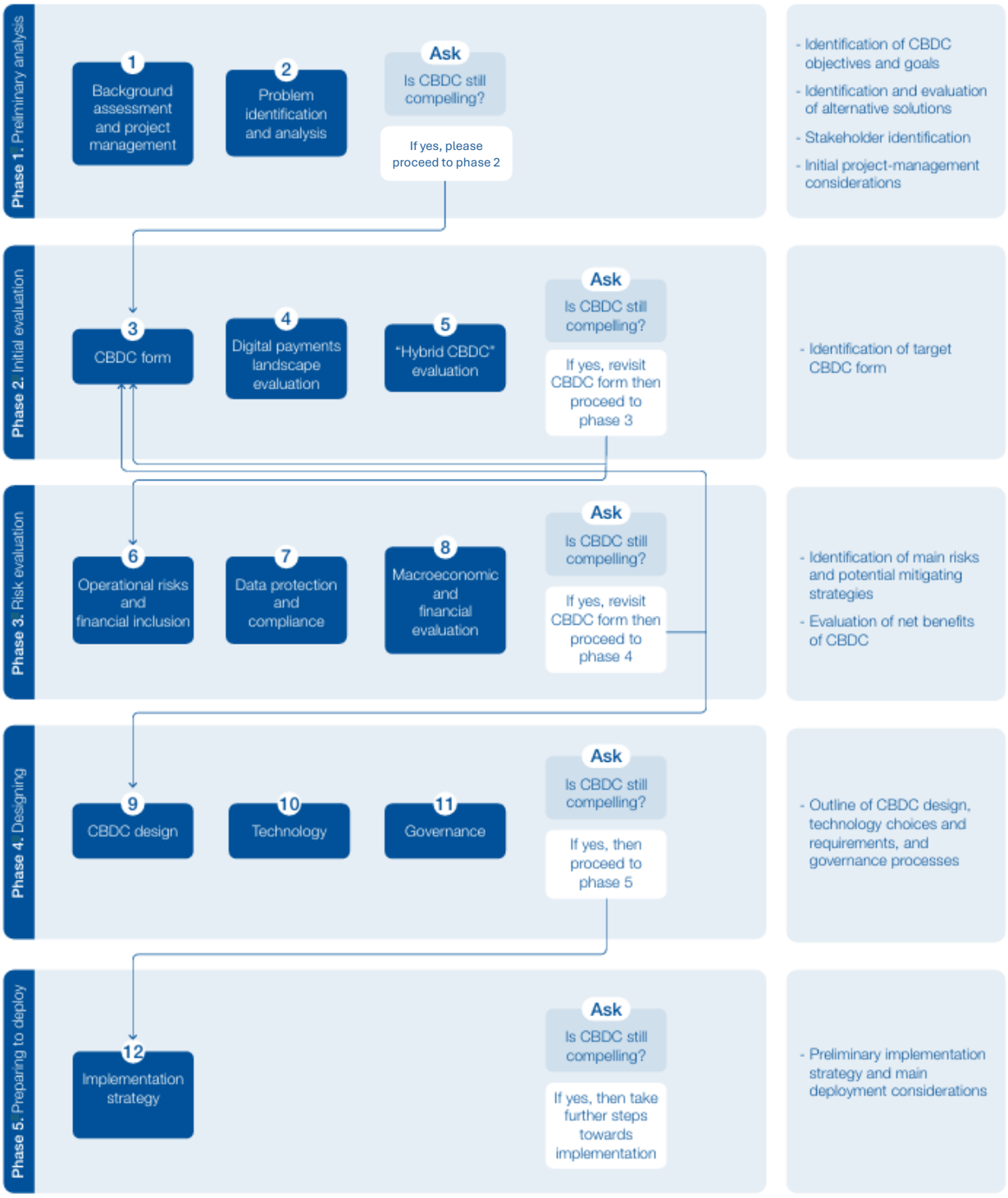
Central banks considering a digital currency should therefore evaluate these use cases early in the design process, linking them to broader strategic outcomes laid out in national development plans, public financial management reforms and digital transformation agendas.

Customizing objectives for national contexts

Several country-specific geographic, political, economic and technological conditions could affect the usefulness or desirability of digital currency.[5] Policymakers should therefore assess specific socioeconomic needs, such as rural financial inclusion, youth unemployment or informal sector integration, as well as regional objectives, such as interoperability with neighbouring countries or alignment with regional trade agreements (see figure III).

[5] World Economic Forum, "Central bank digital currency policy-maker toolkit", Insight Report (Geneva, Switzerland, 2020).

Figure III
Central bank digital currency decision flow chart



Source: World Economic Forum, "Central bank digital currency policy-maker toolkit", Insight Report (Geneva, Switzerland, 2020).

Abbreviation: CBDC, central bank digital currency.

Consulting stakeholders

Stakeholder consultation is essential in order to define central bank digital currency objectives that are inclusive, practical and aligned with national and regional priorities, thereby ensuring the successful exploration, design and implementation of a potential digital currency. By engaging diverse voices, central banks can identify opportunities, anticipate challenges and build a broad coalition of support for digital currency initiatives. The stakeholder consultation process not only enhances the design and implementation of a digital currency but also ensures its relevance and sustainability within the broader socioeconomic context.

Importance of consulting stakeholders

- Broad perspective: stakeholders represent diverse sectors, ensuring that digital currency objectives are inclusive and comprehensive.
- Policy coherence: stakeholder insights align digital currency initiatives with existing national and regional policies.
- Risk identification: engaging stakeholders uncovers potential barriers, technical gaps and social challenges.
- Buy-in and trust: early engagement fosters ownership, collaboration and public confidence in digital currency initiatives.

Identifying crucial stakeholders

Digital currency stakeholders span multiple sectors, and each offers a unique perspective and potential use cases that can support motivations and accelerate return on investment.

Government and the public sector

- Finance, trade, digital economy and justice ministries and agencies provide strategic policy alignment.
- Central bank departments, financial regulators and identity and data protection authorities highlight compliance and oversight needs.

Financial and technology sector

- Commercial banks offer insights into operational feasibility and integration with existing banking systems.
- Payment providers and fintech firms provide expertise on digital payment solutions, mobile money integration and technological innovation.
- Telecommunications operators highlight infrastructure readiness and connectivity issues.

Civil society and the public

- Consumer advocacy groups address trust, accessibility and usability concerns.
- Academics and think tanks provide data-driven insights and independent evaluations.

- Representatives of marginalized communities ensure inclusivity for underserved populations, such as rural areas and informal workers.

Structured approach to stakeholder consultation

Preparation

- Objective-setting: define the purpose of the consultation (e.g. gather feedback on priorities, identify risks or assess public perception).
- Mapping stakeholders: create a comprehensive stakeholder map to ensure representation across sectors.
- Customized engagement: tailor consultation methods (e.g. surveys, focus groups, interviews) to each stakeholder group.

Engagement

- Workshops and forums: facilitate interactive discussions to gather qualitative insights.
- Surveys and questionnaires: collect quantitative data on stakeholder priorities and expectations.
- Focus groups: engage specific demographics, such as fintech companies or underserved communities.

Synthesis

- Analyse feedback: consolidate and analyse stakeholder input to identify common themes, opportunities and challenges.
- Refine objectives: use feedback to refine digital currency objectives and align them with stakeholder priorities.

Best practices for effective consultation[6]

- Transparency: communicate the purpose and scope of the consultation clearly to stakeholders.
- Inclusivity: ensure diverse representation, in particular from underserved and/or marginalized groups.
- Iterative feedback: maintain continuous engagement, updating stakeholders on progress and incorporating their input at every stage.

Further reading

To deepen understanding of how digital currencies can align with national development strategies and support inclusive growth, the IMF publication entitled *Central Bank Digital Currency Adoption: Inclusive Strategies for Intermediaries and Users* provides valuable guidance. The report contains practical insights into designing digital currencies that advance

[6] See the digital euro governance and stakeholder engagement approach of the European Central Bank at www.ecb.europa.eu/euro/digital_euro/timeline/stakeholder/html/index.en.html.

financial inclusion while integrating with the broader digital economy and financial sector goals, reinforcing the policy alignment principles outlined in the present section.

3.2 Policy alignment framework

To ensure that digital currency objectives align with national development goals, policymakers can use the policy alignment framework, which is designed to help central banks to clearly map digital currency policy objectives to broader development strategies, ensuring that the initiative is not viewed as a stand-alone technological exercise but as a catalyst for structural transformation.

The framework encourages policymakers to draw direct connections between national priorities, such as financial inclusion, improved governance or regional trade integration, and tangible digital currency policy goals. By highlighting practical use cases, the framework supports strategic decision-making that maximizes the public value of a digital currency.

Table 8 offers an illustrative guide. Central banks are encouraged to adapt and expand this guide to reflect country-specific strategies, such as digital economy blueprints, information and communications technology master plans and financial sector development programmes.

Table 8
Uses of digital currency to forward national development goals

National development goal	Relevant digital currency policy objective	Example
Expand financial inclusion	Enable digital financial access for unbanked populations	Use of a digital currency that is interoperable with mobile money
Modernize payment systems	Increase efficiency and reduce transaction costs	Use of digital currency as a replacement for high-cost cash transactions
Strengthen monetary policy	Improve money-supply management and domestic currency stability	Adoption of digital currency in highly dollarized economies
Enhance fiscal transparency	Increase public sector efficiency and combat illicit flows	Use of digital currency for government payments and automated taxation
Facilitate regional trade	Improve cross-border payment systems and reduce reliance on foreign currencies	Use of digital currency to enable seamless intra-African trade transactions

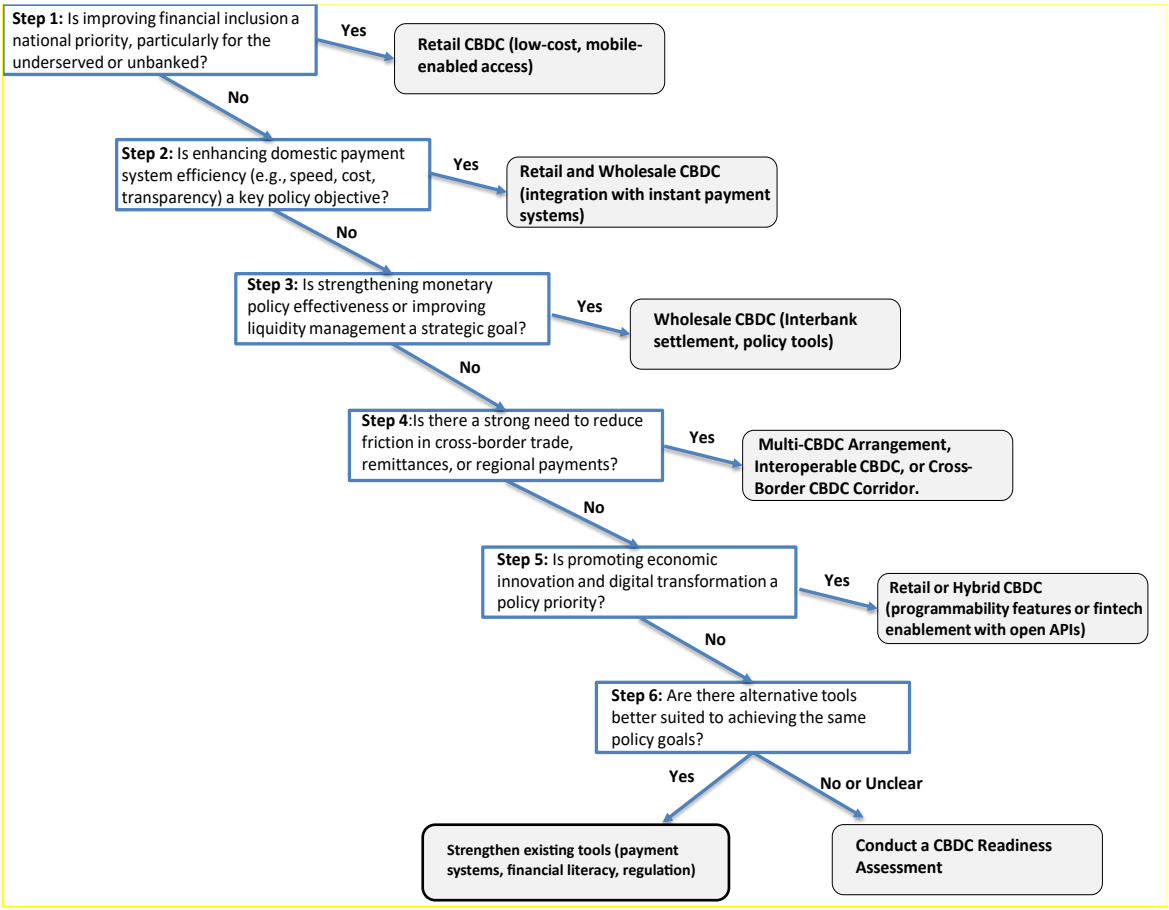
Source: Author's elaboration.

This structured approach allows central banks to evaluate whether a digital currency is the right tool to achieve their policy goals and ensure that design choices align with national priorities.

3.2.1 Strategic decision framework for digital currency exploration

To support strategic decision-making, a central bank digital currency decision tree (see figure IV) can serve as a high-level guide for policymakers to determine whether to pursue a digital currency and, if so, what type of digital currency is most appropriate, depending on their country's policy priorities and financial system context.

Figure IV
Central bank digital currency decision tree



Source: Author's elaboration.

Abbreviations: API, application programming interface; CBDC, central bank digital currency.

The decision process begins with an assessment of whether the central bank's primary goal is to address unmet strategic needs, such as financial inclusion, digital transformation, monetary policy effectiveness or cross-border payment efficiency, or to strengthen existing structures, such as payment infrastructure or financial market resilience. If policy drivers point towards systemic innovation or expanding financial access, the exploration of a retail digital currency may be warranted. If the objective is to enhance interbank settlement, monetary transmission or cross-border efficiency, a wholesale or hybrid digital currency may be more suitable. If current systems already meet critical policy goals, however, or if institutional or technological readiness is unclear, central banks are encouraged to first conduct a digital currency readiness assessment. This assessment helps to identify capability gaps, clarify strategic value and determine whether other digital reforms may offer more immediate benefits. Ultimately, this decision framework ensures that digital currency exploration is justified by

clear national priorities and grounded in the country's institutional capacity and economic context.

3.3 Application of the framework across African archetypes

Since African countries differ in their economic structures, financial maturity and policy priorities, the central bank digital currency policy alignment framework must be adapted to specific national conditions. Table 9 shows examples of how different African archetypes can use digital currencies to advance their respective development goals.

Table 9
African archetypes and considerations for digital currency implementation

Economic archetype	National priority	Digital currency implementation consideration
High mobile money penetration (e.g. Kenya, Ghana, Uganda)	Financial inclusion and interoperability	Digital currencies should integrate with mobile money platforms and support offline transactions
Highly dollarized (e.g. Democratic Republic of the Congo, Liberia, Zimbabwe)	Currency sovereignty and monetary control	Digital currencies should offer a stable, trusted digital alternative to foreign currencies
Fixed exchange rate and currency-pegged (e.g. Morocco, CFA Franc Zone)	Exchange rate stability and cross-border payments	Digital currencies should align with regional monetary union frameworks and promote cross-border payment efficiency
Resource-dependent (e.g. Angola, Chad, Nigeria)	Fiscal transparency and revenue tracking	Digital currencies should use programmable features for automated tax collection and subsidy distribution
High remittance (e.g. Gambia, Lesotho, Somalia)	Reduce remittance costs and strengthen financial resilience	Digital currencies should integrate with international payment networks and enable low-cost cross-border transfers

Source: Author's elaboration.

This archetype-specific approach ensures that central banks develop digital currency strategies that are customized, relevant and aligned with national economic goals.

3.4 Framework for the prioritization of policy objectives

The policy objective prioritization framework provides a structured approach for central banks to:

1. Identify national policy priorities that could be supported by a digital currency
2. Evaluate the strategic importance of each objective
3. Rank and sequence objectives on the basis of national needs, constraints and economic conditions
4. Align digital currency objectives with the country's broader financial and monetary policies

A structured prioritization framework ensures that digital currency development is goal-oriented, evidence-based and aligned with national financial sector strategies.

Step 1. Identify crucial policy drivers for a central bank digital currency

The first step is to identify the policy areas in which a digital currency could have the most significant impact. These may include:

1. **Financial inclusion**. Expanding access to digital financial services for unbanked and underbanked populations
2. **Payment system modernization**. Enhancing efficiency, security and resilience in domestic and cross-border payments
3. **Monetary policy transmission**. Strengthening the central bank's ability to manage liquidity and control interest rates
4. **Currency sovereignty**. Reducing reliance on foreign currencies in highly dollarized economies
5. **Fiscal transparency and efficiency**. Improving tax collection, subsidy distribution and government-to-person payments
6. **Cross-border trade and remittances**. Lowering the cost and increasing the speed of cross-border financial transactions
7. **Financial stability**. Reducing systemic risks associated with privately issued digital money (e.g. stablecoins)

These drivers must be assessed within the broader context of national economic policies to determine their relative importance.

Step 2. Assess policy alignment with national objectives

A central bank should assess the ways in which digital currency objectives align with national economic and financial sector policies, including:

- National financial inclusion strategies
- Digital transformation and modernization of payments
- Monetary policy frameworks
- Fiscal policy and public finance management goals
- Cross-border payment and trade integration policies (e.g. the African Continental Free Trade Area)

Using a central bank digital currency policy alignment matrix, policymakers can assess the degree of alignment between digital currency objectives and national priorities (see table 10).

Table 10
Central bank digital currency policy alignment matrix

National policy goal	Digital currency relevance	Potential policy synergies
Expand financial inclusion	High	Aligns with mobile money regulation and banking sector reforms
Modernize payment infrastructure	High	Supports faster settlement systems and financial sector digitalization
Strengthen monetary policy	Medium	Depends on design choices (retail versus wholesale)
Enhance fiscal transparency	High	Complements e-government initiatives and tax reforms
Reduce dollarization	High	A trusted digital currency restores confidence in local currency
Improve cross-border transactions	Medium	Depends on regional cooperation and interoperability agreements

Source: Author's elaboration.

By mapping objectives to policy goals, central banks can determine areas in which digital currency intervention will have the most impact.

Step 3. Evaluate readiness and constraints for each objective

Each policy objective must be assessed against existing national capabilities and potential barriers to implementation. This includes:

1. **Technical readiness**. Does the country have adequate payment infrastructure, digital identity systems and cybersecurity capacity?
2. **Regulatory readiness**. Are legal frameworks in place to support digital currency issuance, compliance with AML/CFT policies and data privacy?
3. **Market readiness**. What is the level of trust in and demand for digital financial services among citizens and businesses?
4. **Institutional readiness**. Is the central bank capable of overseeing a digital currency system?
5. **Financial system readiness**. Are banks and payment service providers prepared to integrate a digital currency into the financial system?

This evaluation can be structured using a digital currency readiness versus policy impact matrix (see table 11).

Table 11
Digital currency readiness versus policy impact matrix

Policy objective	Readiness level	Potential impact	Priority level
Financial inclusion	High	High	High
Payment system modernization	Medium	High	High
Monetary policy strengthening	Low	Medium	Low
Fiscal transparency enhancement	High	High	High
Dollarization reduction	Low	High	Low
Cross-border transaction improvement	Medium	Medium	Low

Source: Author's elaboration.

This assessment helps to prioritize objectives on the basis of both feasibility and impact.

Step 4. Rank and sequence objectives

Once objectives have been assessed for policy alignment, readiness and impact, they should be ranked and sequenced into short-term, medium-term and long-term priorities.

- **Short-term objectives (1–2 years)**
  - Expanding financial inclusion
  - Increasing payment system interoperability
  - Integrating public sector digital payments
- **Medium-term objectives (3–5 years)**
  - Strengthening monetary policy tools
  - Addressing dollarization challenges
  - Establishing cross-border central bank digital currency frameworks
- **Long-term objectives (5 or more years)**
  - Transitioning to a full-scale retail or wholesale digital currency
  - Integrating with global digital financial systems

Central banks should focus first on high-impact, high-readiness objectives, ensuring that early-stage digital currency efforts build foundational capabilities for long-term goals.

Step 5. Develop a policy road map for implementation

To translate prioritized objectives into an actionable implementation timeline, a central bank digital currency policy road map may be used. The road map should include:

1. **Defined phases of exploration**

    - Phase 1. Feasibility study (assess objectives, readiness and regulatory gaps)
    - Phase 2. Proof-of-concept (test digital currency design in a limited environment)
    - Phase 3: Pilot deployment (launch small-scale digital currency in targeted use cases)
    - Phase 4. Gradual expansion (integrate digital currency into the broader financial landscape)

2. **Measurable milestones**

    - Define key performance indicators for financial inclusion, payment efficiency and adoption
    - Establish "go/no-go" criteria for transitioning between phases

3. **Continuous policy alignment**

    - Conduct periodic reviews of the impact of central bank digital currency on policy goals
    - Adapt the digital currency design as financial and economic conditions evolve

The prioritization framework serves as a foundation for the subsequent steps in digital currency exploration, ensuring that central banks make informed, strategic decisions tailored to their national contexts. The following sections of the *Central Bank Digital Currency Handbook for Africa* will delve more deeply into digital currency readiness assessments, design considerations, cost-benefit analyses and phased implementation strategies.

4 Infrastructure and policy gaps

Before embarking on a central bank digital currency initiative, central banks must conduct a comprehensive readiness assessment to evaluate their financial system's capacity to support the adoption of a digital currency. A well-structured readiness assessment ensures that a digital currency can be introduced safely, efficiently and in alignment with national economic priorities.

The framework provides a structured evaluation across five critical dimensions:

1. **Technical readiness**. Infrastructure, security and operational capacity
2. **Legal and regulatory readiness**. Compliance with existing laws and policies
3. **Financial system readiness**. Banking sector preparedness and market integration
4. **Social and cultural readiness**. Public trust, digital literacy and adoption challenges

Each dimension represents a critical pillar for digital currency success and requires a detailed assessment before moving on to pilot testing or implementation.

4.1 Technical readiness

Technical readiness is a foundational aspect of digital currency exploration and implementation and determines the feasibility of deploying digital currencies within existing infrastructure and technological landscapes. The present section provides a comprehensive methodology for evaluating technical readiness, covering essential subcategories, dependencies and global best practices.

Importance of technical readiness

- **Critical role**. The technological infrastructure underpins the operational viability, efficiency and security of a digital currency.
- **Broader scope**. Beyond core payment systems, readiness assessments must take into consideration supporting infrastructure, such as digital identity systems, connectivity and cybersecurity frameworks.
- **Global considerations**. Lessons from global digital currency projects highlight the need for scalable, secure and inclusive technical solutions.

4.1.1 Digital infrastructure

The foundation of any digital currency implementation is a robust digital infrastructure that ensures reliable and inclusive access to the currency. Crucial components of digital infrastructure include Internet penetration, mobile network coverage and data centre availability. High-speed Internet and extensive mobile network reach are critical, in particular in rural and underserved areas where financial inclusion efforts are often focused. For example, in Kenya, mobile networks have facilitated the widespread adoption of such mobile money systems as M-PESA, demonstrating how a robust infrastructure can transform financial accessibility. Data centres and cloud services play an equally vital role, ensuring secure and scalable hosting for digital currency systems. Central banks must evaluate the resilience of their infrastructure, including the availability of energy and measures for redundancy, to mitigate risks associated with power outages and natural disasters. Expanding and modernizing digital infrastructure, in particular in remote areas, should be prioritized to maximize the reach and reliability of a digital currency.

4.1.2 Payment system capabilities

Payment system capabilities are at the heart of digital currency operations, enabling secure, efficient and scalable transactions. One critical factor is interoperability: the ability of the digital currency system to integrate seamlessly with existing financial infrastructure, including mobile money platforms, banks and payment processors. For instance, the eCedi pilot in Ghana prioritized interoperability with existing digital payment systems to ensure widespread adoption and functionality. Scalability is another essential consideration, as digital currency systems must handle increased transaction volumes without compromising performance. Real-time payment systems, which enable instantaneous transactions, are a crucial component of modern financial systems and should be evaluated for their readiness to support digital currency operations. Central banks should assess whether their payment systems have the capacity to support digital currency use cases, identify potential bottlenecks and plan upgrades to ensure future scalability and resilience.

4.1.3 Advanced technologies

Advanced technologies offer opportunities to enhance the functionality, security and accessibility of digital currencies. Distributed ledger technology, for example, provides transparency and immutability, making it a popular choice for digital currency pilots in such countries as China and in the European Union. Distributed ledger technology may not be suitable in all contexts, however, in particular in regions that have limited computational capacity or high energy costs. Offline functionality is another crucial technological consideration, especially in areas that have intermittent connectivity. The SandDollar of The Bahamas has successfully incorporated offline wallets to facilitate transactions in island communities that have limited access to telecommunications networks. Digital identity infrastructure also plays a pivotal role, providing a secure and efficient means to verify users. Such countries as India have demonstrated the transformative potential of digital identification systems, such as Aadhaar, in enabling secure and inclusive financial services. Advanced technologies must be carefully evaluated to ensure that they align with national priorities, technical feasibility and user needs.

4.1.4 Cybersecurity and resilience

The integrity and security of a digital currency system depend heavily on robust cybersecurity measures and operational resilience. Central banks must evaluate their existing risk management frameworks to identify vulnerabilities and develop strategies to mitigate cyberthreats, fraud and data breaches. This includes implementing end-to-end encryption, using multifactor authentication and conducting regular security audits. Operational resilience is equally important, as it ensures that the digital currency system can withstand such disruptions as power outages, natural disasters or cyberattacks. Disaster recovery mechanisms, redundancy in critical infrastructure and real-time monitoring systems are essential components of a resilient digital currency framework. Lessons can be drawn from the e-CNY pilot in China, which integrated advanced cybersecurity features and redundancy measures to ensure continuous operation. Central banks should prioritize cybersecurity and resilience planning to safeguard public trust and the seamless functioning of the digital currency landscape.

Leading global cybersecurity frameworks provide a structured approach to managing and mitigating cyber risks. Among the most widely recognized are the National Institute of Standards and Technology cybersecurity framework 2.0, the International Organization for Standardization/International Electrotechnical Commission (ISO/IEC) 27001 standard and the Center for Internet Security critical security controls. In addition, sector-specific guidance offers tailored insights for the financial sector. These frameworks guide organizations in identifying, protecting, detecting, responding to and recovering from cyberthreats, while also establishing strong governance mechanisms.

A comprehensive cybersecurity framework for digital currency must encompass several critical components. First, robust governance and risk management should be foundational, ensuring that cybersecurity is integrated into the digital currency's overall strategy, and clear accountability and continuous oversight should be provided by leadership. Second, establishing the framework requires a thorough identification of all critical assets, an understanding of their vulnerabilities and an assessment of the specific threats they face, including those posed by third-party dependencies within the supply chain. Third, effective protection involves implementing such safeguards as stringent access controls, data encryption, system hardening measures and mandatory cybersecurity training for all personnel. Fourth, advanced detection capabilities, utilizing continuous monitoring and threat intelligence, are crucial in order to promptly identify anomalous activities. Finally, well-defined response and recovery plans are

essential to contain incidents, mitigate their impact and rapidly restore operations, thus ensuring business continuity and resilience.

Advanced encryption standards, data anonymization and a broad array of privacy-enhancing techniques are indispensable tools for safeguarding sensitive financial data and maintaining trust. “Privacy enhancing techniques” refers to a suite of tools and methodologies designed to minimize the exposure of sensitive information throughout its life cycle while still enabling data processing and analysis. Examples include pseudonymization (replacing identifiers with artificial labels, which can sometimes be reversible), homomorphic encryption (allowing computations on encrypted data without decrypting them), differential privacy (adding statistical noise to data to protect individual records within a data set) and secure multiparty computation (enabling multiple parties to jointly compute a function over their private inputs, ensuring that each of the parties receives only its own output and no additional information, keeping those inputs private). For digital currencies, privacy enhancing techniques are paramount in order to achieve “privacy by design”. They foster secure data-sharing and collaboration within the highly interconnected financial landscape, support ethical artificial intelligence and machine-learning initiatives and ensure compliance with evolving privacy regulations while preserving the utility of financial data for innovation and risk management.

For policymakers, the adoption of these frameworks when considering a digital currency enhances systemic resilience. It promotes a consistent, risk-based approach to cybersecurity that can adapt to evolving threat landscapes and technological advancements. Policymakers should encourage the use of frameworks that prioritize continuous improvement, facilitate cross-border information-sharing and rigorously address third-party risks.

By fostering an environment in which a digital currency adheres to internationally recognized best practices, regulators can strengthen the collective security posture of the financial system, safeguard its integrity and ensure its smooth functioning even in the face of sophisticated cyberchallenges.

4.1.5 Supporting infrastructure

Supporting infrastructure extends beyond the immediate technological components of a digital currency system to encompass telecommunications, energy systems and physical infrastructure. Telecommunications networks are the backbone of digital payments, and their robustness has a direct impact on the accessibility and reliability of a digital currency. Central banks should assess the capacity and coverage of telecommunications providers, in particular in underserved and rural areas. Reliable energy systems are another critical factor, as the high energy requirements for such technologies as distributed ledger technology can pose challenges in regions that have unstable power supplies. Moreover, the resilience of infrastructure to climate-related disruptions is becoming increasingly important as digital currencies are integrated into global financial systems. Supporting infrastructure should also align with broader digital economy goals, including e-governance and sustainable development. Ensuring that these foundational systems are robust and inclusive will enhance the overall effectiveness and adoption of digital currencies, in particular in diverse and challenging environments.

4.1.6 Technical readiness assessment

Methodology for technical readiness assessment

The technical readiness assessment consists of a structured evaluation of a central bank’s digital infrastructure and the technological capabilities required for digital currency

implementation. This assessment helps to identify strengths, gaps and areas that require investment before the launch of a digital currency initiative.

Assessment approach

The assessment is divided into five core dimensions, each of which is a critical technical readiness area:

- Digital infrastructure: availability of broadband, mobile penetration and cloud computing
- Payment system capabilities: strength of real-time payment systems, interoperability and mobile money networks
- Advanced technologies: adoption of distribution ledger technologies, artificial intelligence and smart contracts
- Cybersecurity and resilience: existing frameworks to protect financial infrastructure from cyberthreats
- Supporting infrastructure: national digital identity, data protection laws and Internet access

Scoring framework

Each dimension is scored from 1 to 5:

1 = Not ready (significant gaps, major infrastructure investment needed)

2 = Low readiness (basic infrastructure exists but has major limitations)

3 = Moderate readiness (partial infrastructure available, improvements required)

4 = High readiness (most systems in place, minor enhancements needed)

5 = Fully ready (meets international standards, can support digital currency deploy ment)

Gap identification and recommendations

After scoring, a readiness gap analysis highlights priority areas requiring policy interventions, infrastructure upgrades or regulatory enhancements (see table 12).

Table 12

Example of a technical readiness assessment matrix

Technical dimension	Assessment criteria	Score (1-5)	Gaps identified	Recommendations
Digital infrastructure	Broadband and mobile penetration	4	Limited rural broadband coverage	Expand mobile connectivity in underserved regions
	Cloud computing and data centres	3	Limited reliance on cloud-based financial systems	Invest in secure cloud hosting for digital currency platforms
Payment system capabilities	Real-time gross settlement infrastructure	5	Fully operational real-time gross settlement system	No immediate action required
	Interoperability across banks and fintech firms	3	Mobile money integration challenges	Mandate application programming interface (API)-based open banking standards
	Cross-border payment infrastructure	2	High remittance costs and slow transaction times	Explore harmonization of regional payment system
Advanced technologies	Use of blockchain/distribution ledger technology	2	Limited blockchain experimentation in banking	Conduct digital currency pilot using both distribution ledger technology and centralized systems
	Artificial intelligence and automation in financial services	3	Low adoption of artificial intelligence with regard to risk management	Develop artificial-intelligence-based fraud detection and anti-money-laundering tools
	Smart contracts and programmability	1	No regulatory framework for automated payments	Define legal basis for smart contracts in digital currency transactions
Cybersecurity and resilience	National cybersecurity framework	4	Some gaps in coordination with private banks	Strengthen cybersecurity collaboration between regulators and banks
	Fraud prevention and AML/CFT controls	3	Limited capabilities to detect digital fraud	Enhance digital fraud monitoring using artificial intelligence
	Disaster recovery and backup systems	4	Strong infrastructure but dependent on centralized networks	Implement decentralized backup solutions for resilience
Supporting infrastructure	Digital identity system	4	Digital identification widely used but not fully integrated with payments	Enable seamless digital currency and know-your-customer integration with national identification
	Data protection and privacy laws	3	Privacy laws exist but lack fintech-specific provisions	Update data regulations to cover digital currency user data
	Internet penetration and financial access	2	Low financial access in rural communities	Expand offline digital currency payment options using near field communication and short message service (SMS)-based solutions

Source: Author's elaboration.

The technical readiness assessment is a vital tool for evaluating a country's preparedness for digital currency deployment. By systematically reviewing digital

infrastructure, payment capabilities, advanced technologies, cybersecurity and supporting frameworks, central banks can develop a targeted action plan to address weaknesses and enhance digital financial resilience before rolling out a digital currency.

The next section of the handbook will explore the policy considerations and regulatory adjustments necessary to support digital currency implementation within existing legal frameworks.

4.2 Regulatory and legal readiness

The regulatory and legal readiness of a central bank digital currency is essential to ensure its successful exploration, development and deployment. The present section sets out the critical regulatory considerations that central banks must address, the gaps often observed in African contexts and approaches to bridge these gaps. Global insights and actionable outputs offer practical guidance for adapting regulatory frameworks to support digital currency initiatives.

Importance of regulatory and legal readiness

- **Supportive legal framework**. A robust legal framework ensures that digital currencies are legally recognized, enforceable and aligned with existing national and international laws.
- **Policy coherence**. Regulatory alignment with broader financial and economic policies promotes consistency and minimizes conflicts.
- **Trust and stability**. Effective regulation builds public trust, ensures financial stability and mitigates risks associated with the implementation of a digital currency.

4.2.1 Data privacy and protection

Data privacy and protection are critical for any digital currency system, in particular in such regions as Africa, where comprehensive data protection laws may still be evolving. Digital currencies inherently involve sensitive user information, making it essential to establish legal and technical safeguards. Robust data protection frameworks must clearly define ownership, access and the permissible usage of user data, ensuring that it is not misused by Governments, financial institutions or third parties. Lessons can be drawn from the General Data Protection Regulation of the European Union, which offers a strong foundation for creating privacy laws that are adaptable to digital currencies. In addition, implementing such advanced technologies as encryption and anonymization can strike a balance between user privacy and regulatory requirements. For African countries, prioritizing the development or enhancement of data privacy laws is essential to fostering public trust and supporting regional interoperability.

4.2.2 Compliance with anti-money-laundering and countering the financing of terrorism policies

Compliance with AML/CFT policies is central to the implementation of central bank digital currencies, as it ensures alignment with global standards and mitigates illicit activities. African countries often face challenges in adapting traditional AML/CFT frameworks to cover digital financial transactions. To address this, central banks can develop enhanced know-your-customer protocols tailored to digital currencies and integrate real-time monitoring tools to detect suspicious activities. Collaboration with such international bodies as the Financial

Action Task Force ensures compliance with global standards while addressing region-specific risks. For instance, real-time transaction reporting systems, combined with machine-learning tools, can enhance the detection of unusual patterns and strengthen regulatory oversight. Aligning these efforts with regional payment initiatives can further bolster cross-border security and compliance.

4.2.3 Consumer protection

Digital currencies present unique challenges and opportunities for consumer protection. As digital currencies become integrated into everyday transactions, ensuring the safety and rights of users will be paramount. A robust consumer protection framework should address fraud prevention, the resolution of errors and unauthorized access to digital currency wallets. Transparency is vital: users must be provided with clear information on digital currency terms of use, potential risks and redress mechanisms. Public education campaigns are also critical to enhance digital literacy and build trust in the system. Tailored protections for vulnerable groups, such as rural populations or the elderly, ensure inclusivity. Examples from the implementation by The Bahamas of the SandDollar highlight the importance of educating users and establishing dispute resolution mechanisms to ensure wide acceptance and trust.

4.2.4 Interoperability and settlement

Interoperability is a cornerstone of digital currency functionality, ensuring seamless integration with existing financial systems and cross-border frameworks. African countries, which are often characterized by fragmented payment systems, require robust legal adjustments to facilitate digital currency interoperability. This includes updating the laws that govern property rights and settlement finality to reflect the digital nature of digital currencies. Regional collaboration, such as alignment with the policies of the African Continental Free Trade Area and the Pan-African Payment and Settlement System, is critical for enabling cross-border transactions. By developing standardized legal definitions for digital currencies, central banks can create a unified framework that supports local and regional economic activity. Furthermore, partnerships with telecommunications and payment providers can strengthen the integration of digital currencies into mobile money systems and traditional banking infrastructure.

4.2.5 Governance and oversight

A clear governance structure is vital to ensuring the smooth operation and accountability of a digital currency. The clear delineation of roles and responsibilities among central banks, regulators and other stakeholders is necessary to prevent conflicts and inefficiencies. Central banks should lead digital currency governance and be supported by oversight committees that monitor compliance with regulatory standards and address emerging risks. Regular audits and transparent reporting mechanisms will foster trust and confidence in the system. Governance frameworks should also be dynamic, allowing for updates to address technological advancements and evolving user needs. Lessons from the e-CNY pilot in China underscore the importance of central governance in effectively managing digital currency issuance, distribution and operations.

4.2.6 Regulatory and legal readiness assessment

Methodology for legal and regulatory readiness assessment

A legal and regulatory readiness assessment evaluates the ability of a central bank to implement a digital currency within the existing legal and regulatory framework. This process

identifies gaps, inconsistencies and necessary reforms to ensure a digital currency's compliance with financial regulations, consumer protection laws and international best practices.

Regulatory and legal readiness checklist

- Does the country have a comprehensive data protection law?
- Are AML/CFT regulations adaptable to digital transactions?
- Are there clear legal definitions for digital currencies and settlement processes?
- Is there a framework for consumer protection in digital finance?

Assessment approach

The assessment is divided into five core dimensions, each of which is a critical legal and regulatory readiness area:

- Data privacy and protection: ensuring compliance with national and international data protection laws
- AML/CFT compliance: meeting AML/CFT obligations
- Consumer protection: safeguarding users from fraud, mismanagement and financial abuse
- Interoperability and settlement: legal recognition of digital currency as a valid payment instrument in national and cross-border transactions
- Governance and oversight: defining the role of the central bank and other regulators with regard to digital currency issuance and management

Scoring framework

Each dimension is scored from 1 to 5:

1 = Not ready (significant legal gaps, no clear regulations)

2 = Low readiness (regulations exist but are outdated or insufficient)

3 = Moderate readiness (partial legal framework in place, requires updates)

4 = High readiness (comprehensive regulations, minor refinements needed)

5 = Fully ready (complete and internationally aligned regulatory framework)

Gap identification and recommendations

After scoring, a legal and regulatory gap analysis is conducted to highlight areas requiring new policies, amendments or regulatory guidance (see table 13).

Table 13
Example of a regulatory and legal readiness assessment matrix

Regulatory and legal dimension	Assessment criteria	Score (1–5)	Gaps identified	Recommendations
Data privacy and protection	Alignment with national data protection laws	4	No fintech-specific provisions in data laws	Update privacy regulations to cover digital currency transactions
	Compliance with international standards (e.g. General Data Protection Regulation)	3	No explicit cross-border data-sharing rules	Introduce digital currency data-sharing agreements for regional transactions
AML/CFT compliance	Know-your-customer/anti-money-laundering integration with digital currency transactions	3	Lack of clear identity verification standards for digital currency wallets	Implement tiered know-your-customer policies for retail digital currency users
	Transaction monitoring and fraud prevention	2	No real-time transaction monitoring system	Deploy artificial-intelligence-driven anti-money-laundering screening tools
	Compliance with the Financial Action Task Force "travel rule"	3	No framework for cross-border digital currency reporting	Align with Financial Action Task Force digital asset guidelines
Consumer protection	User rights and liability protections	3	No dedicated digital currency consumer protection laws	Develop a digital-currency-specific financial consumer protection framework
	Redress mechanisms for digital currency disputes	2	No established complaint resolution process for digital payments	Create regulatory oversight for digital currency complaints
	Transparency in fees and terms of use	4	Existing laws apply but lack enforcement mechanisms	Strengthen enforcement for digital payment disclosures
Interoperability and settlement	Recognition of digital currency as legal tender	3	No explicit legislation recognizing digital currency	Amend national currency laws to include digital currency
	Digital currency usage in the private sector and commercial banks	4	Uncertainty over the impact of digital currency on financial stability	Define the role of digital currency in comparison to traditional banking
	Cross-border settlement legality	2	No agreements with regional monetary unions	Negotiate legal frameworks for digital currency cross-border use
Governance and oversight	Central bank authority over digital currency issuance	5	Clear legal mandate for monetary control	No immediate action required
	Coordination among regulators (central bank, financial and telecommunications authorities)	3	Fragmented oversight of digital assets	Establish a cross-sectoral regulatory body for digital finance
	Contingency planning and crisis management	2	No legal provisions for digital currency failure or cybersecurity breaches	Introduce digital currency operational risk and recovery policies

Source: Author's elaboration.

The regulatory and legal readiness assessment is essential to ensuring that a digital currency operates within a secure, transparent and well-regulated financial environment. By systematically evaluating data privacy, compliance with anti-money-laundering and consumer protection policies, interoperability and governance, central banks can develop a comprehensive legal framework to support the implementation of digital currencies.

4.3 Financial system readiness

Financial system readiness is a cornerstone of digital currency implementation, reflecting the ability of a jurisdiction's financial landscape to integrate and support a digital currency. The present section includes the components of financial system readiness, the challenges commonly faced in Africa and globally relevant insights. It serves as a framework for assessing readiness across both formal and informal systems, including cross-border payment infrastructure, and concludes with actionable tools to address gaps.

A well-prepared financial system ensures that digital currencies can be seamlessly integrated into existing payment infrastructures while driving innovation and inclusion. In Africa, financial systems often consist of a mix of traditional banking, mobile money services and informal networks. Ensuring that all components can support a digital currency is critical in order to maximize adoption and achieve such policy objectives as financial inclusion, payment efficiency and cross-border trade facilitation.

4.3.1 Integration with traditional banking systems

Traditional banking institutions remain the backbone of most financial systems, providing the infrastructure for savings, loans and large-scale transactions. Assessing their readiness requires evaluating:

- **Core banking systems**. Are they capable of integration with a digital currency platform? Do they have real-time processing capabilities and application programming interface compatibility to ensure interoperability?
- **Interbank settlement systems**. Can digital currencies be incorporated into existing interbank payment mechanisms, such as real-time gross settlement systems?
- **Commercial bank participation**. Are banks willing and able to adopt and integrate digital currency systems into their operations?

For example, both the eNaira in Nigeria and the e-CNY in China successfully engaged commercial banks as intermediaries, leveraging their existing infrastructure for digital currency distribution and user interaction.

4.3.2 Integration with mobile money systems

Mobile money plays a significant role in many African countries, providing financial services to unbanked and underserved populations. Critical considerations for readiness include:

- **Interoperability**. Can digital currencies operate seamlessly with such mobile money platforms as M-PESA in Kenya or Mobile Telephone Networks-Mobile Money (MoMo) in Ghana?

- **Agent networks**. Are mobile money agents equipped to support such digital-currency-related activities as onboarding and cash-in/cash-out transactions?
- **Technical standards**. Do mobile money systems adhere to technical and regulatory standards that facilitate digital currency integration?

The eNaira of Nigeria continues to have a focus on enabling integration with mobile payment platforms, enhancing financial inclusion.

4.3.3 Informal financial systems

Informal financial networks, including savings groups and community lending systems, remain vital in many African contexts. For digital currencies to succeed, the following issues need to be considered:

- **Incentives for transition**. How can digital currencies provide value, such as lower transaction costs or increased security, to users of informal systems?
- **Ability to bridge the gap**. Can central banks partner with non-governmental organizations and community groups to bridge the divide between formal and informal systems?
- **Digital literacy and access**. Are users in informal systems equipped with the tools and knowledge to adopt digital currencies?

Case studies from Ghana highlight the importance of using community leaders and organizations to promote trust and adoption among users of informal systems.

4.3.4 Cross-border payment infrastructure

African trade corridors are critical conduits for regional integration, accounting for most of intracontinental commerce; however, they are constrained by fragmented currencies, informal settlements and weak cross-border payment infrastructure. Over 70 per cent of intra-African trade is still settled in United States dollars or euros, and informal trade accounts for an estimated 30 to 60 per cent of total intra-African flows, depending on the corridor.[7] Digital currencies can eliminate inefficiencies by enabling real-time, cost-effective and traceable cross-border payments (see box 2). When they are strategically implemented along high-volume corridors, digital currencies reduce reliance on foreign currency, support trade formalization and align with the objectives of the Agreement Establishing the African Continental Free Trade Area to accelerate economic convergence and financial inclusion. Cross-border trade and remittances are significant drivers of economic activity in Africa, but current systems are often expensive and inefficient. Assessing readiness requires examining:

- **Regional initiatives**. Are there frameworks in place, such as the Pan-African Payment and Settlement System, that can support digital currency interoperability?
- **Settlement mechanisms**. Can digital currencies streamline cross-border settlement processes, reducing costs and delays?

[7] ECA, "Data of information trade is key in accurate monitoring of intra-Africa trade says Karingi of ECA", 21 April 2022. Available at www.uneca.org/stories/data-of-informal-trade-is-key-in-accurate-monitoring-of-intra-african-trade-says-karingi-of?utm.

- **Collaboration**. Are partnerships in place with neighbouring countries or regional organizations to harmonize digital currency standards?

The wholesale digital currency pilot of South Africa[8] demonstrated the potential of cross-border digital currencies to enhance trade efficiency and reduce settlement times.

Box 2

Priority African trade corridors that could be strengthened by a digital currency

- The Abidjan-Lagos corridor, spanning Côte d'Ivoire, Ghana, Togo, Benin and Nigeria, facilitates more than $20 billion in annual trade,[a] making it one of the most active trade routes in West Africa. Despite its volume, the corridor is marked by currency fragmentation and significant informal cross-border flows. A cross-border digital currency could streamline transactions by reducing foreign exchange conversion, improving settlement efficiency and reinforcing regional value chains.
- The North-South corridor connects the Democratic Republic of the Congo, Zambia, Zimbabwe, Botswana and South Africa and facilitates more than $14 billion in trade annually.[b] It is a crucial route for mineral exports, agriculture and regional logistics, and settlements are predominantly conducted in United States dollars and South African rand. Introducing a digital currency along this corridor could reduce reliance on foreign currencies, enable programmable trade finance solutions and strengthen regional clearing and settlement mechanisms.
- The Lagos-Kano-Niamey-Algiers corridor links Nigeria, the Niger and Algeria, supporting an estimated $5.6 billion in trade annually.[c] Characterized by limited banking infrastructure, high cash dependence and informal trade routes, the corridor faces persistent settlement inefficiencies. A digital currency could enable traceable transactions, reduce cash leakage and enhance financial inclusion across remote and underserved areas.

These three strategic trade corridors, Abidjan-Lagos, North-South and Lagos-Kano-Niamey-Algiers, collectively represent high-volume, high-potential routes critical to regional integration and trade expansion in Africa. Despite their importance, they are constrained by currency fragmentation, reliance on foreign currencies, informal trade flows and weak financial infrastructure. Implementing cross-border digital currencies in these corridors could address these challenges by improving settlement efficiency, reducing foreign exchange dependence and formalizing trade, thereby advancing the goals of economic convergence under the Agreement.

[a] International Trade Administration, "Ghana design and construction: Abidjan-Lagos corridor highway project", 21 March 2025. Available at www.trade.gov/market-intelligence/ghana-design-and-construction-abidjan-lagos-corridor-highway-project.

[b] Southern African Development Community, "SADC launches the time release study report for the North South trade corridor to bolster trade and ease bottlenecks in goods movement", 8 February 2025. Available at www.sadc.int/latest-news/sadc-launches-time-release-study-report-north-south-trade-corridor-bolster-trade-and.

[c] United Nations Conference on Trade and Development, "The trans-Saharan road corridor". Available at https://unctad.org/publication/trans-saharan-road-corridor.

4.3.5 Financial inclusion and accessibility

Financial inclusion is a priority for many African central banks. Readiness in this area involves:

- **Reaching underserved populations**. Are there strategies to use digital currencies to provide services to rural, low-income or marginalized communities?
- **Reducing costs**. Can digital currencies lower transaction fees compared with existing mobile money or remittance services?

[8] The project is described in detail in section 9.7.1.

- **Ensuring accessibility**. Are there offline payment capabilities and accessible interfaces for users who have limited literacy or connectivity?

The SandDollar of The Bahamas offers lessons on tailoring digital currencies to improve access for underserved communities, including offline functionality for remote areas.

4.3.6 Financial system readiness assessment

Methodology for financial system readiness assessment

This methodology provides a structured approach to evaluating the preparedness of financial systems to support development objectives, digital transformation and inclusive financial services. It combines qualitative and quantitative techniques to assess infrastructure, regulatory frameworks, institutional capacity and user adoption. The framework aims at identifying strengths, gaps and interdependencies across critical components, thereby informing policy decisions and targeted interventions.

- Data collection: conduct surveys and stakeholder consultations to gather insights into financial infrastructure and adoption challenges.
- Scoring framework: assess readiness across major components using a scoring system (1 to 5) to evaluate strengths and gaps.
- Dependency analysis: identify areas in which readiness is contingent on regulatory, technical or social dimensions (see table 14).

Table 14
Example of a financial system readiness assessment matrix

Financial system dimension	Score (1–5)	Strengths	Gaps identified	Recommendations
Traditional banking systems	4	Established real-time gross settlement and commercial bank networks	Limited application programming interface compatibility for digital currencies	Develop application programming interface standards for interoperability
Mobile money landscapes	3	Widespread adoption in urban areas	Low rural agent density	Expand agent networks and rural outreach
Informal financial systems	2	High usage in underserved communities	Limited digital literacy	Provide targeted education campaigns
Cross-border infrastructure	3	Regional initiatives, such as the Pan-African Payment and Settlement System	Lack of harmonized standards	Collaborate on cross-border digital currency pilots
Financial inclusion	4	Strong focus on underserved populations	Limited offline capabilities	Invest in offline payment technologies

Source: Author's elaboration.

Financial system readiness is essential for the successful implementation of digital currencies. By addressing integration with traditional banking and mobile money systems, engaging informal networks and aligning with cross-border infrastructure, central banks can ensure a cohesive financial system that supports digital currency objectives. Such tailored tools

as the financial system readiness assessment matrix and actionable recommendations provide a road map for strengthening financial readiness across diverse contexts.

4.4 Social and cultural readiness

Social and cultural readiness is a critical dimension in the exploration and implementation of central bank digital currencies. It reflects the degree to which a population is prepared to adopt and trust digital currency systems, taking into account societal norms, attitudes and cultural diversity. The present section is focused on crucial social and cultural factors, with a priority on the unique challenges in the African context, while incorporating global examples. Actionable outputs provide tools and recommendations to address readiness gaps effectively.

The adoption of a digital currency depends not only on its technical functionality but also on societal acceptance and cultural alignment. Public trust, digital literacy, inclusivity and attitudes towards financial and government institutions all play significant roles in determining the success of a digital currency initiative. In Africa, where informal economies and cultural diversity are prevalent, social and cultural considerations are particularly vital.

4.4.1 Public trust in financial and government institutions

Public trust is foundational for the adoption of digital currencies, as individuals must believe in the reliability and fairness of both the currency and its issuer. In some African countries, historical distrust of Governments and financial systems can hinder the adoption of a digital currency.

Approaches to building trust

- Promote transparency in digital currency operations through public consultations and regular updates.
- Engage trusted local leaders and influencers to disseminate information and build confidence.
- Draw lessons from the launch of e-CNY in China, where extensive public education campaigns helped to address concerns about data privacy and government control.

4.4.2 Digital literacy and access

Digital literacy is essential for users to understand and effectively use digital currencies. In Africa, where digital literacy rates vary widely, in particular between urban and rural areas, this is a significant challenge.

Approaches to enhancing literacy

- Implement targeted education campaigns focusing on how to use digital currencies safely and effectively.
- Partner with schools, community organizations and non-governmental organizations to provide training programmes.
- Develop user-friendly interfaces that cater to varying literacy levels, such as voice-guided apps in local languages.

4.4.3 Societal attitudes towards digital and cash economies

Many African societies rely heavily on cash, viewing it as secure and universally accepted. Shifting these attitudes requires addressing cultural preferences for tangible assets and concerns about the intangibility of digital currencies.

Approaches to addressing attitudes

- Highlight the benefits of digital currencies, such as reduced transaction costs and enhanced security.
- Offer transitional solutions, such as digital currencies that can operate offline and mimic cash-like functionalities.
- Learn from the launch of the SandDollar in The Bahamas, which emphasized offline capabilities to ease the transition from cash to digital.

4.4.4 Inclusivity for marginalized groups

Inclusivity ensures that digital currencies cater to all segments of the population, including women, rural communities and individuals in informal economies. Marginalized groups often face such barriers as a lack of access to technology, education or identification.

Approaches to enhancing inclusivity

- Design digital currencies that have features that address the needs of underserved populations, such as offline payments and mobile phone compatibility.
- Partner with grass-roots organizations to reach marginalized communities and provide support during onboarding.
- Draw inspiration from the Aadhaar-enabled payment systems in India, which significantly expanded financial access for underserved groups.

4.4.5 Cultural diversity and language

The continent's linguistic and cultural diversity requires digital currencies to be accessible and relevant to a broad spectrum of users.

Approaches to addressing diversity

- Provide digital currency interfaces in multiple local languages to ensure accessibility.
- Tailor public-awareness campaigns to resonate with cultural values and norms.
- Leverage local insights to design culturally appropriate user experiences.

4.4.6 Social and cultural readiness assessment

Methodology for social and cultural readiness assessment

This assessment framework is designed to evaluate the social and cultural factors that influence the adoption, acceptance and effective use of financial systems and digital solutions. It focuses on dimensions such as trust in institutions, behavioural norms, inclusion and digital literacy levels. By capturing both perceptions and lived experiences, the methodology provides

insights into potential barriers and enablers of adoption, ensuring that policies and interventions are socially responsive and culturally appropriate.

- Data collection: use surveys, focus groups and interviews to assess public perceptions, trust levels and digital literacy.
- Scoring framework: evaluate readiness across critical factors using a scoring system (1 to 5) to identify strengths and gaps.
- Dependency analysis: examine linkages with other readiness dimensions, such as technical infrastructure and regulatory frameworks (see table 15).

In addition, the assessment highlights disparities across population groups (e.g. gender, age, geographic location) to ensure that inclusion considerations are fully integrated into the analysis and subsequent policy recommendations.

Table 15
Example of a social and cultural readiness assessment matrix

Social and cultural dimension	Score (1–5)	Strengths	Gaps identified	Recommendations
Public trust	3	Trust in mobile money platforms	Distrust of government-led systems	Increase transparency and engage local leaders
Digital literacy	2	High smartphone penetration	Low literacy in rural areas	Develop targeted education campaigns
Societal attitudes	3	Positive view of mobile money	Preference for cash	Highlight security and cost benefits
Inclusivity	2	Growing fintech solutions for women	Barriers for participants in the informal economy	Design offline and accessible systems
Cultural diversity	4	Strong local engagement mechanisms	Limited language diversity in platforms	Expand multilingual interfaces

Source: Author's elaboration.

Social and cultural readiness checklist

- Are public trust levels in financial and government institutions sufficient to support digital currency adoption?
- Are digital literacy programmes in place to educate users about the functionalities and benefits of digital currencies?
- Do digital currency interfaces and outreach campaigns reflect local languages and cultural values?
- Are there mechanisms to ensure inclusivity for women, rural populations and participants in the informal economy?

Recommendations

- **Short-term**. Conduct baseline surveys to understand public perceptions and readiness.
- **Medium-term**. Roll out education campaigns in local languages, focusing on digital literacy and financial security.
- **Long-term**. Collaborate with community leaders and organizations to build trust and ensure widespread adoption.

Social and cultural readiness is as crucial to digital currency success as technical or regulatory preparedness. By addressing public trust, digital literacy, societal attitudes, inclusivity and cultural diversity, central banks can ensure that digital currencies are not only functional but also widely accepted and impactful. The tools and recommendations provided here offer a road map for integrating social and cultural considerations into digital currency planning and implementation.

4.5 Assessment methodology

Step 1. Collect data and insights

- Conduct surveys, interviews and focus groups with stakeholders.
- Use national data on digital penetration, financial inclusion and infrastructure.

Step 2. Use scoring and weighting tools

- Score each dimension (1–5) on the basis of readiness level:

    1. Not ready: there are significant gaps that require foundational improvements
    2. Partly ready: some elements are in place; however, there are notable gaps and inconsistencies that limit effectiveness, and targeted improvements and capacity-building measures are required to reach an operational level
    3. Moderately ready: moderate gaps still exist that need targeted action
    4. Largely ready: most key components are well established and functioning effectively, with only minor gaps or inefficiencies; limited adjustments and refinements are needed to achieve full readiness
    5. Ready: a strong foundation exists that has minimal gaps

- Apply weights to dimensions on the basis of their importance to prioritized objectives (established in step 1).

Step 3. Generate a readiness matrix

- Visualize strengths and gaps across dimensions.

Linkages to other readiness dimensions

Legal and regulatory readiness cannot be assessed in isolation; it is deeply intertwined with other dimensions of digital currency readiness:

- Technical readiness: data privacy laws must align with technical infrastructure, such as encryption and secure storage, to ensure seamless compliance.
- Financial system readiness: regulatory frameworks should accommodate the integration of digital currencies with mobile banking and payment systems, in particular in regions where informal financial systems play a significant role.
- Social and cultural readiness: public-awareness campaigns must address regulatory aspects to build trust and ensure adoption. For instance, data privacy regulations should be communicated clearly to reassure users about their rights and protections.

These interdependencies underscore the need for a holistic approach in which regulatory considerations are integrated with technical, financial and social strategies to ensure the cohesive and effective implementation of a digital currency (see table 16).

Table 16
Example of an overall readiness matrix

Dimension	Score (1–5)	Strengths	Gaps identified	Recommendations
Technical readiness	3	Interoperable payment systems	Limited rural connectivity	Expand digital infrastructure in rural areas
Regulatory readiness	2	Data protection framework in place	Lack of digital-currency-specific legal provisions	Amend regulations to address digital currencies
Financial readiness	4	Strong mobile banking adoption	Informal sector largely excluded	Develop incentives for informal sector inclusion
Social readiness	3	Growing trust in digital platforms	Low digital literacy in rural regions	Conduct targeted education campaigns

Source: Author's elaboration.

Step 4. Tailor findings to central bank digital currency stages

- Exploratory stage: highlight foundational gaps requiring early action.
- Pilot stage: focus on targeted interventions to test systems and features.
- Implementation stage: identify opportunities to fine-tune policies to enhance scalability and adoption.

4.6 Minimum readiness thresholds prior to pilot testing

Central banks must ensure foundational preparedness across four critical readiness dimensions – technical, legal and regulatory, financial system and social and cultural – before embarking on the pilot phase for a digital currency.

Each dimension functions as a pillar supporting the successful deployment and evaluation of a digital currency. Without minimum baseline conditions in place, pilot programmes risk yielding inconclusive results, which can reinforce exclusion or create reputational and operational setbacks.

The present section introduces a minimum readiness threshold framework, offering a structured method to help central banks to determine whether conditions are sufficiently mature to proceed to pilot testing.

Purpose of minimum thresholds

Minimum readiness thresholds (see table 17) serve as predefined criteria that must be met or actively addressed before progressing to the pilot phase. These thresholds help to:

- Prevent premature deployment of pilots that may fail owing to foundational weaknesses
- Align pilot design with achievable objectives on the basis of actual capabilities
- Improve the quality and usefulness of feedback from pilot testing
- Promote responsible innovation and minimize financial, reputational or political risks

Table 17
Example of minimum readiness threshold factors

Readiness dimension	Minimum threshold before pilot phase	Critical indicators
Technical readiness	Core infrastructure (e.g. connectivity, sandbox or distribution ledger technology environment) in place to simulate or run a pilot	Existence of test environment, payment switch access, secure infrastructure layers
Legal and regulatory readiness	Legal clarity or exemptions to enable digital currency testing within a safe environment	Temporary legal sandbox authorizations, data privacy protocols, AML/CFT controls
Financial system readiness	Engagement from major intermediaries and payment service providers secured	Signed memorandums of understanding or participation agreements, system integration pathways
Social and cultural readiness	Basic user research completed to identify barriers to adoption, and trusted communication channels established	Awareness campaigns piloted, stakeholder mapping complete, user readiness studies conducted

Source: Author's elaboration.

While the readiness thresholds provide a universal framework for responsible digital currency exploration, their application must be contextualized to reflect each country's economic realities, institutional capacity and digital landscape. The handbook's archetype-based approach (see section 4.9) allows central banks to adapt these thresholds in ways that are both practical and proportionate.

4.7 Output: infrastructure and policy gaps

The identification of infrastructure and policy gaps is a critical deliverable of the digital currency readiness assessment and scorecard process. By systematically evaluating preparedness across technical, regulatory, financial, social and cultural dimensions, central banks can pinpoint weaknesses that may hinder the successful implementation of a digital currency. The present section contains an expanded overview of critical gaps, insights into their implications and recommendations for addressing them effectively.

1. Importance of identifying gaps

Addressing infrastructure and policy gaps ensures that digital currency initiatives are built on a solid foundation. These gaps, whether in technical capacity, legal frameworks or societal acceptance, represent potential barriers to achieving such objectives as financial inclusion, payment system efficiency and economic resilience. Identifying and bridging these gaps enables central banks to allocate resources strategically and engage stakeholders in designing solutions that are tailored to national and regional contexts.

2. Critical infrastructure and policy gaps across readiness dimensions

Technical gaps

Infrastructure limitations. Many jurisdictions, in particular in rural and underserved areas, struggle with limited Internet penetration, weak mobile network coverage and inadequate data centres. These deficiencies restrict the accessibility and reliability of digital currency platforms, especially for marginalized populations. To address this, central banks must conduct granular assessments of digital infrastructure to identify bottlenecks and prioritize investments in expanding network coverage and data-storage capabilities.

Cybersecurity and resilience. Robust cybersecurity frameworks are often lacking in existing financial infrastructure, leaving systems vulnerable to breaches and fraud. In addition, operational resilience, which involves ensuring that systems remain functional during disruptions, is frequently overlooked. Central banks should prioritize the building of secure, scalable infrastructure that includes end-to-end encryption, multifactor authentication and disaster recovery mechanisms.

Interoperability. Many existing payment systems lack the ability to integrate seamlessly with a digital currency framework. Without standardized protocols, digital currencies may struggle to connect with mobile money platforms, banking networks and cross-border payment systems. Interoperability should be addressed by developing common standards and collaborating with technology providers to ensure compatibility with existing systems.

Regulatory and legal gaps[9]

Data privacy and protection. A significant gap in many jurisdictions, in particular in Africa, is the absence of comprehensive data privacy laws. As digital currencies involve vast amounts of sensitive user data, robust frameworks are required to ensure user protection and build trust. Central banks should focus on creating or updating legislation to define data ownership, access rights and usage policies, drawing on such models as the General Data Protection Regulation of the European Union.

Consumer protection. Current regulatory frameworks often fail to provide adequate safeguards against fraud, errors and unauthorized transactions in the digital space. A lack of clear mechanisms for dispute resolution and user recourse can further exacerbate

[9] See Marianne Bechara and others, "Private law aspects of token-based central bank digital currencies", Fintech Notes No. 2025/003 (Washington, D.C., IMF, 2025).

the problem. Central banks should draft and implement consumer protection regulations specific to digital currencies, ensuring clarity on user rights and responsibilities.

Cross-border interoperability. Regulatory fragmentation across borders limits the potential of digital currencies to facilitate regional trade and remittances. Legal frameworks must be harmonized to enable cross-border settlement, compliance with AML/CFT standards and seamless interoperability. Regional collaboration through such initiatives as the African Continental Free Trade Area can support these efforts.

Financial system gaps

Integration challenges. Traditional banking systems and mobile money platforms often operate in silos, creating barriers to the adoption of a digital currency. Banks may lack the technical capability to integrate with a digital currency, while mobile money operators may face challenges in aligning with digital currency standards. Central banks should work to establish application programming interface protocols and incentivize collaboration among stakeholders to ensure seamless integration.

Limited accessibility. Access to financial services is uneven, and rural areas and marginalized groups are often underserved. While mobile money platforms have made significant strides in increasing accessibility, gaps remain, in particular in the informal financial sector. Digital currencies should be designed to complement existing systems and extend services to previously excluded populations.

3. Using the digital currency readiness scorecard to identify gaps

The readiness scorecard provides a structured approach to identifying strengths and weaknesses across dimensions. By scoring each category from 1 (low readiness) to 5 (high readiness), central banks can visualize gaps and prioritize interventions (see table 18).

Table 18
Example of a digital currency readiness scorecard analysis

Dimension	Score (1–5)	Gaps identified	Priority actions
Technical readiness	2	Limited rural connectivity, weak cybersecurity	Expand network infrastructure, develop cybersecurity frameworks
Legal and regulatory readiness	3	Lack of consumer protection laws	Draft tailored regulations for digital currency users
Financial system readiness	4	Limited interoperability with informal systems	Create incentives for informal system integration
Social and cultural readiness	2	Low digital literacy in rural areas	Implement digital literacy campaigns

Source: Author's elaboration.

4. Recommendations for addressing infrastructure and policy gaps

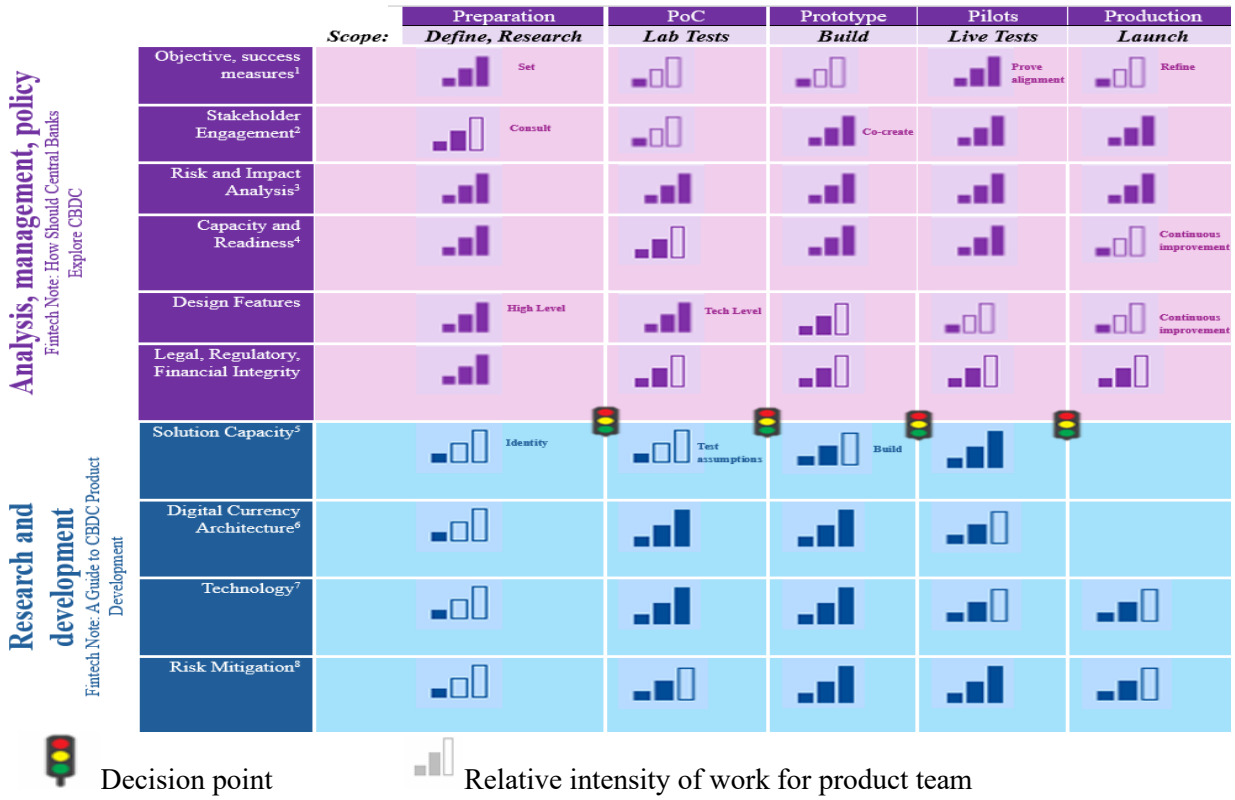
Prioritize critical gaps. Efforts should focus on addressing the most critical gaps that have a direct impact on the feasibility and user adoption of digital currencies. For example, expanding rural connectivity and establishing robust data privacy frameworks should take precedence, as these are foundational to inclusive and secure digital currency operations.

Collaborate with stakeholders. Stakeholders, including financial institutions, telecommunications providers, regulators and community leaders, play a crucial role in closing readiness gaps. Engaging these groups early and often ensures that solutions are practical, context-sensitive and widely supported.

4.8 Implementing phased approaches

A phased implementation strategy allows central banks to address gaps iteratively, starting with pilot projects that test solutions in controlled environments. Feedback from these pilots can inform broader roll-out strategies (see figure V).

Figure V
Example of a phased implementation strategy



Source: Author's elaboration.

[1] Includes review of existing payment systems, alternative solutions.
[2] Includes users, merchants, banks, payment service providers, telecommunications providers, government services.
[3] Includes monetary policy implementation impact, financial stability risks.
[4] Includes institutional capacity, infrastructure foundations.
[5] Includes people, processes, technology management.
[6] Solution architecture; technologies, platforms, infrastructure used to build and operate the solution; standards; interoperability.
[7] Includes ledger technology, databases, platforms, wallets, cybersecurity.
[8] Cybersecurity, resilience.

Central banks should review and consider the IMF project on the five phases of central bank digital currency,[10] which outlines the scope of each phase of developing a central bank digital currency and provides example questions that a technology research team should typically answer.

4.9 Tools for gap analysis

4.9.1 Infrastructure and policy gap checklist

- Have major infrastructure components (Internet, mobile networks, data centres) been evaluated for scalability and inclusivity?
- Are existing regulations aligned with the requirements for data privacy, AML/CFT compliance and consumer protection?
- Does the financial system accommodate both formal and informal players, and is it interoperable with digital currency infrastructure?
- Have public trust and digital literacy levels been assessed as part of readiness planning?

A model checklist is shown in table 19.

Table 19
Example of an infrastructure and policy gap checklist

Gap identified	Description	Impact	Action plan	Time frame
Cybersecurity framework	Insufficient safeguards for digital assets	High	Develop national cybersecurity strategy for digital currencies	Short-term
Rural network connectivity	Limited access in underserved regions	High	Expand mobile network coverage and affordability	Medium-term
Consumer protection laws	Lack of dispute resolution mechanisms	Medium	Draft specific consumer protection policies	Short-term

Source: Author's elaboration.

Addressing infrastructure and policy gaps is essential for building a robust digital currency system. By leveraging readiness assessments and scorecards, central banks can systematically identify and prioritize areas for improvement. Through targeted actions, phased implementation and stakeholder collaboration, these gaps can be effectively bridged, ensuring that digital currency initiatives are secure, inclusive and aligned with national priorities.

4.9.2 Archetype-specific considerations

Digital currency readiness varies significantly across African countries owing to differences in economic structure, financial sector maturity and policy priorities. A one-size-fits-all approach to infrastructure and policy readiness is not effective; instead, central banks

[10] Hervé Tourpe, Ashley Lannquist and Gabriel Soderberg, "A guide to central bank digital currency product development: 5P methodology and research and development", Fintech Notes No. 2023/007 (Washington, D.C., IMF, 2023).

must tailor their readiness assessments and implementation strategies on the basis of their unique national conditions.

The present section contains a detailed breakdown of digital currency readiness considerations for each archetype, highlighting vital infrastructure, policy gaps and dependencies. The goal is to help central banks to identify the most critical factors that may have an impact on the feasibility and adoption of digital currency within their respective archetype.

1. Large economies (e.g. Egypt, Morocco, Nigeria, South Africa)

Significant readiness considerations

Advanced financial infrastructure: these economies have well-developed interbank payment systems, real-time gross settlement networks and mobile banking penetration, making integration with a digital currency easier.

High regulatory complexity: owing to diverse and mature financial landscapes, large economies require strong regulatory coordination across banks, fintech companies and payment providers.

Market competition and stakeholder buy-in: the presence of multiple financial intermediaries means that a digital currency must coexist with existing financial products without creating disintermediation risks.

Infrastructure and policy gaps

There may be a need to:

- Ensure interoperability across diverse financial platforms
- Address regulatory concerns from commercial banks regarding deposit displacement
- Develop risk frameworks for a digital currency's impact on monetary policy transmission

Readiness thresholds

Large economies often have complex financial systems, multiple payment rails and diverse user populations.

- Technical readiness: thresholds may include integration with national switches, application programming interfaces for commercial banks and modular architecture for scalability
- Social readiness: requires user segmentation research to test different channels (smartphone, Unstructured Supplementary Service Data, Quick Response (QR) codes)
- Pilot focus: multi-sector pilots across urban and rural areas, including merchant and government transactions

Best practices

- The e-CNY model in China provides insights into tiered digital currency distribution through commercial banks to minimize disruptions.
- The eNaira in Nigeria serves as a regional case study on challenges to digital currency roll-outs and barriers to adoption.

2. High-mobile-money-penetration economies (e.g. Ghana, Kenya, Uganda, United Republic of Tanzania)

Significant readiness considerations

Widespread adoption of mobile money: digital currency implementation must operate with existing mobile money platforms to ensure adoption and avoid redundancy.

Reliance on telecommunications-based payment networks: any digital currency framework must integrate smoothly with mobile network operators to avoid infrastructure duplication.

Regulatory framework for non-bank financial institutions: mobile money is often regulated separately from traditional banking; central bank digital currency policies must address this dual system.

Infrastructure and policy gaps

There may be a need to:

- Ensure interoperability between digital currency and mobile money platforms
- Define the role of telecommunications operators and payment service providers in digital currency distribution
- Ensure low-cost transactions to maintain financial inclusion benefits

Readiness thresholds

These economies can leverage widespread digital finance habits but must align digital currency with mobile system incentives.

- Financial system readiness: active engagement of mobile money operators and wallet providers in test environments
- Legal readiness: clarifying roles and responsibilities among e-money and digital currency providers
- Pilot focus: mobile-first pilots that have offline functionality and are integrated with agent networks

Best practices

The United Payments Interface model developed by India demonstrates how a centralized payments interface can enable seamless integration between banks, fintech companies and mobile wallets.

The JAM-DEX in Jamaica and the eNaira in Nigeria provide insights into digital currency design for mobile-first economies.

3. Fixed exchange rate and currency peg (e.g. Morocco, CFA Franc Zone)

Significant readiness considerations

Monetary policy constraints: central banks in countries where systems are pegged to a currency have limited flexibility when making adjustments to their money supply, which makes the issuance of a digital currency more complex.

Dependence on foreign exchange reserves: a digital currency must not disrupt foreign reserve stability, which is crucial to maintaining the peg.

Regulatory harmonization within monetary unions: countries within regional currency unions (e.g. the West African Economic and Monetary Union) need coordinated digital currency policies to avoid fragmentation.

Infrastructure and policy gaps

There may be a need to:

- Ensure that a digital currency does not disrupt the currency peg mechanism
- Develop a harmonized approach for subregional monetary unions
- Address cross-border digital currency transactions within the peg framework

Readiness thresholds

Monetary policy considerations and regional coordination will shape readiness thresholds.

- Regulatory readiness: coordination with subregional central banks or currency boards
- Governance readiness: joint supervisory protocols and data-sharing mechanisms
- Pilot focus: domestic use with clear guidelines to prevent the disruption of the peg or convertibility

Best practices

The digital currency project of the Eastern Caribbean Central Bank (DCash) serves as an example of a multi-country digital currency within a pegged exchange rate system.

4. Highly dollarized economies (e.g. Democratic Republic of the Congo, Liberia, Zimbabwe)

Significant readiness considerations

Public trust in local currency is weak: digital currency design must focus on stability and credibility to compete with foreign currency alternatives.

Foreign exchange restrictions and capital controls: policy frameworks must define the role of a digital currency in domestic versus international transactions.

Liquidity and convertibility risks: central banks must carefully manage the transition from reliance on foreign currency to the use of a local digital currency.

Infrastructure and policy gaps

There may be a need to:

- Design digital currency incentives to rebuild trust in local currency
- Develop a clear framework for foreign exchange conversion between a digital currency and foreign currencies
- Ensure digital-currency-backed financial instruments to support adoption

Readiness thresholds

Digital currency may be used to restore monetary sovereignty, but trust is a major hurdle.

- Social and institutional readiness: public education campaigns and cross-institutional trust-building are prerequisites
- Technical readiness: offline capabilities are critical in informal, cash-heavy economies
- Pilot focus: targeted sectors (e.g. public transport, social transfers) that have high-volume local currency use

Best practices

The former digital currency project of Ecuador (Dinero electrónico) highlights lessons on public adoption challenges in dollarized economies.

Monetary experiments in Argentina provide insights into alternative monetary tools for economies that are dependent on foreign exchange rates.

5. Resource-dependent economies (e.g. Algeria, Angola, Chad, Nigeria, Zambia)

Significant readiness considerations

Heavy reliance on commodity exports: digital currency implementation must take into account the impact on foreign currency reserves and fiscal stability.

High levels of government subsidies and fiscal transfers: digital currencies can improve transparency in government disbursements for oil, gas or mineral revenues.

Macroeconomic volatility from commodity price swings: digital currency design should include stability mechanisms to manage inflation risks.

Infrastructure and policy gaps

There may be a need to:

- Ensure digital currency programmability for direct government transfers (e.g. distribution of oil revenue)
- Develop a mechanism to ensure digital currency stability when commodity prices are volatile
- Address cross-border payment needs for extractive industry transactions

Readiness thresholds

Volatile revenues and commodity shocks can constrain institutional readiness and budget flexibility.

- Financial and institutional readiness: simplified pilot governance structures and donor-funded technical assistance
- Pilot focus: public sector use cases (e.g. tax or royalty payments) test transaction reliability and reporting efficiency

Best practices

The sovereign wealth fund model used by Norway offers insights on transparent resource revenue management.

The digital currency pilot in Kazakhstan explores digital finance solutions for resource-exporting countries.

6. Fragile-State economies (e.g. Central African Republic, Chad, Somalia, South Sudan)

Significant readiness considerations

Weak financial infrastructure and banking penetration: digital currencies must be designed to operate in low-technology environments and include offline capabilities.

Security risks and political instability: implementing a digital currency in fragile States requires robust safeguards against cyberthreats and operational disruptions.

Heavy reliance on humanitarian aid and remittances: digital currency adoption must be compatible with non-governmental organization disbursement models and cross-border remittance channels.

Infrastructure and policy gaps

There may be a need to:

- Develop offline-capable digital currency solutions for areas that have poor connectivity
- Ensure security resilience against cyberthreats in fragile States

- Facilitate non-governmental organization and humanitarian payments using digital currency

Readiness thresholds

Security constraints, weak institutions and infrastructure deficits require phased and cautious approaches.

- Minimum technical readiness: emphasis on offline, low-bandwidth designs and modular deployment
- Governance readiness: require third-party technical oversight and operational safeguards
- Pilot focus: closed-loop systems (e.g. food voucher distribution) with predefined users and capped volumes

Best practices

Digital payment initiatives in Haiti provide insights into the use of digital currency in fragile economies.

United Nations-based digital cash programmes highlight humanitarian disbursement models using digital currencies.

7. Small island economies (e.g. Cabo Verde, Comoros, Seychelles)

These countries are often reliant on high remittances and lack scale for domestic innovation.

- Technical readiness: cloud-based or shared subregional platforms may reduce cost thresholds
- Financial readiness: participation of money transfer operators and subregional settlement systems
- Pilot focus: cross-border retail pilots (e.g. tourist payments, diaspora remittances) with neighbouring countries

8. Economies that have high debt and financial vulnerabilities (e.g. Sudan, Tunisia, Zambia)

Budget constraints and macroeconomic instability demand careful fiscal coordination and gradual roll-outs of digital currencies.

- Legal and financial readiness: conduct a debt sustainability analysis to guide digital currency financing strategies
- Pilot focus: cost-efficient digital government disbursements and improved expenditure tracking

9. High-remittance economies (e.g. Gambia, Lesotho, Nigeria)

Reducing remittance costs is a crucial objective; interoperability and foreign exchange solutions are central.

- Technical readiness: standards-based application programming interfaces for integration with remittance service providers
- Regulatory readiness: alignment with the Financial Action Task Force “travel rule”, e-know-your-customer policies and licensing regimes
- Pilot focus: test corridors that have high diaspora populations and real-time retail foreign exchange settlement

Summary of readiness

Digital currency readiness depends heavily on a country’s economic structure, financial system maturity and digital infrastructure development. By evaluating archetype-specific readiness factors, central banks can:

- Identify the most critical infrastructure and policy gaps for digital currency adoption.
- Tailor digital currency implementation strategies on the basis of their country’s unique constraints.
- Learn from global best practices relevant to their economic and financial archetype.

The next section of the handbook will explore ways to address identified infrastructure and policy gaps through targeted capacity-building, phased implementation strategies and subregional cooperation frameworks.

Further reading

For a more detailed exploration of how digital currencies interact with monetary policy frameworks and transmission mechanisms, readers are encouraged to consult the IMF publication entitled *Implications of Central Bank Digital Currencies for Monetary Policy Transmission*. The publication contains insights into how different digital currency design choices can affect interest rate pass-through, liquidity management and the overall effectiveness of monetary policy.

5 Central bank digital currency design and technology choices

Evaluating design choices is a pivotal phase in the digital currency exploration process. This step integrates insights from step 1 (policy objectives and prioritization) and step 2 (readiness assessment) of chapter 2, linking potential design options to national priorities and readiness constraints. By carefully analysing the trade-offs and implications of various digital currency designs, central banks can ensure that their chosen path aligns with strategic goals, addresses readiness gaps and supports long-term success.

The success of a digital currency hinges on design choices that reflect policy objectives, readiness levels and existing realities. These decisions encompass technical, operational and governance frameworks that have a direct impact on functionality, scalability and public trust. Design choices must balance innovation with feasibility, ensuring that solutions are both ambitious and attainable given existing infrastructure and regulatory landscapes.

5.1 General design framework and technology frameworks

Digital currency design choices must align with national policy objectives, financial infrastructure maturity and economic conditions. A well-structured design framework ensures that central banks will select a digital currency model that optimally supports financial inclusion, payment efficiency, monetary policy effectiveness and financial stability.

The present section outlines the major digital currency design options, including:

- **Operating models**. Retail, wholesale and general-purpose digital currencies
- **Governance structures**. Centralized, decentralized and public-private partnerships
- **Design considerations**. Privacy, interoperability, programmability and offline functionality

Central banks must carefully evaluate each option on the basis of the banks' readiness, regulatory environment and technological capabilities to ensure the implementation of a fit-for-purpose digital currency.

5.1.1 Operating model: retail, wholesale or general-purpose digital currency

The operating model determines the scope and functionality of a digital currency.

- **Retail central bank digital currency**. Designed for direct use by individuals and businesses, retail digital currencies promote financial inclusion and payment efficiency. They require robust infrastructure to handle high transaction volumes and user interaction. For example, in Nigeria, the eNaira is designed to focus on retail functionality to improve access to financial services in underserved areas.
- **Wholesale central bank digital currency**. Intended for financial institutions, wholesale digital currencies facilitate interbank settlements and cross-border trade. They are easier to run, manage and maintain in practice than retail digital currencies but do not directly enhance financial inclusion. The wholesale digital currency pilot in South Africa has shown how this model can streamline subregional trade processes.
- **General-purpose central bank digital currency**. Combining retail and wholesale features, this hybrid model offers comprehensive capabilities but demands advanced infrastructure and governance frameworks. Its adaptability makes it suitable for jurisdictions seeking to address diverse objectives.

Significant consideration

Central banks must evaluate which model best aligns with their policy objectives, such as inclusion, efficiency or trade facilitation.

5.2 Governance framework

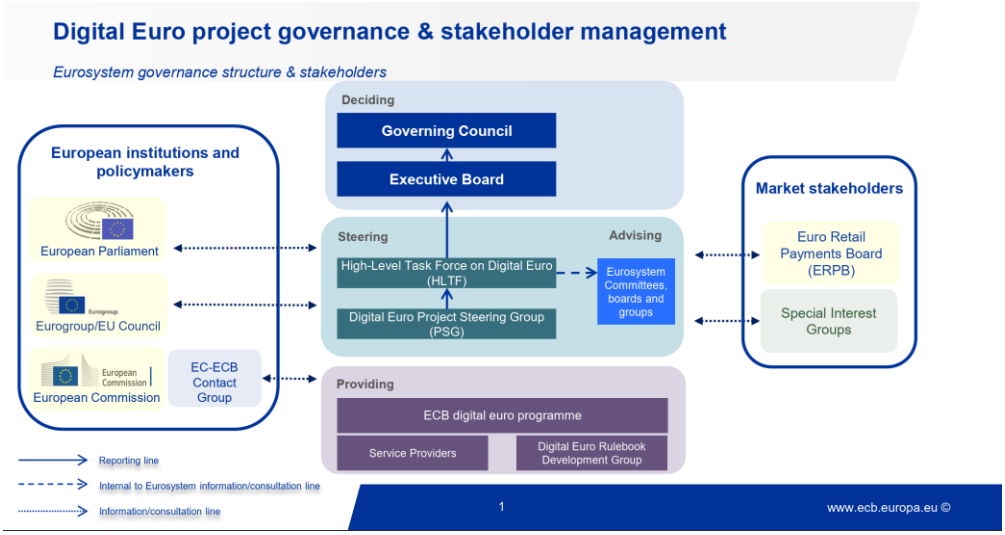
Governance frameworks define how a digital currency is managed, ensuring accountability and stability. A well-defined governance framework is essential for ensuring that a digital currency system remains secure, transparent, accountable and aligned with evolving policy objectives over time. The governance structure determines not only who operates and

maintains the digital currency infrastructure but also how decisions relating to upgrades, interoperability, compliance, privacy and dispute resolution are made (see box 3).

Box 3

Digital euro governance model

The figure entitled "Digital euro project governance and stakeholder management" outlines the multilayered governance framework used by the European Central Bank, which includes oversight by the Governing Council and Executive Board, guidance from the High-Level Task Force on Digital Euro and coordination with such external committees as the European Partnership for Responsible Minerals and the European Parliament.



This governance structure can serve as a global benchmark for digital currency delivery, emphasizing structured oversight, stakeholder involvement, institutional clarity and flexibility during the design and experimentation phases. It can offer valuable lessons for structuring robust and transparent governance in the African digital currency context.

Source: www.ecb.europa.eu/euro/digital_euro/timeline/profuse/shared/pdf/ecb.deprep240517_governance.en.pdf.

Abbreviations: EC, European Commission; ECB, European Central Bank; EU, European Union.

Governance for a digital currency must address both the operational management of the platform and overarching policy and regulatory oversight. In centralized models, the central bank may retain full control over issuance, rules and operations. In contrast, more distributed models, in particular those involving public-private partnerships, may require shared governance arrangements involving regulated financial institutions, fintech firms or mobile network operators.

- **Centralized governance**. The central bank maintains full control over digital currency issuance, distribution and oversight. This model offers clarity and strong regulatory alignment but may face scalability challenges in decentralized financial systems.
- **Decentralized governance**. Employing distributed technologies, this model enhances transparency and resilience; however, it requires sophisticated cybersecurity measures and clear regulatory definitions.
- **Hybrid governance**. Combining centralized oversight with decentralized operations, this model balances control and innovation. It is well suited for

jurisdictions seeking to leverage private sector expertise while retaining regulatory authority.

- **Public-private partnership model**. Central banks collaborate with private sector players (banks, fintech firms, telecommunications companies) to co-develop and distribute the digital currency.

Essential governance decisions include defining the authority responsible for setting technical standards, enforcing compliance with AML/CFT rules, adjudicating transaction disputes and adapting the digital currency to accommodate future innovations. Transparent governance mechanisms can also help to build public trust, especially in systems where programmability or data usage may raise ethical or privacy concerns.

African central banks should consider institutional readiness when determining governance models. Jurisdictions that have strong regulatory bodies and supervisory capacity may opt for tiered or federated models that distribute operational roles, while others may retain tighter central oversight. Moreover, integration with existing financial infrastructure, such as national payment switches or identity systems, may require coordinated governance bodies in order to align stakeholder priorities.

As digital currencies evolve beyond the pilot phase, central banks should also prepare for governance continuity through appropriate legal frameworks, inter-agency coordination protocols and feedback mechanisms that ensure ongoing alignment with monetary policy, financial stability and inclusion objectives.

Significant consideration

Governance choices should reflect the regulatory environment, stakeholder roles and public trust levels identified in readiness assessments.

For additional institutional coordination guidance in the context of regional monetary unions or federated governance models, refer to chapter 9 on central bank digital currencies in subregional monetary unions.

5.3 Important design considerations

Central banks must navigate a range of technical, functional and policy-sensitive design trade-offs. These decisions influence a digital currency's effectiveness in achieving national policy objectives, its user adoption and the resilience of the broader financial system. In addition to such core features as privacy, interoperability, programmability and offline capability, further critical considerations include the wallet model, the identity and transaction framework and the recoverability mechanisms built into the user experience.

5.3.1 Interoperability

Interoperability is a foundational design consideration for any digital currency system that is aimed at supporting a seamless, inclusive and efficient digital financial system. Interoperability ensures that digital currencies integrate seamlessly with existing systems and facilitate cross-system transactions.

- **Domestic interoperability**. Digital currencies must connect with existing banking, mobile money and payment systems to ensure widespread usability. This is especially important in African economies that have high mobile money

penetration, where digital currency adoption depends heavily on integration with such widely used channels as Unstructured-Supplementary-Service-Data-based mobile wallets and agent networks. The digital currency should be designed with open application programming interfaces, standard communication protocols and inclusive access layers that allow both legacy systems and emerging digital services to interface without friction. The Bank of England's Digital Pound Lab is used to explore potential design models for a digital pound, including a so-called "platform model", in which the central bank would provide the core infrastructure that payment interface providers could access to deliver services to users. External service interface providers could also offer value-added services. By testing this separation between core infrastructure and service provision in a simulated environment, the Lab helps industries to assess how such a model could support integration with existing financial and fintech systems, promoting competition, innovation and broader access. For African central banks, especially in economies that have dynamic mobile money markets, adopting a similar architecture can enable interoperability with existing mobile platforms while preserving the central bank's role in ensuring trust, resilience and settlement finality.

Examples of domestic interoperability include:

Nigeria. The eNaira was designed for integration into the banking and fintech system of Nigeria, allowing users to access the digital currency using commercial bank applications and mobile wallets. To expand reach among the unbanked and underbanked populations, the Central Bank of Nigeria later introduced Unstructured Supplementary Service Data functionality, enabling users who have mobile phones with limited features to gain access to and transact in eNaira without smartphones or Internet connectivity.

Ghana. The eCedi pilot in Ghana emphasized mobile-first integration by enabling offline transactions through the use of smart cards and Unstructured-Supplementary-Service-Data-enabled wallets. The Bank of Ghana tested interoperability with mobile money providers to ensure seamless transfers between eCedi and existing mobile wallets, which was critical for financial inclusion, given the country's strong mobile money penetration.[11]

India. The pilot for the digital rupee (e₹-R) of the Reserve Bank of India supports wallet access through partner banks' mobile applications and is being designed to integrate with the Unified Payments Interface, a highly interoperable real-time payment system developed by India. This ensures ease of use and scalability across the country's vast digital landscape.[12]

- **Cross-border interoperability**. Digital currencies designed for international trade require alignment with regional and global standards. Such initiatives as the Pan-African Payment and Settlement System demonstrate the importance of collaboration for cross-border functionality. Moreover, establishing shared standards for messaging (e.g. ISO 20022), identity verification and compliance protocols will be critical to enabling secure and efficient international flows.

Readers interested in the practical implications of cross-border interoperability and subregional collaboration are encouraged to refer to chapter 7 on central bank digital currency

[11] Bank of Ghana, *The eCedi Report: A Report on Bank of Ghana's Central Bank Digital Currency Pilot Project* (Accra, 2024).
[12] Reserve Bank of India, "Frequently asked questions". Available at www.rbi.org.in/commonman/English/scripts/FAQs.aspx?Id=3686.

in subregional monetary unions, which contains more in-depth insights into such multi-digital currency projects as Project mBridge and Project Dunbar, including legal, governance and technical considerations relevant to African monetary unions.

Significant consideration

Interoperability supports scalability and user adoption, making it a critical design priority. Digital currency systems must be structured to integrate seamlessly with established domestic payment infrastructures, including mobile money platforms, banking networks and real-time payment systems. In addition, when applicable, these systems should facilitate cross-border interoperability to support efficient regional and international transactions.

5.3.2 Offline capabilities

In areas that have limited or intermittent connectivity, which is common in many rural and underserved areas across Africa, offline functionality is not merely a feature but a foundational requirement for digital currency inclusivity. Offline digital currency capabilities ensure that users can make payments without relying on continuous Internet or mobile network access, thus extending digital financial services to communities that remain beyond the reach of traditional infrastructure.

Digital currencies piloted in such countries as Nigeria (eNaira) and Ghana (eCedi) have explored forms of offline payments using both smartcard wallets and mobile interfaces. These trials demonstrated that offline functionality could directly support financial inclusion while building user trust in digital currency reliability. Robust design, risk mitigation and clear transaction limits (e.g. value or frequency caps) are essential in order to safeguard the offline landscape.

For further discussion on implementation costs and readiness, readers may refer to section 4.1, on technical readiness, and section 6.1, on cost-benefit evaluations.

- **Card-based systems**. Offline cards allow users to make transactions without Internet access, mirroring the convenience of cash.
- **Mobile wallets with offline storage**. These systems store value locally on devices, enabling transactions even during network outages. The SandDollar of The Bahamas incorporates offline capabilities to serve remote island communities.

Significant consideration

Offline options must align with broader technical readiness and policy objectives, balancing technological feasibility with user accessibility to support underserved populations. Central banks should assess the cost implications, in particular for hardware distribution, and support and ensure interoperability between offline and online environments.

5.3.3 Stakeholder access and inclusivity

Stakeholder access frameworks define who can interact with the digital currency system and under what conditions. A core design consideration for any digital currency is ensuring broad and equitable access across diverse stakeholder groups, including the unbanked, underbanked and those in rural or marginalized communities. Central banks must determine which stakeholders, ranging from individual users and merchants to financial institutions and

fintech companies, can gain access to the digital currency directly, and under what conditions. This involves defining the access framework (e.g. direct versus indirect access) and onboarding mechanisms, and designing a user interface adapted to local realities, such as low literacy or digital unfamiliarity.

Inclusivity must also be embedded in the choice of technology and distribution models. For mobile-first economies, digital currencies should support Unstructured Supplementary Service Data, access by mobile phones with limited features, and agent networks. For infrastructure-heavy economies, integration with digital banking and fintech companies may be more relevant. In addition, gender, age and ability-related disparities in digital finance access should be addressed through tailored onboarding processes, simplified user interfaces and multilingual support.

Policy frameworks must promote fairness and competition, preventing market concentration while enabling public-private partnerships that extend reach. Digital currency initiatives that succeed in broad stakeholder inclusion, such as the onboarding of rural users of eNaira through the microfinance bank agents of the Nigeria Incentive-Based Risk Sharing System for Agricultural Lending,[13] illustrate the need for thoughtful stakeholder segmentation and targeted roll-out strategies that facilitate broader inclusion, especially in areas that are beyond the reach of traditional banking.

- **Tiered access models**. Varying levels of access for individuals, businesses and financial institutions, balancing inclusivity with security, should be defined.
- **Inclusivity measures**. Ensure that digital currencies cater to diverse user needs, including multilingual interfaces, accessible designs and targeted outreach; for example, the Aadhaar-enabled payment systems in India leveraged digital identity infrastructure to promote financial inclusion across rural and urban populations.

Significant consideration

Inclusivity measures must address the gaps in accessibility and digital literacy identified in readiness assessments. Inclusivity is not only a moral imperative but also a practical necessity for digital currencies, as it enables the achievement of the intended public policy objectives. A lack of inclusive access risks reinforcing existing inequities in the financial system and undermining the legitimacy and adoption of a digital currency.

5.3.4 Tiered wallet and know-your-customer frameworks

To maximize financial inclusion while preserving financial integrity, many jurisdictions are exploring tiered wallet and know-your-customer frameworks. These systems allow for varying levels of access and functionality on the basis of the level of identity verification (i.e. know-your-customer steps) completed by the user.

- Tiered know-your-customer structures can enable the opening of low-value wallets with minimal or no formal identification. This can be beneficial when onboarding unbanked populations, especially in rural or informal economies, while higher-value wallets would require more stringent identity checks and offer access to broader services.

[13] https://www.cbn.gov.ng/Currency/eNaira.html.

- Such a model was piloted in the eNaira roll-out in Nigeria, where users could register with basic information and later upgrade their wallet tier by linking a bank account or presenting national identification credentials.
- This approach supports a central bank's dual goals of promoting financial inclusion and maintaining compliance with AML/CFT obligations, and it can be aligned with national digital identification strategies where available.

Significant consideration

An effective digital currency wallet model should incorporate tiered know-your-customer frameworks that balance financial inclusion with risk management, enabling broader access through simplified onboarding while maintaining compliance for higher transaction tiers.

5.3.5 Token-based versus account-based models[14]

Digital currencies can be designed as token-based instruments, resembling digital cash that is bearer-held and validated through possession, or as account-based instruments, in which access and transfer depend on identity-linked accounts managed by intermediaries or the central bank.

- Token-based digital currencies may be more suitable for promoting privacy-preserving, cash-like payments and are often favoured for offline and small-value transactions. Their use, however, may be limited when enforcing compliance requirements or implementing sophisticated user controls.
- Account-based digital currencies allow for real-time identity verification, programmability and improved oversight, and they may better suit economies that have strong digital identification and financial institution networks.

Significant consideration

The choice between token-based and account-based digital currency models should reflect national infrastructure capacity and user needs: token-based systems offer greater privacy and offline potential, while account-based models support stronger identity verification and oversight.

5.3.6 Wallet recoverability and user protection

To foster trust and usability, in particular among first-time digital finance users, digital currency systems must include robust wallet recovery mechanisms that strike a balance between security, privacy and accessibility.

- Custodial wallets, in which intermediaries manage recovery, offer more straightforward solutions for users with low digital literacy but may raise concerns over control and surveillance.

[14] Another, perhaps more precise, taxonomy for classifying digital currencies is outlined by the International Telecommunication Union in a novel framework based on two dimensions (ledger tracking at the platform layer (event-based versus balance-based) and digital currency form at the asset layer (object-based versus entry-based)) that can better allow for the classification of the technical implementation and operational characteristics of digital currencies. See www.itu.int/en/ITU-T/extcoop/dcgi/Documents/DCGI%20CBDC%20Reference%20Architecture%20Report-20-01-2025.pdf, box 1.

- Non-custodial wallets (i.e. user-controlled) provide autonomy but require more advanced recovery tools, such as social recovery, biometric back-up or integration with national identification recovery mechanisms.
- Best practices include offering multilayered recoverability options, from password resets using mobile authentication to biometric or national identification verification, to avoid exclusion due to the loss of a device or forgotten credentials.
- A growing number of pilots, such as the eCedi trials in Ghana, have explored user-centric approaches to recoverability, especially in areas that have limited access to smartphones or formal identity services.[15]

Significant consideration

Wallet recoverability frameworks must ensure secure user access, balancing ease of use and strong protection, especially for users in low-infrastructure or low-literacy areas.

5.4 Framework for linking design choices to objectives and readiness

Central banks can use the following matrix to evaluate design choices on the basis of their alignment with policy objectives and effectiveness within the context of readiness levels (see table 20).

[15] See Ashley Lannquist and Brandon Tan, "Central bank digital currency's role in promoting financial inclusion", Fintech Notes No. 2023/011 (Washington, D.C., IMF, 2023).

Table 20

Example of a central bank digital currency design matrix

Design choice	Policy objective alignment	Readiness requirements	Expected effectiveness	Recommendation
Retail central bank digital currency	Financial inclusion	Robust digital payments infrastructure, digital identification systems, cybersecurity capacity and legal framework	High where a large share of the population remains unbanked	Recommended with infrastructure investment
Wholesale central bank digital currency	Trade efficiency	Advanced interbank payment systems, infrastructure and institutional capacity	High for financial market efficiency and large-value payments	Suitable for advanced readiness
General-purpose central bank digital currency	Comprehensive objectives	Strong technical capacity, scalable infrastructure and phased implementation capability	Highly versatile, supporting both retail and wholesale use cases while addressing multiple policy objectives	Requires phased implementation
Centralized governance	Stability and control	Strong regulatory and supervisory capacity	High with regard to maintaining monetary control and regulatory compliance	Align with robust regulatory oversight
Offline capabilities	Inclusivity	Secure offline authentication technology and distribution mechanisms	High in rural and underserved areas	Essential for rural-focused objectives
Cross-border interoperability	Regional trade and remittances	Regional payment agreements, harmonized standards and cross-border regulatory cooperation	Moderate to high for regional integration	Develop through pilot programmes

Source: Author's elaboration.

5.5 Aligning central bank digital currency design with national digital infrastructure

Digital currency design cannot be separated from a country's broader digital landscape. National digital infrastructure, including Internet connectivity, mobile network coverage, device penetration, digital identity systems and payment rails, forms the foundation upon which technical and operational choices for digital currency deployment must rest. The present section provides a strategic framework to help central banks to align digital currency design decisions with central banks' infrastructure readiness and policy priorities, ensuring inclusivity, resilience and relevance.

5.5.1 Why infrastructure alignment matters

The digital infrastructure landscape varies significantly across African countries. Some economies have advanced banking and digital payment systems, while others rely heavily on mobile money and mobile phones with limited features to drive financial access. These differences shape the feasibility and effectiveness of digital currency design choices. A digital currency that requires high connectivity or digital literacy in a country with low mobile penetration or fragile information and communications technology infrastructure risks reinforcing exclusion, rather than overcoming it.

Table 21
Example of how policy objectives and infrastructure affect design choices

Policy objective	Infrastructure context	Design considerations
Financial inclusion	Low smartphone penetration, patchy connectivity	Token-based digital currency, Unstructured Supplementary Service Data access, SIM toolkit apps, offline functionality
Payment system efficiency	Real-time systems, modern real-time gross settlement system, high digital transaction volumes	Account-based models, interoperability with banks and fintech companies, ISO 20022 compliance
Monetary policy efficacy	Well-regulated banking sector, data-rich environment	Centralized ledger, direct central bank account access for intermediaries
Cross-border trade	Dependence on remittances, regional corridors	Interoperability layers, cross-border sandbox pilots, regional standards integration
Digital innovation	Strong fintech landscape, digital identification framework	Smart contract capability, application programming interface access, open-source software development kits, programmable money functionality

Source: Author's elaboration.

Moreover, policy objectives, whether focused on financial inclusion, digital innovation or monetary sovereignty, must be achievable within existing infrastructural constraints or informed by clear investment pathways. Aligning design to infrastructure readiness ensures that a digital currency project is not only technically feasible but also scalable and sustainable.

5.5.2 Infrastructure assessment as a design input

Central banks should begin their exploration of digital currency with a comprehensive infrastructure readiness assessment, including both technical infrastructure (e.g. payment systems, Internet bandwidth, distribution ledger technology capabilities) and supporting infrastructure (e.g. electricity access, digital identification, agent networks). This assessment enables the identification of capabilities, limitations and areas requiring investment or partnership.

The digital currency readiness scorecard, outlined in section 4.7 (see also table 18), can be used to diagnose the maturity level across these domains and map them against potential design pathways.

5.5.3 Linking infrastructure to design choices and policy objectives

These links ensure that digital currency choices are purposeful and that each technology component or access channel directly contributes to a measurable policy outcome (see table 21).

5.5.4 Country archetypes and infrastructure pathways

Countries can be broadly situated along a spectrum between infrastructure-heavy (e.g. Egypt, Morocco, South Africa) and mobile-first/low-infrastructure (e.g. Democratic Republic of the Congo, Kenya, Sierra Leone). While infrastructure-heavy economies may support advanced retail and wholesale use cases through smartphone-based wallets and application programming interface-integrated platforms, mobile-first economies require inclusive design, such as Unstructured-Supplementary-Service-Data-based interfaces, offline transaction functionality and agent-assisted onboarding, to ensure digital currency accessibility and trust.

Hybrid models that accommodate both high-technology and low-technology user bases are especially relevant in jurisdictions that have uneven infrastructure distribution (e.g. urban/rural divides).

5.5.5 Strategic recommendations for alignment

To successfully align digital currency design with national digital infrastructure, central banks should:

- Map national infrastructure capabilities early in the exploration process, using both quantitative and qualitative tools
- Tailor ledger architecture and access modalities to fit user device capabilities and connectivity conditions
- Leverage such existing digital infrastructure as mobile money rails, agent networks and government platforms to extend digital currency reach and minimize deployment costs
- Design modular and scalable systems that allow the future integration of more sophisticated features as infrastructure improves
- Coordinate with telecommunications regulators and information and communications technology ministries to identify synergies with digital economy strategies and plan for infrastructure expansion

5.6 Archetype-specific design guidance

Digital currency design must be tailored to the specific economic structures, financial landscapes and policy priorities of each country. While the general design framework provides broad options for digital currency implementation, it is essential that central banks adapt these choices on the basis of their unique national conditions. The present section offers archetype-specific design recommendations, helping central banks to align digital currency features with their country's readiness levels, financial system maturity and strategic objectives.

5.6.1 Large economies (e.g. Egypt, Morocco, Nigeria, South Africa)

Large economies typically have well-developed financial systems, including diverse payment infrastructures, advanced banking networks and strong regulatory oversight. Given their size and complexity, these economies require a digital currency design that balances financial sector stability with innovation. A general-purpose digital currency that serves both retail and wholesale use cases is most suitable, ensuring seamless integration across various payment systems. Governance should follow a decentralized, two-tier model, in which the central bank issues the digital currency but relies on commercial banks, fintech companies and payment service providers for distribution. Interoperability with existing payment rails, such as real-time gross settlement systems, instant payments and mobile wallets, is crucial to maintaining system efficiency. Large economies must also implement tiered know-your-customer requirements, allowing for more anonymous transactions at lower values while enforcing stricter compliance for larger transactions. This ensures a balance between privacy and financial integrity.

5.6.2 High-mobile-money-penetration economies (e.g. Ghana, Kenya, Uganda, United Republic of Tanzania)

In countries where mobile money dominates the financial landscape, a digital currency must be designed to complement, rather than compete with, existing digital payment systems. Retail digital currencies that integrate seamlessly with mobile money platforms can provide greater financial inclusion while maintaining trust in established digital financial services. A public-private partnership model, in which telecommunications operators and mobile money providers act as intermediaries in digital currency distribution, is ideal. To ensure accessibility, digital currency transactions should be compatible with Unstructured-Supplementary-Service-Data-based systems, QR codes and near-field communications technologies, allowing users who have phones with limited features to participate. Privacy considerations should focus on embedding know-your-customer compliance within mobile wallets to streamline user onboarding. A successful implementation with regard to this archetype requires close collaboration among the central bank, telecommunications regulators and mobile payment operators to prevent market disruption and encourage widespread adoption.

5.6.3 Fixed exchange rate and currency-pegged economies (e.g. Morocco, CFA Franc Zone)

Countries that operate within a fixed exchange rate system or are pegged to a currency face unique challenges when introducing a digital currency. The central bank must ensure that digital currency issuance does not disrupt foreign exchange reserves or the stability of the currency peg. A wholesale digital currency model, primarily used for interbank settlements and cross-border transactions, is best suited to this archetype. Governance should follow a centralized approach, in which the central bank directly manages issuance and transaction processing to maintain tight control over monetary flows. Blockchain-based settlement systems could enhance efficiency in cross-border transactions within subregional monetary unions. Ensuring interoperability between domestic and subregional financial infrastructures will be critical to avoiding fragmentation. In addition, regulatory coordination among member countries will be necessary to establish harmonized rules for digital currency issuance and usage, ensuring that these frameworks align with broader monetary policy goals.

5.6.4 Highly dollarized economies (e.g. Democratic Republic of the Congo, Liberia, Zimbabwe)

In economies where the United States dollar or another foreign currency dominates local transactions, public trust in domestic money is often low. A digital currency initiative in such environments must focus on rebuilding confidence in local currency while ensuring that digital money offers real advantages over foreign cash. The preferred model would be a retail digital currency that contains embedded stability mechanisms, such as programmed exchange-rate adjustments or interest-bearing features. Governance should follow a hybrid approach, in which the central bank guarantees digital currency stability while allowing controlled integration with foreign currency systems. Smart contracts could be used to automate exchange-rate adjustments, reducing volatility concerns. Privacy design should support anonymous small-value transactions to encourage user adoption while implementing robust anti-money-laundering measures for high-value transactions. To foster public trust, central banks may consider offering digital-currency-backed savings accounts that have stability incentives, ensuring that the digital currency provides tangible benefits over cash alternatives.

5.6.5 Resource-dependent economies (e.g. Algeria, Angola, Chad, Nigeria, Zambia)

For economies that are heavily reliant on commodity exports, such as oil, gas and minerals, digital currency design should incorporate fiscal policy tools that enhance transparency and efficiency with regard to the management of government revenues. A general-purpose digital currency that has programmable features is recommended, allowing the central bank to automate tax collection, subsidy distribution and revenue-sharing agreements. The governance structure should be decentralized, enabling government agencies to use digital currency for direct fiscal operations while ensuring central bank oversight. Technology choices should include smart contracts for automatic revenue transfers, tax compliance enforcement and transparent public sector payments. Privacy considerations should balance regulatory transparency with data protection to ensure secure fiscal management. Implementing a digital-currency-based system for government disbursements and extractive industry taxation could improve public financial accountability while reducing leakage in fiscal transfers.

5.6.6 High-remittance economies (e.g. Gambia, Lesotho, Senegal, Somalia)

Countries that rely heavily on remittances as a major source of income must design a digital currency that facilitates low-cost, efficient cross-border transactions. A retail digital currency with cross-border payment functionality is the most effective model, allowing migrants to send funds home instantly and at lower fees. A public-private governance model, in which central banks partner with global remittance providers and foreign financial institutions, can ensure broad adoption. The technology must support interoperability with international remittance networks and foreign exchange trading platforms, enabling the smooth conversion of currency. Privacy measures should incorporate AML/CFT compliance requirements while maintaining user-friendly onboarding for migrants and their families. Establishing digital-currency-based remittance corridors with instant exchange mechanisms could significantly reduce reliance on expensive informal money transfer systems while improving financial transparency.

5.6.7 Small island economies (e.g. Cabo Verde, Comoros, Mauritius, Seychelles)

For small island countries, in which tourism and foreign exchange inflows are critical economic drivers, a digital currency must cater to both domestic users and international visitors. A retail digital currency with built-in foreign exchange capabilities would enable tourists to make digital payments seamlessly while ensuring that local businesses receive funds in local currency. The governance model should be hybrid, allowing integration with mobile payment networks and digital banking services. Technology choices should prioritize multi-currency compatibility, ensuring easy conversion between digital currency and foreign currencies. Privacy considerations should align with cross-border compliance requirements, preventing illicit financial activity while maintaining user convenience. A successful use case for this archetype would be a digital-currency-enabled tourism payment system, which would reduce reliance on cash-based transactions and provide a secure digital alternative for visitors.

5.6.8 Fragile-State economies (e.g. Central African Republic, Chad, Somalia, South Sudan)

In fragile States that have weak financial and telecommunications infrastructures, designing a digital currency requires prioritizing accessibility, resilience and security. A retail digital currency that has offline functionality is essential to ensure its usability in areas that have unreliable access to electricity and the Internet. The governance model should be centralized, with the central bank overseeing issuance while partnering with non-governmental organizations and humanitarian agencies for distribution. The technology must support Unstructured Supplementary Service Data/text-based transactions and hardware wallets to enable financial inclusion for users who do not have smartphones or bank accounts. Privacy considerations should allow for minimal know-your-customer onboarding for small-value transactions, which would encourage informal sector participation while maintaining compliance for high-value users. A practical implementation would be to use digital currency as a vehicle for humanitarian aid disbursement, ensuring that funds reach recipients efficiently, securely and without risk of diversion.

Further reading

Central banks seeking to deepen their understanding of digital currency system design and interoperability considerations may consult two crucial resources. The publication entitled "Central bank digital currencies: system design and interoperability", the second in a series of publications by the Bank for International Settlements and a group of central banks, provides guidance on foundational technical architectures and connectivity models for domestic and cross-border integration. In addition, the report to the Group of 20 by the Bank for International Settlements, entitled Options for *Access to and Interoperability* of Central Bank Digital Currencies for *C*ross-*B*order *P*ayments, explores practical design and policy options to enhance cross-border functionality, offering valuable context for the matters discussed in the present section, including technology selection, infrastructure alignment and cross-jurisdictional cooperation.

6.1 Framework for cost-benefit evaluation

A central bank digital currency is a strategic policy tool that has potential macroeconomic, financial and technological implications. Before committing to its implementation, central banks must conduct a comprehensive cost-benefit evaluation to

determine whether a digital currency aligns with national economic priorities and financial system capacities.

The cost-benefit evaluation framework provides a structured methodology for assessing the financial, operational and economic impacts of digital currency deployment. This approach enables central banks to make data-driven decisions, ensuring that digital currency adoption delivers net positive benefits while minimizing unintended risks.

6.1.1 Main components of the cost-benefit evaluation framework

A structured cost-benefit evaluation must assess:

- Direct financial costs: infrastructure investment, operational expenses and ongoing maintenance
- Indirect economic costs: disruptions to banking intermediation, financial stability and monetary policy transmission
- Quantifiable economic benefits: efficiency improvements, cost reductions and enhanced financial inclusion
- Qualitative benefits: increased transparency, innovation and resilience in the financial system
- Risk-adjusted considerations: hidden costs, implementation risks and regulatory uncertainties

By weighing costs against expected benefits, central banks can justify digital currency adoption or determine whether alternative solutions (e.g. enhancing existing payment systems) may be more effective.

6.1.2 Step 1. Identifying direct financial costs

Digital currency implementation requires substantial investment in digital infrastructure, security and operational management. The major cost categories include the following.

Infrastructure investment costs

Infrastructure investment represents one of the most significant upfront financial commitments in a digital currency project. It includes the development, procurement and deployment of the core systems, hardware and operational tools necessary to launch and scale a digital currency. For African central banks, these costs must be evaluated not only in terms of technical robustness and security, but also accessibility and equity, in particular in areas where digital connectivity is limited.

Primary components of infrastructure investment include:

- Core ledger infrastructure: this encompasses distributed ledger systems or centralized databases, consensus mechanisms and system-redundancy layers to ensure operational resilience.
- Integration middleware: investments are often needed to develop application program interfaces, service orchestration layers and cross-platform interfaces in

order to integrate the digital currency system with banks, mobile money operators, fintech companies and public services.

- Offline capability infrastructure: a critical but frequently underestimated component in the African context, offline functionality enables digital currency transactions in areas that have unreliable electricity or network coverage. This may involve:
    - Hardware wallets (e.g., smart cards, secure near-field communications-based devices)
    - Unstructured Supplementary Service Data gateway infrastructure, enabling access by mobile phones with limited features
    - Point-of-sale terminals and agent networks designed to support delayed settlement or peer-to-peer syncing, which are especially pivotal in rural areas and fragile States, where offline capability directly supports financial inclusion and equitable access
- Cybersecurity and resilience systems: investments in this regard may include firewalls, intrusion-detection systems, zero-trust architecture elements and resilience testing environments.
- Operational support infrastructure: this includes hosting environments, monitoring dashboards, help-desk systems and disaster recovery frameworks that are necessary for ongoing operations.
- Data management and compliance: this requires the establishment of secure data storage systems and compliance frameworks for AML/CFT and know-your-customer policies.
- Technology development: this includes building a centralized or decentralized ledger system that is integrated with real-time gross settlement systems, instant payment platforms and mobile money networks.

Central banks should ensure that infrastructure investment decisions, especially with regard to offline capability, are aligned with the geographic, socioeconomic and technological diversity of their jurisdictions. A readiness-driven approach should guide whether such investments are front-loaded, phased or piloted with targeted populations before they are scaled up.

Operational and maintenance costs

- Software and platform maintenance: continuous updates, bug fixes and performance optimizations
- Regulatory oversight: supervisory costs related to consumer protection, financial stability monitoring and reporting
- Public engagement and education: ensuring widespread awareness, trust and adoption through marketing and literacy programmes

Cost of transitioning from cash-based to digital systems

- Adjustments to automatic teller machine and cash-handling infrastructure

- Training of staff at financial institutions, merchants and consumers

Example. In The Bahamas (SandDollar), operational costs included nationwide awareness campaigns and merchant onboarding incentives to drive adoption.

6.1.3 Step 2. Assessing indirect economic costs

Beyond financial outlays, a digital currency could introduce broader economic trade-offs that must be evaluated, which include the following.

Impact on commercial banks and financial intermediation

- Deposit disintermediation: if consumers shift funds from bank deposits to digital currency wallets, banks may face reduced lending capacity.
- Credit market adjustments: central banks may need to introduce liquidity buffers or lending incentives to maintain credit supply.

Monetary policy transmission risks

- Interest-bearing digital currency dilemmas: if a digital currency accrues interest, it could disrupt money-market operations and monetary policy tools.
- Capital flight considerations: in highly dollarized economies, a digital currency could influence foreign exchange demand and external balances.

Cybersecurity and financial stability risks

- Increased fraud and cyberthreats: a digital currency system could be targeted by hackers and financial criminals and draw State-sponsored cyberattacks.
- Risk of operational failure: ensuring that digital currency remains functional during financial crises, power outages or technical failures is crucial.

Example. The developers of the e-Krona project in Sweden assessed the potential risks of digital currency substitution on bank deposits, concluding that a hybrid, two-tier model would mitigate disintermediation risks.

6.1.4 Step 3. Estimating quantifiable economic benefits

A central bank digital currency can drive long-term economic efficiencies and provide measurable benefits for financial systems, including the following.

Payment system cost savings

- Lower cash-handling costs: a digital currency would reduce expenses related to printing, transporting and securing cash.
- Faster and cheaper digital transactions: a digital currency can lower remittance costs, benefiting high-remittance economies.

Financial inclusion gains

- Enabling digital financial access for unbanked populations: digital access would be particularly effective in economies that have high mobile money penetration but low participation in formal banking.
- Increasing government-to-person payment efficiency: a digital currency would enable faster and more secure welfare disbursements, pensions and stimulus payments.

Cross-border trade and payment efficiencies

- Reducing dependence on correspondent banking networks: a digital currency would enable faster, cheaper international transactions.
- Enhancing regional and subregional economic integration: a digital currency would support trade within African free trade zones (African Continental Free Trade Area, West African Economic and Monetary Union, East African Community, Southern African Development Community).

Example. The Eastern Caribbean Central Bank's DCash project demonstrated a 50 per cent reduction in cross-border remittance costs using digital-currency-based transactions.

6.1.5 Step 4. Evaluating qualitative benefits

Some digital currency advantages are difficult to quantify but contribute significantly to economic resilience. They include the following.

Transparency and anti-corruption benefits

- Reducing illicit financial flows: digital currency transactions can be traceable, helping authorities to combat money-laundering, tax evasion and corruption.
- Improving fiscal efficiency: Governments can track and automate social payments, reducing leakages.

Strengthening financial system resilience

- Alternative payment infrastructure: a digital currency could reduce dependence on international credit/charge card networks.
- Crisis preparedness: a digital currency can serve as a fallback payment system during banking crises or financial instability.

Example. The e-CNY pilot in China introduced programmable payments for government subsidies, reducing fraud risks in distribution chains.

6.1.6 Step 5. Risk-adjusted cost-benefit analysis

After evaluating costs and benefits, central banks must conduct a risk-adjusted analysis, which includes the following.

Weighing costs against realized benefits

If a digital currency's projected costs exceed its economic benefits, alternative solutions (e.g. improving real-time gross settlement, expanding digital banking) should be considered.

Scenario planning for unintended consequences

Central banks should run stress tests, economic simulations and pilot studies to assess adoption rates, market reactions and unintended policy effects.

Comparative analysis against alternative digital payment innovations

Evaluate whether stablecoins, enhanced mobile money frameworks or existing digital payment rails offer lower-cost alternatives with similar benefits. See the IMF publication entitled "Positioning central bank digital currency in the payments landscape".[16]

6.2 Linking costs to policy drivers

The implementation of a digital currency is a strategic policy decision, not just a technological upgrade. While a cost-benefit evaluation provides an economic and financial assessment, central banks must also ensure that the costs incurred align with broader policy priorities. The true value of a digital currency should be assessed on the basis of how effectively it addresses national economic objectives, rather than simply financial return on investment.

The present section provides a structured approach to linking digital currency costs to policy drivers, ensuring that central banks can weigh costs and benefits in relation to national priorities, such as financial inclusion, monetary sovereignty and the modernization of payment systems.

6.2.1 Step 1. Mapping central bank digital currency costs to policy objectives

Each country has distinct economic and financial policy goals, and a digital currency must be assessed on the basis of its ability to advance these objectives. Table 22 below illustrates how different policy drivers can justify digital-currency-related costs.

[16] Manisha Patel, Safari Kasiyanto and André Reslow, "Positioning central bank digital currency in the payments landscape", Fintech Notes No. 2024/006 (Washington, D.C., IMF, 2024).

Table 22

Potential digital currency costs and expected long-term benefits

Policy objective	Potential digital currency costs	Expected long-term benefits
Enhancing financial inclusion	Cost of developing low-cost digital currency wallets, expanding mobile connectivity and enabling offline transactions	Increased financial participation, improved access for unbanked populations
Strengthening monetary policy transmission	Investment in digital currency liquidity management tools, anti-money-laundering/know-your-customer compliance systems and regulatory oversight mechanisms	Greater control over money supply, improved interest rate transmission
Modernizing payment systems	Infrastructure costs for real-time transactions, interoperability with mobile money and cross-border functionality	Faster and cheaper transactions, reduced reliance on cash and on private payment networks
Reducing cash management costs	Expense of phasing out high-cost cash distribution networks and incentivizing merchants to accept digital payments	Lower government expenses related to printing, distributing and securing cash
Improving fiscal transparency and government payments	Costs of integrating digital currency with government payment systems, developing automated tax collection frameworks	Reduced corruption, improved efficiency in welfare disbursements and tax compliance
Reducing dependence on foreign currencies (dollarized economies)	Costs of public trust campaigns, creating stable value mechanisms for local digital currency adoption	Greater monetary sovereignty, reduced exchange rate volatility
Strengthening subregional and/or regional and cross-border trade	Investment in interoperability with subregional and/or regional financial systems, regulatory alignment with trade partners	Faster and more cost-effective international payments, reduced reliance on correspondent banking

Source: Author's elaboration.

This mapping exercise helps central banks to justify investment in digital currency infrastructure, regulation and security on the basis of long-term policy benefits.

6.2.2 Step 2. Weighing costs and benefits against policy priorities

Not all policy objectives carry equal weight. Some goals, such as financial inclusion and payment modernization, may be immediate priorities, while others, including monetary policy innovation or subregional and/or regional integration, may be longer-term considerations.

Framework for prioritizing digital currency costs by policy impact

To ensure a rational decision-making process, central banks can assess costs using the following three-step evaluation model.

Policy relevance assessment

- Does the digital currency directly support a critical policy goal?
- Would alternative financial solutions achieve the same outcome at a lower cost?

- *Example*: If a country already has a successful mobile money system, would expanding mobile money regulations achieve financial inclusion more effectively than a digital currency?

Implementation cost versus expected benefit

- Are the upfront investment costs proportionate to long-term policy gains?
- Does the digital currency reduce systemic inefficiencies (e.g. lowering cash-handling costs, improving tax collection)?
- *Example*: In high-remittance economies, a digital currency that reduces remittance fees from 8 to 2 per cent could justify investment costs by returning billions in savings to the economy.

Short-term versus long-term trade-offs

- Is the digital currency implementation financially and operationally sustainable over time?
- Does the policy objective justify a phased roll-out to manage costs?
- *Example*: Instead of launching a full-scale digital currency, a central bank could start with a wholesale interbank settlement model, reducing risk while modernizing payment infrastructure.

By applying this framework, central banks can ensure that digital currency costs are aligned with national priorities, are economically justified and are phased in a way that minimizes fiscal strain.

6.2.3 Step 3. Using cost-sensitivity analyses to optimize digital currency design

Since policy objectives differ across countries, central banks must conduct sensitivity analyses to determine how different digital currency design choices affect costs and benefits (see table 23).

Table 23
Costs and benefits of various digital currency designs, and alternative approaches

Digital currency design choice	Cost considerations	Policy impact	Alternative approaches
Retail versus wholesale central bank digital currency	Retail requires extensive infrastructure and user adoption costs, while wholesale is cheaper and easier to implement	Retail digital currency directly supports financial inclusion, while wholesale enhances interbank efficiency	If financial inclusion is a priority, a hybrid model that integrates a central bank digital currency with mobile money may be more cost-effective
Centralized versus Distribution-ledger-technology-based architecture	Distribution ledger technology is more expensive to develop but may offer better resilience and transparency	Centralized digital currencies align with monetary policy control, while distribution-ledger-technology-based models support financial transparency	Countries that have a strong banking infrastructure may benefit more from centralized ledgers
Offline versus online central bank digital currency functionality	Offline support requires investments in hardware wallets and near-field communication/text-based solutions	Offline functionality is critical in rural and areas that have low connectivity	Countries that have high mobile phone penetration can focus on Unstructured-Supplementary-Service-Data-based solutions
Programmability features (e.g. smart contracts)	Adding programmability increases development costs and regulatory complexity	Programmability supports automated tax collection, conditional payments and subsidy disbursements	A phased roll-out could start with basic digital currency functions, introducing programmability in later stages

Source: Author's elaboration.

Cost-sensitivity analysis: design trade-offs

Cost-sensitivity analysis allows central banks to tailor digital currency design choices to policy priorities while controlling unnecessary expenditures.

6.2.4 Establishing a cost-justification model for digital currency investment

To ensure transparency and accountability, central banks can adopt a cost-justification model, balancing costs with expected policy benefits over a 5-year to 10-year horizon. To create a cost-justification model, central banks should:

- Break down all digital-currency-related expenses into one-time costs (infrastructure setup, legal amendments) and recurring costs (system maintenance, cybersecurity, regulatory oversight, customer support)
- Estimate economic returns from improved financial inclusion, payment efficiency, reduced remittance fees and cost savings from the reduced usage of cash
- Use key performance indicators, such as digital currency adoption rates, cost reductions related to government payments and banking sector stability metrics, to measure success

- Conduct a comparative analysis with alternative financial innovations (e.g. mobile money expansion, regulatory reforms in digital banking) to ensure that digital currency is the most efficient tool for achieving national objectives

6.3 Archetype-specific cost-benefit considerations

The cost-benefit dynamics of a central bank digital currency vary significantly depending on a country's economic structure, financial infrastructure and policy priorities. While some economies may realize higher efficiency gains and cost savings, others may face greater implementation challenges owing to weak financial systems, regulatory constraints or infrastructure gaps.

The present section examines the ways in which different archetypes influence the cost-benefit equation, highlighting critical trade-offs, expected returns and potential risks. By considering archetype-specific factors, central banks can make better-informed investment decisions and optimize their digital currency design and implementation strategy.

6.3.1 Large economies (e.g. Egypt, Morocco, Nigeria, South Africa)

Cost-benefit dynamics

Large economies benefit from economies of scale, meaning that the cost of digital currency implementation can be spread across a large population and financial system. High regulatory complexity and competition from existing payment systems, however, may slow adoption.

Potential benefits:

- High transaction volume leads to cost-efficiency in digital payments.
- Cross-border trade facilitation for large export-driven economies is enhanced.
- Wholesale interbank settlements are supported, improving financial market efficiency.

Potential costs and challenges:

- Integration with diverse financial players (banks, fintech companies) increases implementation costs.
- Strong regulatory coordination is required to balance innovation with financial stability.

Strategic cost optimization

- Leverage existing banking and fintech infrastructure to reduce development costs.
- Focus initial roll-out on wholesale digital currency for interbank settlements, minimizing risks.

6.3.2 High-mobile-money-penetration economies (e.g. Ghana, Kenya, Uganda, United Republic of Tanzania)

Cost-benefit dynamics

In economies in which mobile money is dominant, a retail digital currency can build on existing digital payment systems, reducing infrastructure costs. Competition with mobile money providers, however, may limit adoption unless the digital currency is designed for interoperability.

Potential benefits:

- Development costs are lowered owing to the existing digital payment infrastructure.
- Financial inclusion is increased by providing low-cost transactions for underserved communities.
- Government-to-person payments are enhanced, reducing fraud and inefficiencies.

Potential costs and challenges:

- Resistance may occur from telecommunications and mobile money operators, who may see digital currency as a threat.
- Robust offline functionality will be needed to reach rural areas that have limited connectivity.

Strategic cost optimization

- Ensure that digital currency is interoperable with mobile money platforms to encourage adoption.
- Partner with telecommunications operators to reduce distribution costs and leverage existing customer networks.

6.3.3 Fixed exchange rate and currency-pegged (e.g. Morocco, CFA Franc Zone)

Cost-benefit dynamics

In fixed exchange rate economies, a digital currency must be carefully designed to avoid destabilizing the currency peg. While cost savings from cross-border trade efficiencies are attractive, implementation risks are higher owing to monetary policy constraints.

Potential benefits:

- Cross-border payments within subregional monetary unions are enhanced, reducing reliance on foreign correspondent banks.
- Currency control mechanisms are strengthened, preventing capital flight.
- Monetary policy transparency is enhanced for regimes that use pegged currencies.

Potential costs and challenges:

- A high level of regulatory coordination is required across subregional central banks.
- The currency peg may be destabilized if the issuance of the digital currency is not carefully managed.

Strategic cost optimization

- Limit the initial roll-out to wholesale interbank transactions to manage exchange rate risks.
- Collaborate with subregional monetary authorities to ensure harmonized digital currency frameworks.

6.3.4 Highly dollarized economies (e.g. Democratic Republic of the Congo, Liberia, Zimbabwe)

Cost-benefit dynamics

Highly dollarized economies face significant adoption barriers, as the public may prefer foreign currencies over digital currency. A primary challenge is ensuring that digital currency adoption strengthens local currency use rather than increasing capital flight.

Potential benefits:

- Dependence on foreign cash is reduced, improving the stability of the local currency.
- Digital currency can be linked to incentives (e.g. tax breaks, government payments) to promote adoption.

Potential costs and challenges:

- Low public trust in domestic currency may hinder the uptake of digital currency.
- Additional stability mechanisms are required to prevent depreciation risks.

Strategic cost optimization

- Implement such incentives as digital-currency-based savings accounts with stability guarantees.
- Focus initial roll-out on government payments and business transactions to build confidence.

6.3.5 Resource-dependent economies (e.g. Algeria, Angola, Chad, Nigeria, Zambia)

Cost-benefit dynamics

For economies that are reliant on oil, gas and mineral exports, a digital currency could enhance fiscal transparency and improve revenue management. Fluctuations in commodity

prices, however, may create liquidity risks that must be addressed when designing a digital currency.

Potential benefits:

- Leakages in resource revenue distribution are reduced, improving fiscal accountability.
- Programmable smart contracts are enabled for automated tax collection and subsidy disbursement.

Potential costs and challenges:

- Dependence on cyclical commodity revenues may make the management of digital currency liquidity difficult.
- Strong regulatory oversight is required to ensure stability.

Strategic cost optimization

- Implement digital-currency-based fiscal transfer programmes to ensure efficient distribution of resource revenues.
- Develop liquidity buffers to manage volatility risks in commodity-dependent economies.

6.3.6 High-remittance economies (e.g. Gambia, Lesotho, Senegal, Somalia)

Cost-benefit dynamics

Digital currencies can significantly reduce remittance fees and transaction times, benefiting economies in which migrant workers send substantial funds home. Interoperability with international payment networks, however, is critical.

Potential benefits:

- Remittance costs are reduced, increasing household incomes.
- Foreign exchange reserves are strengthened by reducing reliance on informal transfer systems.

Potential costs and challenges:

- Cross-border regulatory agreements with foreign jurisdictions are required.
- Resistance may occur from traditional remittance service providers.

Strategic cost optimization

- Develop digital-currency-based remittance corridors with countries that have large migrant populations.
- Ensure seamless integration with international remittance networks (e.g. Society for Worldwide Interbank Financial Telecommunication (SWIFT), mobile money).

6.3.7 Small island economies (e.g. Cabo Verde, Comoros, Mauritius, Seychelles)

Cost-benefit dynamics

Tourism-heavy economies require digital currencies that facilitate international transactions, but the high cost of multi-currency integration can be a challenge.

Potential benefits:

- Frictionless digital payments for tourists are enabled.
- Reliance on cash-based foreign exchange services is reduced.

Potential costs and challenges:

- There are high costs with regard to developing multi-currency functionality.
- Strong cybersecurity measures are required to prevent financial crime.

Strategic cost optimization

- Develop a digital-currency-based tourist payment system that includes automatic currency conversion.
- Implement risk-based anti-money-laundering schemes and other frameworks to mitigate financial crime risks.

6.3.8 Fragile-State economies (e.g. Central African Republic, Chad, Somalia, South Sudan)

Cost-benefit dynamics

Fragile States face significant infrastructure challenges that make digital currency implementation costly. A well-designed digital currency, however, can enhance financial inclusion and support the distribution of humanitarian aid (see box 4).

Potential benefits:

- A secure and transparent system for aid disbursement is provided.
- Financial services are enabled in areas that have weak banking infrastructure.

Potential costs and challenges:

- Investment costs are high, owing to weak digital infrastructure.
- Political instability and cyberthreats can cause security risks.

Strategic cost optimization

- Implement offline-capable digital currency wallets to ensure usability in low-connectivity areas.
- Partner with international organizations to subsidize infrastructure costs.

Box 4

Design priorities for fragile States: resilience, simplicity and offline access

Fragile States often face such structural challenges as political instability, limited digital infrastructure, weak financial systems and inconsistent energy production and connectivity. For these countries, digital currency design must focus on resilience, usability in low-resource settings and gradual scalability. The costs for secure offline devices, Unstructured-Supplementary-Service-Data-routing and operational deployment activities must be explicitly captured early in budgeting processes. During the deployment of eNaira in Nigeria, early limitations in offline access highlighted the need for broader infrastructure investment.[a] Critical design considerations include:

Offline functionality as a baseline requirement

In the light of patchy or non-existent Internet and energy infrastructure, enabling offline payments through secure hardware wallets, Unstructured-Supplementary-Service-Data-based mobile wallets, or SIM-overlay technology is essential. This ensures inclusivity for rural populations and displaced communities.

Low-tech access channels

Digital currency systems should support phones with limited features, avoid dependence on smartphones and minimize data consumption. Unstructured Supplementary Service Data interfaces, text confirmation and agent-assisted services can drive adoption even in digitally constrained environments.

Minimal viable feature set

A modular digital currency architecture allows for phased feature deployment. Fragile States may benefit from starting with basic peer-to-peer and government-to-person payments before expanding to more complex programmability or cross-border functions.

Risk-resilient infrastructure

Systems should be designed to tolerate disruptions, including the offline queuing of transactions, data redundancy across jurisdictions and fallback protocols in cases of institutional or network failure.

Governance models for transitional contexts

Institutional fragility may necessitate interim governance frameworks, including technical stewardship by external partners or subregional monetary authorities during the initial phases.

[a] See Ozili Peterson, "eNaira central bank digital currency (CBDC) for financial inclusion in Nigeria", 11th IMF Statistical Forum. Available at www.imf.org/-/media/files/news/seminars/2023/11th-stats-forum/11-peterson-ok.pdf?.

6.4 Pilot testing as a cost-benefit validation and data-generation tool

Pilot testing serves as a critical bridge between theoretical cost-benefit modelling and real-world evidence. While earlier phases of digital currency exploration may rely on projections, assumptions and comparative studies, pilots allow central banks to validate expected outcomes, test user and system behaviours and adjust plans on the basis of empirical data. The present section outlines how pilot testing can function both as a validation mechanism and as a tool to generate data, enhancing the precision and credibility of the cost-benefit analysis.

6.4.1 Pilots as validation mechanisms

Pilot programmes validate critical assumptions related to user adoption, transaction costs, system reliability and trust dynamics. They allow central banks to:

- Confirm or recalibrate estimates of direct and indirect implementation costs

- Test the scalability and efficiency of underlying technical infrastructure
- Observe behavioural and market responses to the functionality, incentives and access models related to digital currencies
- Identify hidden operational costs and logistical challenges that may not be visible in desk-based assessments

By incorporating structured evaluation metrics into the pilot, such as transaction volumes, user registration rates, error margins and system uptime, central banks can make data-driven "go/no-go" decisions on broader deployment and associated investments.

6.4.2 Pilots as data-generation tools

Beyond validation, pilot testing provides granular data that are essential to enriching cost-benefit models with real-world behavioural, operational and adoption insights. For example, pilot testing can provide:

- User segmentation analytics: demographics, device-usage patterns and transaction behaviour can inform the inclusivity and accessibility dimensions of the digital currency.
- Trust and perception metrics: surveys and usage rates during the pilot programme help to quantify public confidence in the digital currency.
- Integration challenges: real-time data on interoperability with banks, fintech institutions and mobile money platforms identify the costs and timelines required for system harmonization.

This information helps central banks to revise previously abstract economic projections using grounded, contextual inputs that improve the accuracy of policy-impact forecasting.

6.4.3 Embedding data learned from pilots into the cost-benefit framework

Central banks are encouraged to formally integrate pilot testing into their cost-benefit workflow through:

- Pre-pilot projections: baseline estimates of adoption rates, cost parameters and economic value
- Mid-pilot checkpoints: structured review intervals with real-time feedback loops
- Post-pilot recalibration: adjustment of long-term cost, benefit and risk scenarios using validated pilot data

6.4.4 Archetype sensitivity to pilot insights

The impact of pilot results on cost-benefit models may differ by archetype. For example:

- In large economies, pilots can uncover infrastructure scalability issues that affect cost-efficiency.
- In fragile States, pilots may expose security or offline access needs that raise implementation costs but yield significant inclusion benefits.

- In high-mobile-money-penetration contexts, insights on adoption acquired from pilots can redefine expected benefits by validating wallet integration or interoperability models.

By tailoring pilot structures to archetype characteristics and policy goals, central banks can optimize both the efficiency of pilots and their analytical value.

6.4.5 Main recommendations

- Design pilots to capture both quantitative metrics and qualitative feedback, aligned with cost-benefit dimensions.
- Treat pilots as iterative rather than one-off, allowing for phased adjustments and refinements.
- Incorporate pilot data into the decision frameworks and risk management matrices seen throughout the handbook.
- Use pilots to generate not just adoption metrics but also signals that policies are effective, such as improved payment system reach or reduced transaction frictions.

Further reading

For central banks evaluating the cost-benefit trade-offs and operational considerations of digital currency deployment, two authoritative resources offer critical insights. The publication by the Consultative Group on Risk Management of the Bank for International Settlements, entitled *Central Bank Digital Currency (CBDC) Information Security and Operational Risks to Central Banks*, outlines the risk management frameworks that are essential to ensuring the resilience and security of digital currency systems. Complementing this, the publication by the United States Federal Reserve System, entitled "Money and payments: the U.S. dollar in the age of digital transformation", presents a structured approach to assessing the costs, benefits and potential risks of digital currencies within broader monetary and financial system objectives. These references reinforce the evaluation frameworks discussed in the present section.

7 Developing pathways

7.1 Phased implementation framework

The "developing pathways" stage is the culmination of the digital currency planning process, as it provides a structured road map that allows central banks to move from exploration to full implementation. At this stage, insights are integrated from earlier steps to ensure that each phase of digital currency development is aligned with policy objectives, readiness levels and practical considerations. By breaking down the process into well-defined phases, central banks are equipped with a robust framework to manage complexities, mitigate risks and maximize the potential of digital currency initiatives.

7.1.1 Purpose of a phased action plan

A phased action plan is essential to managing the multifaceted process of digital currency implementation. It ensures that critical decisions and activities are organized into manageable stages, each having clear objectives, measurable outcomes and decision points. By adopting a phased approach, central banks can build stakeholder confidence, address gaps

incrementally and adjust strategies on the basis of the insights gained at each stage. This methodology is particularly valuable for jurisdictions that have varying levels of readiness, as it offers the flexibility to tailor pathways to specific national contexts.

7.2 Phases of the action plan

7.2.1 Phase 1. Exploratory stage

The exploratory stage lays the foundation for a digital currency project, with a focus on understanding national priorities, assessing readiness and building consensus among stakeholders.

Central banks begin by engaging with important stakeholders, such as government agencies, financial institutions, industry leaders and civil society, to define the policy objectives for the digital currency. These consultations help to align expectations and identify use cases that address pressing national challenges, such as financial inclusion or cross-border remittances.

A comprehensive readiness assessment follows, evaluating technical infrastructure, regulatory frameworks, financial systems and social acceptance. Through this assessment, gaps and opportunities are identified, forming the basis for a feasibility report that outlines the scope and potential of the digital currency initiative.

The exploratory phase concludes with a "go/no-go" decision taken on the basis of stakeholder alignment, feasibility findings and resource availability. Success in this phase is marked by a shared vision for the digital currency project and a clear understanding of the steps required to advance to the pilot phase.

7.2.2 Phase 2. Pilot design and testing

In the pilot phase, technical and operational models are validated while testing the digital currency's functionality, usability and impact in a controlled environment. Rather than being limited to a one-off exercise, digital currency pilot programmes benefit most from an iterative approach, allowing for multiple rounds of testing and refinement in response to user feedback, technical challenges and evolving market conditions.

Central banks begin this phase by developing a prototype system that is tailored to the priority use cases identified in phase 1. This may include testing retail payments, wholesale settlements or specific functionalities, such as offline transactions, programmability or cross-border use cases. By intentionally limiting the scope of early iterations, central banks can focus on addressing critical challenges and gathering actionable insights before expanding functionality in subsequent rounds of pilot development.

Successful pilot programmes often employ rigorous monitoring and evaluation frameworks to assess performance, user satisfaction and potential risks. Central banks can leverage pilots to engage diverse stakeholders, ranging from individual users to financial institutions, to collect insights that inform system refinements. For example, feedback on usability may lead to adjustments in interface design, while operational insights may prompt enhancements to security measures.

Importantly, pilots and deployments should be designed with flexibility for phased expansion. This approach has been used effectively in such global initiatives as the Eastern Caribbean Central Bank's DCash, which expanded digital currency functionality and coverage

in stages, and the eNaira in Nigeria, in which additional features and user segments were introduced progressively. Iterative approaches not only accommodate learning but also enable trust-building with users and the financial sector over time.

At the conclusion of each pilot round, central banks assess whether the pilot's outcomes meet predefined success metrics, such as user adoption rates, transaction volumes and system reliability. These assessments inform a "go/no-go" decision for broader deployment or further iteration, ensuring that the digital currency evolves in alignment with policy goals and technical readiness (see box 5).

Box 5
Sveriges Riksbank e-krona project

The e-krona project by the Sveriges Riksbank (the central bank of Sweden) was executed in phases and included multiple rounds of testing and refinement. Throughout these phases, the Riksbank emphasized that the pilot projects were for learning and testing purposes, and that no decision had been made regarding the actual issuance of an e-krona.

Phase 1. This initial phase was focused on developing a technical platform for the e-krona. The solution tested was based on distributed ledger technology, specifically the Corda platform of R3. The e-krona was represented as tokens in a distributed network. This phase was primarily aimed at understanding how a technical solution for the e-krona could work.[a]

Phase 2. This phase built upon the first and continued the testing of the conceptual distribution model and the technical solution.[b] The main areas of investigation included:

- Ways in which the technical solution could be integrated with potential distributors (such as banks) and existing point-of-sale terminals
- The feasibility of offline payments with the e-krona
- Performance testing of the model and technical solution
- Integration with external participants, such as Handelsbanken and Tietoevry, to test how their internal systems interacted with the e-krona network

Phase 3. This phase was focused on exploring the interaction between the Riksbank and other payment market participants to provide public access to and use of the e-krona.[c] It also examined:

- Different levels of governance in a potential collaboration model
- The possibility of conditional payments using the e-krona
- The potential for the e-krona to simplify cross-border payments (through participation in Project Icebreaker with other Nordic central banks and the Bank for International Settlements)

Phase 4. This phase delved deeper into offline payment solutions.[d] An alternative offline solution was developed and tested that involved using a card as a payment instrument, with only the balance and transactions stored on the card. This contrasted with the phase 2 solution, in which mobile phones were used. Tests were focused on transfers to and from cards, payments to stores, transfers between users and synchronization between offline and online systems.

[a] See Sveriges Riksbank, "E-krona pilot phase 1", available at www.riksbank.se/globalassets/media/rapporter/e-krona/2021/e-krona-pilot-phase-1.pdf.

[b] See Sveriges Riksbank, "E-krona pilot phase 2", available at www.riksbank.se/globalassets/media/rapporter/e-krona/2022/e-krona-pilot-phase-2.pdf.

[c] See Sveriges Riksbank, "E-krona pilot phase 3", available at www.riksbank.se/globalassets/media/rapporter/e-krona/2023/e-krona-pilot-phase-3.pdf.

[d] See Sveriges Riksbank, "E-krona pilot phase 4", available at www.riksbank.se/globalassets/media/rapporter/e-krona/2024/e-krona-pilot-phase-4.pdf.

7.2.3 Phase 3. Implementation and scale-up

The implementation phase occurs when the digital currency transitions from a pilot project to a fully operational system, with a focus on nationwide deployment and user adoption.

Central banks implement the digital currency, ensuring its integration with existing financial systems, such as banking networks and mobile payment platforms. To encourage adoption, central banks launch comprehensive public-awareness campaigns that highlight the digital currency's benefits, such as reduced transaction costs and enhanced security. These campaigns address potential concerns, including privacy and usability, to build trust and confidence among users.

Operational governance frameworks are established to oversee the digital currency's functioning, covering such areas as regulatory compliance, cybersecurity and user support. These frameworks ensure that the digital currency operates seamlessly and remains resilient to technical or operational disruptions.

Success in this phase is measured by widespread adoption, stable system performance and effective governance. Central banks monitor these indicators to evaluate the digital currency's impact and guide ongoing improvements.

7.3 Tailoring pathways for different readiness levels

The phased action plan is designed to be flexible, accommodating jurisdictions at various stages of digital currency exploration:

- **Low readiness**. Jurisdictions at this level focus heavily on phase 1 activities, such as building stakeholder consensus and addressing foundational gaps in infrastructure and regulatory frameworks.
- **Medium readiness**. These jurisdictions emphasize pilot design and testing, refining use cases and scaling pilots to prepare for full implementation.
- **High readiness**. These jurisdictions prioritize advanced functionalities, such as cross-border interoperability and innovative use cases, ensuring that the digital currency integrates seamlessly with complex financial systems.

By tailoring pathways to readiness levels, central banks can allocate resources efficiently and advance at a pace that is suited to their capabilities.

7.4 Templates for pathway development

The pathway to digital currency implementation is not uniform across all jurisdictions. Countries differ in terms of financial infrastructure, economic conditions, regulatory frameworks and digital readiness, requiring tailored approaches for different national archetypes. The present section outlines the ways in which each archetype can prioritize and sequence digital currency development, ensuring that implementation pathways align with unique national challenges, policy objectives and infrastructure constraints.

The guidance provided here helps central banks to determine the most effective entry points, whether through wholesale interbank transactions, targeted pilot programmes or full-scale retail deployments.

Template 1. Low-readiness jurisdictions

- **Phase 1**. Engage stakeholders and conduct readiness assessments to identify gaps.
- **Phase 2**. Launch a limited pilot that is focused on a single use case, such as government payments.
- **Phase 3**. Gradually expand the digital currency's functionality, prioritizing underserved regions.

Template 2. Medium-readiness jurisdictions

- **Phase 1**. Rapidly finalize policy objectives and identify use cases.
- **Phase 2**. Conduct multiple pilot programmes to test technical and operational models.
- **Phase 3**. Scale up the digital currency across major economic sectors, supported by targeted adoption campaigns.

Template 3. High-readiness jurisdictions

- **Phase 1**. Focus on refining objectives and addressing minor readiness gaps.
- **Phase 2**. Test retail, wholesale and cross-border functionalities simultaneously.
- **Phase 3**. Deploy the digital currency at scale with advanced features, such as programmable money.

7.5 Success metrics for each phase

Success metrics provide clear benchmarks for evaluating progress and ensuring alignment with policy objectives. To guide progress, central banks should define success metrics at each stage, informed by national policy objectives and operational realities. These metrics should not only assess adoption and performance but also track institutional preparedness, user trust and public value creation.

Phase 1. Exploratory stage

In this foundational phase, success is measured by the extent to which the digital currency concept is aligned with strategic goals, institutional capacity is prepared and stakeholder consensus is built.

Example metrics

- Completion of stakeholder consultations, readiness assessments and feasibility reports
- Completion of a robust policy objective mapping exercise that is linked to national development plans
- Completion of readiness assessments across technical, legal, financial and governance dimensions
- Establishment of a digital currency steering committee or task force
- Evidence of political and institutional buy-in through formal declarations or frameworks

Feedback mechanism

Policy round tables, internal capability reviews and the publication of a feasibility report to collect external perspectives and update strategic direction.

Phase 2. Pilot design and testing

During this stage, important assumptions are validated through operational testing. The aim is not only to assess technology performance but also to refine the user experience, gauge trust and expose system vulnerabilities.

Example metrics

- Functionality testing (e.g. offline payments, interoperability, latency, throughput)
- High satisfaction rates among pilot participants, stable system performance and successful mitigation of identified risks
- Adoption rates within targeted pilot populations (e.g. rural agents, small merchants)
- Results from system stress tests and security audits
- Pilot cost-efficiency and data gained versus planned parameters

Feedback mechanism

Structured pilot debriefs, participant surveys, performance dashboards and iteration cycles, enabling the refinement of technical features, policy rules and roll-out strategies before scale-up.

Phase 3. Implementation and scale-up

At this stage, the digital currency is deployed to broader user segments. Metrics are focused on sustained adoption, system integration, financial inclusion outcomes and macroeconomic alignment.

Example metrics

- Broad user adoption, seamless integration with existing systems and effective governance oversight
- Transaction volumes and wallet activity across demographics and areas; integration rates with mobile money platforms, banks and government systems (e.g. government-to-person)
- Operational resilience, including system uptime and fraud detection; public trust and perception as measured through ongoing surveys or complaints data

Feedback mechanism

Nationwide monitoring frameworks, feedback from financial service providers and open digital literacy campaigns to sustain engagement and fine-tune deployment parameters.

7.6 Recommendations for optimizing pathways

The development of effective pathways requires strategic planning and ongoing adaptation and includes the following actions.

- **Engage stakeholders continuously**. Early and ongoing engagement ensures alignment with national priorities and builds broad-based support.
- **Focus on incremental progress**. Phased implementation allows central banks to address challenges iteratively, reducing risks and enhancing system resilience.
- **Prioritize scalability and flexibility**. Design systems and processes that can adapt to future needs, such as increased transaction volumes or new use cases.

By incorporating these recommendations, central banks can navigate the complexities of digital currency development while maximizing the initiative's potential benefits.

7.7 Success metrics and decision points

A well-structured digital currency implementation road map requires clear success metrics and decision points to evaluate progress and ensure that digital currency development is aligned with policy objectives, technical readiness and market adoption expectations. Establishing "go/no-go" criteria at each phase helps central banks to determine whether to proceed, refine their approach or reconsider digital currency deployment strategies.

The present section provides a framework for measuring success at different stages of digital currency implementation, ensuring that central banks remain accountable, adaptive and aligned with national priorities.

7.7.1 Establishing success metrics

Each digital currency phase should have defined performance indicators that measure progress against:

- Policy objectives: has the digital currency contributed to financial inclusion, payment efficiency or monetary stability?
- Readiness assessments: is the country's financial and technological infrastructure prepared for the next phase?
- Stakeholder engagement: have regulatory bodies, financial institutions and the public demonstrated acceptance and trust?
- Operational viability: has the digital currency performed reliably in pilot tests with no major security risks or technical failures?

Table 24 outlines major success metrics for each phase of digital currency implementation.

Table 24
Success metrics by implementation phase

Phase	Primary objectives	Success metrics	"No-go" criteria
Phase 1. Research and feasibility study	Assess economic, regulatory and technical feasibility of digital currency	Completion of digital currency impact study Readiness assessment score of at least 70 per cent Stakeholder consultation completed	If significant risks (e.g. financial stability, cybersecurity) remain unaddressed If feasibility study does not demonstrate clear policy benefits
Phase 2. Pilot design and prototyping	Develop a small-scale prototype for limited user testing	Successful pilot tests in at least two core use cases (e.g. retail payments, interbank settlements) No major cybersecurity or operational failures in test environments Positive feedback from at least 80 per cent of pilot participants	If infrastructure cannot support stable, scalable transactions If user feedback highlights unresolved accessibility issues
Phase 3. Controlled pilot with real transactions	Test digital currency in a limited real-world environment with selected users	At least 1,000 participants onboarded More than 95 per cent transaction success rate with no major security breaches Demonstrated interoperability with existing payment systems	If pilot reveals disruptive effects on financial stability If transaction success rates fall below 90 per cent as a result of technical issues
Phase 4. Policy and legal review	Align legal and regulatory frameworks for broader deployment	Digital currency legal framework approved by regulatory authorities Compliance with anti-money-laundering/countering the financing of terrorism, data privacy and financial consumer protection laws	If central bank authority over digital currency issuance remains unclear If legal risks related to privacy, fraud or taxation remain unresolved
Phase 5. Gradual expansion and public adoption	Scale digital currency adoption through a nationwide roll-out strategy	At least 30 per cent of target population onboarded within first year Strong private sector participation (e.g. merchants, fintech companies, banks) Demonstrated cost-efficiency compared with cash-based transactions	If public adoption remains below 15 per cent after one year If commercial banks experience excessive deposit disintermediation

Source: Author's elaboration.

7.7.2 Linking success metrics to policy drivers

Since policy priorities differ across countries, success metrics should be customized to align with national objectives. Table 25 illustrates the ways in which different policy drivers influence digital currency success criteria.

Table 25
Policy drivers and their influence on digital currency success

Policy driver	Success metrics	Impact on implementation phase
Financial inclusion	Digital currency adoption rate among unbanked and underserved populations Percentage of low-cost transactions using digital currency	Pilot testing (phase 3) and public roll-out (phase 5)
Payment-system modernization	Reduction in cash-handling costs and transaction fees Digital currency integration with existing mobile money and banking systems	Feasibility study (phase 1) and pilot design (phase 2)
Monetary policy transmission	Effectiveness of digital currency interest rate mechanisms Impact on deposit migration and bank lending	Controlled pilot (phase 3) and legal review (phase 4)
Cross-border trade and remittances	Reduced remittance costs and settlement time Digital currency interoperability with subregional financial systems	Pilot testing (phase 3) and expansion (phase 5)

Source: Author's elaboration.

This assessment ensures that the digital currency remains focused on national priorities, rather than being implemented purely as a technological upgrade.

7.7.3 Decision points: when to proceed, adjust or halt digital currency implementation

Each phase should include a structured evaluation that determines whether to:

- Proceed to the next phase if all success metrics are met
- Adjust design, policies or roll-out strategies if issues emerge that can be corrected
- Halt or reconsider digital currency implementation if risks outweigh benefits

Table 26 illustrates the decision criteria to be considered at each stage.

Table 26
Decision criteria to be considered at each stage

Phase	Proceed if	Adjust if	Halt if
Research and feasibility	Digital currency aligns with policy objectives Readiness assessment confirms feasibility	Some technical or regulatory gaps remain but are solvable	Readiness score is below 50 per cent Economic impact assessment shows high risk to financial stability
Pilot testing	Prototype functions successfully with minimal technical failures	Some user experience issues exist but can be improved	Pilot shows low public interest or major security flaws
Legal review	There is a clear legal mandate for digital currency issuance	Some privacy and compliance risks remain but can be addressed	Legal conflicts exist that prevent the central bank from issuing digital currency
Nationwide roll-out	Public adoption meets or exceeds 30 per cent in the first year	Adoption is slower than expected but improves with targeted incentives	Public adoption is below 15 per cent after a sustained period, signalling lack of demand

Source: Author's elaboration.

These structured decision points ensure that digital currency development remains flexible and responsive to real-world challenges.

7.7.4 Ensuring long-term success beyond initial implementation

Even after launch, digital currencies require continuous evaluation to maintain operational efficiency, public trust and financial stability. Ongoing success metrics should include:

- Transaction volume growth and system uptime stability
- User adoption trends, in particular among unbanked populations
- Assessment of impact on commercial bank liquidity and financial stability
- Assessment of international acceptance and cross-border settlement efficiency

Central banks must also remain adaptable, making adjustments on the basis of real-time data, emerging risks and economic trends.

8 Archetype-based guidance

8.1 Overview of archetypes

Central bank digital currencies are not a one-size-fits-all solution. Each country faces unique economic, financial, regulatory and technological conditions that influence digital currency design and implementation strategies. To address this, the archetype-based approach provides tailored guidance by grouping countries that have shared economic and financial characteristics into defined categories. These archetypes help central banks to identify critical challenges, prioritize policy objectives and select appropriate digital currency features that align with their national contexts and strategic goals.

While archetypes provide a useful reference framework, it is important to recognize that country profiles are not static. As economies evolve, including through digital infrastructure development, regulatory reforms, subregional integration or changes in monetary conditions, digital currency needs and design strategies may also shift. Therefore, digital currency planning must be dynamic, and regular reassessments of the national context must be conducted to ensure alignment with emerging priorities and opportunities. Strategies should allow for phased evolution, enabling central banks to adjust their approach as readiness improves, use cases expand or cross-border considerations emerge.

The present section outlines the primary archetypes relevant to African economies, describing their main characteristics and how these factors affect digital currency design, deployment and policy trade-offs. In situations in which countries exhibit features from multiple archetypes, guidance is offered to help to navigate the interplay between overlapping considerations and ensure a flexible, forward-looking approach to digital currency development.

8.1.1 Large economies (e.g. Egypt, Morocco, Nigeria, South Africa)

Defining characteristics

- Economies have high GDP and large, diversified financial sectors.
- Banking institutions and payment infrastructure are well established.
- Central banks have strong monetary policy and regulatory oversight.

Digital currency considerations

- Digital currencies must be integrated into existing banking and payment systems.
- Wholesale and retail digital currencies may both be viable, depending on policy objectives.
- Interoperability with commercial bank deposits and fintech systems is crucial.
- Financial stability risks should be carefully managed to avoid bank disintermediation.

8.1.2 High-mobile-money-penetration economies (e.g. Ghana, Kenya, Uganda, United Republic of Tanzania)

Defining characteristics

- Mobile money dominates digital transactions and financial inclusion efforts.
- There is limited reliance on traditional banking infrastructure in rural areas.
- Mobile network operators play a significant role in financial services.

Digital currency considerations

- Digital currencies must operate seamlessly with existing mobile money platforms.
- Unstructured-Supplementary-Service-Data-based and offline transactions will enhance accessibility.

- Regulatory clarity on digital currency's role in a mobile-led financial system is required.
- Fair competition should be promoted while leveraging mobile fintech innovations.

8.1.3 Fixed exchange rate and currency-pegged economies (e.g. Morocco, CFA Franc Zone)

Defining characteristics

- Currency stability is maintained through a fixed peg to an external currency.
- There is high reliance on foreign currency reserves for monetary policy.
- Central banks have limited flexibility in terms of domestic monetary expansion.

Digital currency considerations

- A digital currency must not disrupt the exchange rate peg, which requires careful monetary design.
- Digital currencies need to be integrated into subregional payment systems to facilitate trade and remittances.
- Ensuring capital flow stability will be essential to prevent speculative attacks on the peg.

8.1.4 Highly dollarized economies (e.g. Democratic Republic of the Congo, Liberia, Zimbabwe)

Defining characteristics

- There is extensive use of foreign currency in domestic transactions.
- The population has weak confidence in the local currency owing to inflationary history.
- Central banks have limited monetary policy autonomy.

Digital currency considerations

- Digital currency should restore trust in local currency by offering price-stability mechanisms.
- Capital controls and incentives would be required to prevent rapid conversion to foreign currency.
- Interest-bearing digital currency wallets could be used to encourage domestic currency usage.
- Digital currency must be integrated into remittance flows to facilitate liquidity in local markets.

8.1.5 Resource-dependent economies (e.g. Algeria, Angola, Chad, Nigeria, Zambia)

Defining characteristics

- There is high reliance on commodity exports (oil, minerals, agriculture).
- Government revenues fluctuate in line with commodity price volatility.
- Such economies are prone to instability due to external shocks.

Digital currency considerations

- Transparent tax mechanisms could enhance the collection of government revenues.
- Digital currency may be integrated into commodity-backed stablecoins to provide fiscal stability.
- Careful monetary policy coordination would be required to avoid exacerbating economic shocks.
- Digital currency should support cross-border trade settlements in natural resource markets.

8.1.6 High-remittance economies (e.g. Gambia, Lesotho, Senegal, Somalia)

Defining characteristics

- Remittances represent a significant share of GDP.
- There is high dependence on diaspora inflows for household income.
- Traditional remittance channels (wire transfers, online transfers) incur high costs.

Digital currency considerations

- Digital currencies should offer low-cost, real-time remittance solutions to reduce reliance on expensive intermediaries.
- Interoperability with international payments is crucial for efficient diaspora transactions.
- Cross-border digital currency partnerships with major sending countries and regions (e.g. European Union, United States of America) should be explored.
- Multi-currency digital currency solutions may be integrated to facilitate international trade and remittances.

8.1.7 Fragile-State economies (e.g. Central African Republic, Chad, Somalia, South Sudan)

Defining characteristics

- Countries experience political instability and have weak governance structures.
- Financial infrastructure is underdeveloped, and banking penetration is low.

- There is heavy reliance on international aid and humanitarian assistance.

Digital currency considerations

- Digital currencies should prioritize offline functionality to increase financial inclusion in rural and conflict-prone areas.
- Secure and auditable digital aid distribution mechanisms are required.
- AML/CFT protections must be integrated to prevent misuse in unstable environments.
- Regulatory oversight challenges must be addressed through regional and international cooperation.

8.2 Combining archetypes

While the archetype-based approach helps central banks to tailor digital currency strategies to their economic, financial and policy environments, many countries do not fit neatly into a single archetype. For instance, a country may experience significant dollarization, dependence on resources and substantial remittance inflows simultaneously, or it may exhibit traits of both fragile-State and high-mobile-money-penetration economies. In such scenarios, conflicts may arise when policy objectives, such as maintaining monetary stability and enhancing financial inclusion, diverge.

The present section provides guidance for central banks on:

- Supporting strategic clarity in decision-making in order to chart an adaptive course
- Identifying overlapping archetypes and assessing their combined impact on digital currency design
- Addressing potential conflicts between different archetype-driven priorities
- Leveraging synergies to enhance digital currency implementation and maximize benefits

By understanding the interactions among archetypes, central banks can develop digital currency strategies that are both flexible and effective in addressing their unique challenges.

8.2.1 Decision-making framework to resolve archetype trade-offs

1. Map archetype attributes to strategic goals

Identify which archetype attributes (e.g. dollarization, high-mobile-money-penetration infrastructure) correspond to which national priorities (e.g. exchange rate control, digital access).

2. Identify conflicting priorities

Discover potential tensions. For example:

- Token-based, offline-first models may serve fragile-State and high-mobile-money-penetration populations but could undermine AML/CFT controls.

- Account-based, fully regulated systems may reinforce monetary sovereignty but reduce access for the informally employed.

3. Apply national development anchors

Use national development plans, digital economy blueprints and macroeconomic strategies to frame which priorities must take precedence. For instance, if a digital financial inclusion strategy is a top-tier policy, inclusion-enabling design choices should be favoured.

4. Evaluate systemic risk implications

Prioritize choices that avoid destabilizing the financial system or undermining trust. Monetary and financial stability goals typically act as thresholds: if risks to stability are high, they may override other goals in the short term.

5. Implement in stages to balance goals over time

Design the digital currency implementation road map to address more urgent policy objectives first (e.g. financial access) while building capacity to address others in future phases (e.g. interoperable monetary controls).

Prioritization checklist: managing multiple archetype trade-offs

The checklist (see table 27) serves as a practical tool to support decision-making for central banks that are navigating overlapping digital currency archetypes. By systematically assessing such factors as systemic risk, infrastructure readiness, policy urgency and alignment with national development strategies, policymakers can make informed trade-offs when objectives appear to conflict. The checklist is especially useful for jurisdictions that are balancing inclusion with stability, or innovation with regulatory capacity, and ensures that digital currency strategies remain both context-sensitive and forward-looking.

Table 27
Prioritization checklist

Consideration	Guiding question
National vision alignment	Does one policy driver align more directly with stated national development plans?
Time sensitivity	Which objective is more urgent or politically time-bound (e.g. elections, reforms)?
Systemic risk impact	Which pathway presents a lower risk to financial stability or monetary integrity?
Inclusivity and equity	Which design supports the underserved or corrects historic exclusions?
Feasibility with current infrastructure	Which policy goal is more technically and financially viable in the short term?
Phasing potential	Can one policy goal be addressed in early phases and the other in later phases?
Cross-border and subregional commitments	Are there treaty obligations or monetary union commitments that tilt policy prioritization?
Stakeholder support	Which objective has broader support among public and private stakeholders?
Measurement clarity	Can success for this policy goal be clearly measured and tracked during roll-out?

Source: Author's elaboration.

8.2.2 Identifying countries that have overlapping archetypes

Some African countries exhibit multiple defining characteristics that span more than one archetype. Table 28 contains examples of overlapping archetypes and their implications for digital currency design.

Table 28
Overlapping archetypes and implications for digital currency design

Country	Overlapping archetypes	Implications for digital currency design
Kenya	High mobile money penetration; high remittances	Digital currency requires seamless mobile money integration and low-cost cross-border transfers
Morocco	Large economy; fixed exchange rate; high remittances	Requires scalable infrastructure, monetary stability features and cross-border payment integration
Nigeria	Large economy; high mobile money penetration; resource-dependent	Digital currency must balance interoperability with mobile money, scalability and government-revenue transparency
South Sudan	Fragile State; highly dollarized; resource-dependent	Digital currency must focus on offline functionality, monetary stability and transparent public financial management
Zimbabwe	Highly dollarized; high remittances; fragile State	Digital currency should restore confidence in local currency, support remittance inflows and ensure financial stability

Source: Author's elaboration.

Takeaway

Each country's digital currency strategy should be designed using a blended approach that prioritizes the most pressing economic and financial needs while ensuring long-term sustainability.

8.2.3 Managing trade-offs across overlapping archetypes

When a country is included under multiple archetypes, conflicts may arise with regard to defining policy priorities, technical design choices and regulatory approaches. Table 29 illustrates common conflict areas and mitigation strategies.

Table 29
Conflict areas and mitigation strategies

Conflict area	Example	Mitigation strategy
Exchange rate stability versus financial inclusion	Fixed exchange rate economies may want to limit capital flows, while high mobile money penetration requires open access	Implement tiered access policies whereby retail digital currency is freely available but wholesale transactions have controls
Interoperability versus monetary sovereignty	Highly dollarized economies may want limited convertibility, while remittance-dependent countries need cross-border interoperability	Introduce controlled exchange mechanisms to limit United States dollar outflows while supporting remittance inflows
Government revenue transparency versus privacy concerns	Resource-dependent economies may use central bank digital currencies to improve fiscal transparency, while fragile States may have trust deficits with regard to government oversight	Implement privacy-preserving transaction records, balancing transparency with user confidence
Scalability versus financial infrastructure limitations	Large economies require high transaction capacity, but fragile States may lack reliable digital networks	Deploy hybrid solutions, using online digital currency for urban areas and offline digital currency for rural/unstable areas

Source: Author's elaboration.

Takeaway

By understanding and mitigating conflicts, central banks can balance competing priorities and ensure a stable digital currency framework that is aligned with both economic stability and financial inclusion goals.

8.2.4 Leveraging synergies across archetypes

While conflicts exist, overlapping archetypes also create opportunities for synergistic central bank digital currency features that enhance policy outcomes. Table 30 sets out examples of synergies and design approaches that benefit countries that have characteristics of multiple economic archetypes.

Table 30
Synergies and digital currency strategies

Synergy opportunity	Countries and monetary unions	Digital currency strategy
High-mobile-money-penetration and high-remittance economies	Kenya, Ghana	Digital currency should be interoperable with mobile wallets and remittance platforms for seamless payments
Large economies and resource-dependent States	Angola, Nigeria	Use digital currency for government revenue collection, commodity-backed settlements and payment automation
Fixed exchange rate and high-remittance economies	Morocco, West African Economic and Monetary Union	Establish cross-border digital currency corridors to stabilize remittance flows without disrupting exchange rate pegs
Fragile States and highly dollarized economies	South Sudan, Zimbabwe	Digital currency should have offline functionality, stability mechanisms and gradual de-dollarization strategies
Small island and high-remittance economies	Mauritius, Seychelles	Enable multi-currency digital currency wallets for cross-border tourism and trade settlements

Source: Author's elaboration.

Takeaway

Leveraging synergies helps central banks to design digital currencies that maximize efficiency, financial inclusion and economic resilience.

8.2.5 Developing a hybrid central bank digital currency implementation strategy

For countries that belong to multiple archetypes, a hybrid implementation strategy is essential. Step-by-step guidelines for central banks that are developing a hybrid approach are as follows.

Step 1. Prioritize policy objectives

- Identify which policy goals carry the highest priority (e.g. monetary stability, financial inclusion, payment efficiency).
- Align digital currency design with national development plans and economic policies.

Step 2. Conduct a multilayered readiness assessment

- Assess digital infrastructure, regulatory frameworks and financial market maturity.

- Identify areas of synergy and potential conflicts among different archetype-driven priorities.

Step 3. Select a central bank digital currency design approach

- Define the core digital currency use cases (e.g. wholesale versus retail, online versus offline).
- Ensure that digital currency models are able to adapt over time to reflect economic and financial shifts.

Step 4. Implement phased roll-out strategies

- Start with targeted pilot programmes in specific sectors (e.g. government payments, cross-border trade, remittances)
- Gradually scale up digital currency adoption in accordance with performance, user adoption and regulatory adjustments.

Step 5. Monitor and adjust in line with market responses

- Establish real-time monitoring frameworks to track digital currency adoption, stability and impact.
- Adapt digital currency policies on the basis of feedback from financial institutions, businesses and consumers.

Takeaway

A hybrid strategy ensures that digital currency deployment remains dynamic, scalable and adaptable to evolving national priorities.

9 Central bank digital currency in subregional monetary unions

The present section is intended to support policymakers, central banks and subregional institutions that are in monetary unions, such as the West African Economic and Monetary Union, the Central African Economic and Monetary Community and the Common Monetary Area, by setting out practical guidance on ways to explore or implement a digital currency within a multi-country framework, in which monetary sovereignty is pooled and cross-border coordination is critical.

9.1 Overview of African monetary unions

The African continent is home to several unique examples of monetary integration, where multiple sovereign countries share a common currency and a centralized monetary authority. These formal monetary unions are characterized by treaty-based arrangements, uniform monetary policy and cross-border currency circulation. The present section contains an overview of the three existing legally constituted monetary unions in Africa and a foundation for understanding the implications of central bank digital currency exploration within such frameworks.

Why monetary unions matter when exploring digital currencies

The introduction of a digital currency within a monetary union introduces additional layers of complexity and opportunity beyond those faced by single-country jurisdictions. A digital currency issued by a shared central bank must serve the economic needs, legal systems, digital readiness and financial landscapes of multiple independent countries. It requires subregionally coordinated decision-making, harmonized regulation and interoperable technology solutions to ensure inclusive and stable adoption.

Understanding the structure and characteristics of monetary unions in Africa is therefore a critical step towards designing and implementing any digital currency intended to circulate across borders within a shared currency zone.

9.1.1 West African Economic and Monetary Union

The West African Economic and Monetary Union is composed of eight West African countries: Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, the Niger, Senegal and Togo.

These countries share a common currency, the West African CFA franc, and a single central bank, the Central Bank of West African States, headquartered in Dakar.

Primary characteristics:

- Monetary policy is centralized and governed by the Central Bank of West African States under the oversight of the Council of Ministers, which is composed of the finance ministers of each member State.
- The Central Bank of West African States issues the West African CFA franc, which is pegged to the euro under a monetary cooperation agreement with France.
- The West African Economic and Monetary Union area has a relatively integrated payments system and a growing interest in financial technology and cross-border mobile payments.

Digital currency relevance:

- A retail digital currency issued by the Central Bank of West African States could advance financial inclusion and cross-border interoperability within the Union.
- Any digital currency would require consensus among all member States and, potentially, a revision of the Central Bank of West African States' statutes and of the Treaty on the West African Economic and Monetary Union, in order to accommodate the issuance of a digital currency and ensure its status as legal tender.

9.1.2 Central African Economic and Monetary Community

The Central African Economic and Monetary Community is composed of six Central African countries: Cameroon, the Central African Republic, Chad, the Congo, Equatorial Guinea and Gabon.

These countries share the Central Africa CFA franc, issued by the Bank of Central African States, headquartered in Yaoundé.

Primary characteristics:

- Like the West African Economic and Monetary Union, the Central African Economic and Monetary Community operates under a centralized monetary authority and common currency, with the Central Africa CFA franc also pegged to the euro.
- The Bank of Central African States governs the monetary and exchange rate policy and works with national regulators to implement consistent banking supervision.
- The legal and regulatory harmonization across the Central African Economic and Monetary Community is more limited compared with that of the West African Economic and Monetary Union, although subregional integration efforts are ongoing.

Digital currency relevance:

- The Bank of Central African States has publicly acknowledged an interest in exploring a digital currency for the Central African Economic and Monetary Community area, with emphasis on improving financial-system modernization and transparency.
- Digital currency design must address such challenges as low digital penetration, varying legal frameworks and limited interoperability between digital payment platforms.

9.1.3 Common Monetary Area

The Common Monetary Area in Southern Africa links four countries: Eswatini, Lesotho, Namibia and South Africa.

In this arrangement, the South African rand serves as the legal tender across all four countries, with Eswatini, Lesotho and Namibia also issuing their own national currencies (pegged at par to the rand): the Swazi lilangeni, the Lesotho loti and the Namibia dollar.

Primary characteristics:

- The central bank of South Africa, the South African Reserve Bank, acts as the anchor institution, and Common Monetary Area monetary policy is effectively set by South Africa.
- Member countries maintain their own central banks, but their monetary policy autonomy is limited owing to the fixed exchange rate and common currency peg.
- The Common Monetary Area does not operate under a single legal or institutional framework in the same way as the West African Economic and Monetary Union or the Central African Economic and Monetary Community do.

Digital currency relevance:

- The South African Reserve Bank has conducted extensive digital currency research and experimentation, including Project Khokha (a wholesale digital currency) and retail digital currency feasibility studies.
- Any extension of a digital rand to other Common Monetary Area countries would require subregional coordination, integration with local financial infrastructure and policy alignment.
- Common Monetary Area members may also explore national digital currencies pegged to the rand, which would raise additional questions about convertibility, legal tender status and monetary sovereignty.

9.1.4 Critical differences from proposed monetary unions

It is important to distinguish the three monetary unions above from other subregional blocs, such as the Economic Community of West African States and its proposed eco currency, and the East African Community and its road map towards a monetary union.

These initiatives involve proposed common currencies and central banks, but, as at the time of writing, they remain aspirational and are not yet operational. The present handbook is focused only on the functioning, treaty-based unions in which monetary integration is already in place.

9.2 Governance and institutional coordination

Implementing a central bank digital currency within a subregional monetary union introduces a level of institutional complexity not present in single-country systems. Because monetary unions are composed of multiple sovereign States operating under a shared monetary authority, any decision to issue, design or regulate a digital currency must be made collectively, taking into account national interests, subregional obligations and legal mandates.

Effective governance and institutional coordination are foundational requirements for digital currency success in monetary unions. Unlike national digital currencies, currencies in subregional contexts must navigate a more complex decision-making environment, balancing union-wide policy alignment with national sovereignty.

To be effective, governance arrangements should be:

- Inclusive of all member States perspectives
- Transparent in their process and communication
- Harmonized across regulatory and operational domains
- Adaptable to evolving policy priorities, technologies and risk environments

The present section of the handbook contains a framework that is intended to guide central banks and policymakers in designing and managing the institutional arrangements necessary to successfully explore, implement and scale a digital currency within a monetary union.

9.2.1 Shared governance structures in monetary unions

In a monetary union, monetary policy and currency issuance are centralized, but economic governance and fiscal policy remain under the jurisdiction of individual member States. The existing governance architecture typically includes:

- A common central bank that has the exclusive authority to issue the shared currency and implement monetary policy.
- A council of ministers or equivalent body composed of finance ministers from each member country who are responsible for approving monetary policy and macroeconomic strategies.
- Committees tasked with supervising banking, ensuring financial stability, overseeing payment systems and harmonizing economic policy.

For example:

- In the West African Economic and Monetary Union, the Central Bank of West African States is governed by the Board of Directors, the Monetary Policy Committee and the Council of Ministers, with each member country represented.
- In the Central African Economic and Monetary Community, the Bank of Central African States operates under similar structures, with additional oversight from a commission.
- The Common Monetary Area, while not governed by a single supranational central bank, is anchored and governed by the South African Reserve Bank, and coordination is achieved through bilateral agreements and technical committees.

For a digital currency to be successfully introduced into these unions, it must be endorsed and guided by these collective governance structures, in particular in areas related to:

- Legal authorization of digital legal tender
- Institutional oversight of issuance and distribution
- Harmonized policy frameworks across all member States

9.2.2 Multi-country decision-making when launching a central bank digital currency

Launching a digital currency in a monetary union demands multiple tiers of decision-making, involving technical, legal and political stakeholders across several jurisdictions.

Primary decision-making challenges include:

- Achieving consensus across diverse economies that have different levels of digital readiness, financial inclusion and regulatory capacity
- Balancing the need for subregional uniformity with national economic priorities and development strategies
- Ensuring that digital currency features, such as limits, interest-bearing capacity, access rules and identity verification, are acceptable to all member States

Recommended approach:

- Establish a central bank digital currency steering committee at the subregional central bank, with representation from:
    - Member State finance ministries
    - National central banks (where applicable)
    - Financial regulators and payment system operators
    - Subregional legal and technical experts
- The steering committee would oversee:
    - The development of a central bank digital currency strategy document for the union
    - Feasibility studies and pilot projects
    - The coordination of legal reforms and regulatory harmonization
    - Stakeholder engagement across the subregion

This governance model should be transparent, inclusive and adaptive and should have clearly defined roles, escalation procedures and communication protocols.

9.2.3 Aligning institutional mandates

Each institution in a monetary union, whether at the subregional or national level, has distinct mandates that must be respected and harmonized to ensure digital currency success (see table 31). Misalignment between institutions may cause delays or regulatory uncertainty.

Table 31
Institutional mandates and their implications for digital currencies

Institution	Mandate	Digital currency implication
Common central bank	Currency issuance, monetary policy, financial stability	Must be legally empowered to issue digital currency and oversee distribution
Council of ministers	Policy approval, macroeconomic governance	Must endorse digital currency design and ensure political alignment across countries
National central banks (if applicable)	Payment systems, banking regulation (in some unions)	May need operational roles in digital currency roll-out and integration with local systems
National regulators	Licensing, AML/CFT supervision, consumer protection	Requires harmonized frameworks for digital currency usage, including know-your-customer, privacy and risk controls.

Source: Author's elaboration.

9.2.4 Cross-institutional coordination mechanisms

Coordination will be essential not only within the central bank but also across national Governments, subregional bodies and private sector stakeholders. This coordination must be structured, frequent and focused on actionable deliverables.

Central banks should consider:

- Creating a digital currency coordination secretariat to provide project management, communication and technical support
- Forming technical working groups for such areas as cybersecurity, interoperability, user interface design and legal harmonization
- Establishing a subregional stakeholder forum to consult with:
    - Commercial banks
    - Mobile money operators and fintech companies
    - Telecommunications regulators
    - Consumer advocacy groups

This cross-functional structure would allow for inclusive input, efficient problem-solving and the timely implementation of digital currency initiatives at scale.

9.2.5 Risk governance and crisis response

Digital currency introduction in a multi-country context must include strong governance protocols with regard to risk monitoring and crisis response.

Recommendations:

- Define decision-making triggers and response frameworks for such events as:
    - Cybersecurity breaches
    - Cross-border operational failures
    - Disruptions to offline digital currency usage
    - Political or fiscal crises that affect digital currency trust or usage
- Assign clear roles to subregional and national institutions to ensure accountability
- Embed stress testing, scenario planning and risk dashboards into digital currency governance processes

9.2.6 Institutional readiness assessment

Before launching a digital currency, monetary unions should conduct an institutional readiness assessment to identify gaps in:

- Legal authority for digital currency issuance
- Human and technical capacity within central and national banks

- Inter-agency coordination mechanisms
- Decision-making transparency and stakeholder engagement

9.3 Interoperability among member States

A defining feature, and one of the greatest potential benefits of a central bank digital currency in a subregional monetary union, is the ability of a digital currency to function seamlessly across borders between member States. Because monetary unions, such as the West African Economic and Monetary Union, the Central African Economic and Monetary Community and the Common Monetary Area, are built on a foundation of common currency and centralized monetary policy, a digital currency must extend this subregional coherence to the digital domain.

Interoperability among member States is the backbone of a successful digital currency in a monetary union. It ensures that the digital currency can serve as a truly subregional public good, unlocking the full economic and policy benefits of monetary integration.

To achieve this, central banks and stakeholders must collaborate on:

- A robust, scalable technical architecture
- Harmonized compliance frameworks
- Integrated national and subregional payment systems
- Coordinated institutional governance

The present section contains information on the technical, policy and institutional considerations that must be taken into account to ensure that a digital currency can operate across all countries in a monetary union, enabling efficient, secure and inclusive transactions regardless of national borders. It also contains actionable guidance for ensuring that interoperability supports broader policy goals, such as financial integration, cross-border trade and financial inclusion.

1. Why interoperability matters in monetary unions

Monetary unions are, by nature, integrated currency areas. A physical banknote of the West African CFA franc is accepted equally in Dakar, Ouagadougou or Abidjan, Côte d'Ivoire. A digital version of the same currency must maintain, if not enhance, this standard. Interoperability is therefore not just a feature but also a requirement for a digital currency operating in a multi-country monetary union.

Main reasons why interoperability is critical:

- It ensures uniform access to the digital currency across the subregion, preventing fragmentation of the common currency.
- It facilitates cross-border retail and business transactions within the union, enhancing subregional trade and mobility.
- It promotes economies of scale for digital infrastructure, reducing duplication and operational costs.
- It supports policy coherence by allowing the common central bank to monitor digital currency usage and liquidity flow across the union.

2. Technical interoperability considerations

The central bank digital currency infrastructure must be capable of functioning consistently and reliably across multiple national payment landscapes. This may be achieved through either a single subregional platform or interconnected national platforms governed by a shared architecture.

Primary technical considerations include:

- Common standards and protocols: all member States must adopt shared payment messaging standards (e.g. ISO 20022), data formats, encryption protocols and authentication requirements.
- Platform integration: national financial institutions, mobile money operators and fintech companies must be integrated into the central bank digital currency system using standard application programming interfaces, enabling seamless wallet-to-wallet transfers regardless of geography.
- User identity and know-your-customer interoperability: to ensure AML/CFT compliance and promote trust, the digital currency system must incorporate interoperable digital identity systems or, at a minimum, harmonized know-your-customer standards across borders.
- Offline functionality: in many member States, especially those that have limited connectivity, offline digital currency transactions will be essential. These should be designed to reconcile accurately across borders once connectivity is restored.
- Settlement finality and ledger synchronization: the digital currency ledger must ensure real-time or near real-time settlement and ledger consistency across jurisdictions. This is particularly critical in situations where different member countries host different nodes or ledgers.

Selecting and implementing an interoperability framework

When developing a digital currency for a subregional monetary union, the architecture of the underlying digital currency infrastructure serves as a foundational decision, in particular in relation to how interoperability among member States will be achieved. The interoperability framework must ensure the seamless operation of the digital currency across borders, financial institutions and user platforms while aligning with the union's policy, operational and regulatory needs.

Three primary models for interoperability architecture are presented in the present section, and guidance is provided on how to assess and select the most appropriate option for a subregional digital currency.

1. Centralized architecture

Description

Under a centralized architecture, the entire digital currency system, including issuance, transaction processing, validation and ledger management, is maintained by a single central platform, typically operated by the common central bank (e.g. the Central Bank of West African States for the West African Economic and Monetary Union; the Bank of Central African States for the Central African Economic and Monetary Community).

Main features

- One authoritative ledger for all digital currency transactions across the subregion
- Uniform governance and operational control by the subregional central bank
- Direct integration with financial institutions and service providers from all member States

Advantages

- Simplifies system maintenance and security management
- Ensures the consistent application of rules and policies across the subregion
- Eases implementation of subregion-wide updates and risk controls

Challenges

- Potential bottlenecks or single points of failure
- May raise concerns among member States over the centralization of operational authority
- Limited flexibility with regard to country-specific integration needs or infrastructure realities

Recommended use cases

- Subregions that have strong central bank capacity and highly centralized governance structures
- Member States that have relatively homogenous infrastructure and digital maturity levels

2. Decentralized federated architecture

Description

The decentralized federated architecture model distributes operational responsibilities across national nodes, with each country's central bank or designated institution managing a local ledger that is interoperable with the others. The digital currency is issued centrally, but ledger operations are federated.

Main features

- Multiple interoperable ledgers operate at the country level
- Common standards for data formats, messaging, compliance and settlement
- Coordination through a shared governance layer to ensure policy alignment and dispute resolution

Advantages

- Empowers member States with operational autonomy while maintaining policy coherence
- Allows customization of digital currency access and integration to fit national infrastructures
- Improves resilience through distributed architecture

Challenges

- Greater complexity in managing interoperability protocols and ledger synchronization
- Requires robust coordination mechanisms and mutual trust among national operators
- May involve longer roll-out timelines and greater resource requirements

Recommended use cases

- Subregions that have strong national institutions and variable digital infrastructure
- Unions in which member States desire greater sovereignty over implementation without diverging from a shared monetary framework

3. Hybrid architecture

Description

A hybrid architecture blends the centralized and decentralized approaches. The digital currency is issued and governed centrally, but the access layer (e.g. wallets, interfaces, application programming interfaces) is decentralized, allowing national institutions or regulated providers to manage distribution and front-end services.

Main features

- Central issuance and policy enforcement through a common ledger or centralized core
- Decentralized access and integration points for national actors
- May support optional national subledgers or middleware layers

Advantages

- Balances operational efficiency with national flexibility
- Simplifies compliance and updates through a single source of record while maintaining country-specific adaptations
- Facilitates rapid roll-out and inclusion of legacy systems through application layers

Challenges

- Requires careful delineation of roles and responsibilities at different layers
- Must ensure that decentralization of access does not create inconsistencies in user experience or policy enforcement

Recommended use cases

- Subregions that have significant private sector innovation and mobile money systems
- Monetary unions pursuing phased digital currency implementation with pilot testing in select countries

Guidance for framework selection

To select an appropriate interoperability framework, subregional central banks should assess the following (see table 32).

Table 32
Assessment questions to determine an appropriate interoperability framework

Criteria	Assessment questions
Institutional capacity	Does the central bank have the technical and operational capacity to manage a fully centralized system?
Member State preferences	Are member States aligned in their willingness to cede operational control, or do they prefer autonomy?
Infrastructure maturity	Is digital infrastructure uniformly advanced among member States?
Security and resilience	What level of fault tolerance and cyberresilience is required?
Scalability and evolution	Will the framework support future expansion, innovation or external interoperability?

Source: Author's elaboration.

The choice of interoperability framework is not only a technical decision but also a strategic choice that affects sovereignty, risk distribution, user experience and future scalability. Whether through a centralized platform, a federated network or a hybrid solution, subregional monetary unions must adopt an architecture that reflects their governance structures, policy priorities and infrastructure realities while ensuring that the digital currency remains a truly shared and trusted subregional instrument.

4. Payment system and infrastructure alignment

Even within a monetary union, digital payment systems are often fragmented across national boundaries. Different member States may have different:

- Real-time gross settlement systems
- Instant payment platforms
- Mobile money services
- Banking application programming interfaces and fintech landscapes

To enable digital currency interoperability, these systems must either be integrated with the digital currency platform or upgraded to a common baseline standard that allows for seamless interconnection.

For a digital currency to operate seamlessly among member States in a monetary union, it must be anchored in a robust and harmonized digital payments system. Even within a shared currency area, differences in national infrastructure, payment system maturity and institutional readiness can create barriers to effective digital currency interoperability. The present section contains essential steps to ensuring alignment between national infrastructure and the subregional digital currency platform.

Conducting a subregional infrastructure audit

A foundational step is to undertake a comprehensive audit of existing payment systems across member States. This audit should assess:

- Maturity and capacity of real-time gross settlement systems
- Integration capabilities of national instant payment platforms
- Mobile money and fintech adoption
- Gaps in identity systems, cybersecurity and connectivity

The findings will help to identify interoperability bottlenecks and guide investment priorities for infrastructure upgrades that support the deployment of digital currency and cross-border functionality.

Aligning national payment strategies with subregional digital currency objectives

National Governments and central banks should review and, where needed, adapt their digital payment strategies to ensure consistency with the subregional digital currency road map. This includes:

- Harmonizing payment system development goals
- Integrating national platforms with regional and international standards (e.g. ISO 20022, application programming interface specifications)
- Establishing a shared vision for digital currency utility across public and private sectors

Policy coordination across national jurisdictions will enable synergistic progress, avoiding duplication and fragmentation.

Creating a common payment switch or interoperability layer

A critical enabler of digital currency cross-border functionality is the establishment of a shared subregional switch or interoperability layer. This component will:

- Route and settle digital currency transactions in real time between participants across borders
- Serve as a technical bridge between different national systems
- Standardize messaging formats and ensure transaction consistency

Such an infrastructure can be managed by the subregional central bank or a designated consortium, with service-level agreements covering participation and governance.

Establishing a subregional testing environment or sandbox

To validate technical integration and policy assumptions, central banks can create a subregional digital currency testing environment or regulatory sandbox. This sandbox should:

- Allow for cross-border transaction simulation using prototype wallets or platforms
- Facilitate collaboration among national regulators, fintech companies and payment providers
- Test interoperability, compliance, user experience and data-protection protocols

Insights from the sandbox can inform adjustments to digital currency design, regulatory frameworks and technical standards prior to subregional deployment.

5. Regulatory and compliance interoperability

Technical interoperability is not sufficient without aligned regulatory and compliance frameworks. Transactions that originate in one country and are received in another must:

- Comply with harmonized AML/CFT standards
- Protect user data under mutually recognized privacy laws
- Be monitored by appropriate supervisory authorities without duplication or conflict

Central banks can consider the adoption of:

- A mutually recognized supervisory perimeter across the union, coordinated by the common central bank
- Harmonized customer due diligence and reporting rules across all member States
- Cross-border data-sharing agreements among financial intelligence units and regulators

This alignment will reduce compliance friction, minimize regulatory arbitrage and ensure that digital currency transactions are safe, legal and trusted throughout the subregion.

6. Institutional and operational coordination

Digital currency interoperability must be backed by strong institutional coordination mechanisms to manage system governance, resolve technical disputes and ensure operational consistency.

Recommendations:

- Define a digital currency interoperability governance framework, including:
    - The operational roles of the central bank, national regulators and service providers

- Incident management and risk-escalation procedures
- Service-level agreements across countries

- Establish a subregional interoperability committee to oversee implementation, updates and user feedback
- Coordinate subregional training and capacity-building programmes to ensure uniform understanding of the system

7. Use cases for interoperability within the union

Digital currency interoperability should support a wide range of cross-border use cases, including:

- Retail payments: a trader in Bamako receiving digital currency from a customer in Abidjan
- Remittances: a migrant worker in Cameroon sending funds to family in Chad without intermediary delays or costs
- Government transfers: a subregional development programme disbursing digital aid to citizens in multiple member States
- Business-to-business: a logistics company in Namibia paying a supplier in South Africa using digital rand

To fully leverage the benefits of a digital currency in a monetary union, central banks should design and implement workflows specific to use cases that reflect the economic realities, transaction patterns and user needs within and among member States. While the digital currency platform must remain interoperable and uniform in terms of its core functionality, customized transaction flows and policy configurations can enhance relevance, uptake and impact.

Tailoring use cases to subregional needs

Common use cases that can drive early adoption and demonstrate value include:

- Cross-border retail payments between individuals or small businesses operating in different member countries
- Intraunion remittances, enabling instant, low-cost transfers for workers supporting families across borders
- Government-to-person disbursements, such as digital social transfers and disaster relief or subregional development funding
- Business-to-business settlements across national supply chains within the union

Each of these use cases may require tailored features, such as limits, user verification rules, foreign-remittance tagging or offline capabilities, all while maintaining a consistent transaction format, security standard and user experience.

Maintaining interoperability and user standards

As use cases evolve, it is critical to:

- Adhere to common technical and compliance standards (e.g. ISO 20022, harmonized AML/CFT protocols)
- Ensure consistency in user interface and user experience for end users, even when features differ by country or sector
- Maintain real-time visibility and settlement finality across jurisdictions
- Support language, device and connectivity diversity in regional wallet design

By balancing customization with core interoperability, digital currency use cases can be optimized to drive adoption, efficiency and subregional integration while preserving the digital currency's role as a unifying and trusted subregional digital currency.

8. Interoperability beyond the union

While intraunion interoperability is the priority, the digital currency should also be designed with future integration at the regional and global levels in mind. This includes potential participation in:

- Pan-African payment systems (e.g. Pan-African Payment and Settlement System)
- Multi-central-bank digital currency platforms (e.g. Project Dunbar and Project mBridge of the Bank for International Settlements)
- Digital remittance corridors between African subregions or with diaspora countries

Using a design that has open architecture and international standards from the outset will future-proof the digital currency system and enhance its strategic value.

Multi-central-bank digital currency platforms

When exploring the use of digital currencies for cross-border payments within a shared settlement platform, multi-central-bank digital currency projects offer essential lessons relevant to monetary unions that are considering the implementation of a subregional digital currency.

Project mBridge was a multi-central-bank digital currency initiative resulting from extensive collaboration, beginning in 2021, among the Innovation Hub of the Bank for International Settlements and the Digital Currency Research Institute of the People's Bank of China, the Hong Kong Monetary Authority, the Bank of Thailand and the Central Bank of the United Arab Emirates. The Saudi Central Bank joined the initiative in 2024.

Project Dunbar was a collaborative multi-digital currency initiative led by the Innovation Hub of the Bank for International Settlements and the central banks of Australia, Malaysia, Singapore and South Africa.

Legal and governance considerations

A clearly defined, cross-jurisdictional legal framework is critical in order to outline roles, responsibilities and regulatory compliance across participating central banks.

In Project mBridge, participating central banks established a joint governance body and clearly defined legal agreements to manage responsibilities, liabilities and regulatory compliance across jurisdictions. A comprehensive legal framework was established under the project, including a detailed rulebook, to ensure compliance with each participating jurisdiction's laws and regulations. Each central bank retained control over its digital currency issuance and redemption processes, maintaining monetary sovereignty and adhering to domestic legal requirements. This approach provided clarity and operational certainty, facilitating cooperation despite differing national laws.

Project Dunbar developed structured governance frameworks that explicitly defined decision-making processes, data-sharing rules and regulatory alignment protocols among member countries. The project emphasized the importance of clear jurisdictional boundaries and regulatory compliance, enabling participating banks to transact directly in multiple digital currencies under a common legal framework. By proactively agreeing on cross-border legal principles, Project Dunbar reduced complexity and built trust, setting a clear precedent for effective international digital currency collaboration.

Operational and technical considerations

Project mBridge developed a common blockchain-based platform (mBridge Ledger) designed to seamlessly facilitate multi-currency digital currency transactions. By leveraging technology compatible with widely used standards, it ensured adaptability, interoperability and scalability across jurisdictions. Operational and technical approaches included:

- **Custom blockchain platform**. The mBridge Ledger was a bespoke distributed ledger technology platform designed to facilitate real-time, peer-to-peer cross-border payments and foreign exchange transactions using multiple digital currencies.
- **Decentralized architecture**. Each participating central bank operated a validator node, ensuring decentralized control and compliance with jurisdiction-specific regulations.
- **Interoperability and flexibility**. The platform's compatibility with widely used standards allowed for the integration of smart contracts and interoperability with other systems, enabling the testing of new use cases and add-on solutions.
- **Real-value pilot transactions**. A pilot was conducted that involved over 160 real-value transactions totalling more than $22 million, demonstrating the platform's capability to handle substantial cross-border payments efficiently.

Project Dunbar experimented with multiple technical prototypes, highlighting the practical importance of flexible technology architectures. The project demonstrated that accommodating diverse national payment infrastructures through modular design helped to manage complexity, supported integration and enabled smooth cross-border transactions. Operational and technical approaches included:

- **Multiple prototype platforms**. Two distinct prototypes using different distributed ledger technology platforms were developed and tested to evaluate various design approaches for a shared multi-central-bank digital currency platform.
- **Direct central bank digital currency access for financial institutions**. Direct access to digital currency enabled participating commercial banks to hold and

transact with multiple digital currencies directly on the shared platform, eliminating the need for intermediaries and reducing transaction costs and times.

- **Governance and access control**. A governance framework was established that allowed each central bank to maintain control over the issuance of its digital currency and access policies, ensuring compliance with domestic regulations while facilitating cross-border interoperability.
- **Utilization of smart contracts**. Smart contracts were leveraged to automate transaction processes, enhancing efficiency and reducing the potential for errors in cross-border settlements.

The operational and technical approaches employed through these projects underscore the importance of ensuring technological adaptability and fostering collaboration among member States to successfully implement cross-border digital currency solutions:

- **Collaborative platform development**. Multiple stakeholders are engaged in the design and governance of shared platforms to ensure that operational and regulatory requirements across jurisdictions are met.
- **Technological flexibility**. Platforms are built that are adaptable and interoperable with existing systems to accommodate diverse technical environments and future innovations.
- **Phased implementation and testing**. Iterative development and testing phases are employed to identify and address technical challenges, ensuring the robustness and reliability of the platform before its full-scale deployment.

Risk management and security

Project mBridge and Project Dunbar demonstrated effective strategies to manage critical risks inherent in cross-border digital currency systems.

Project mBridge

- **Established cross-jurisdictional legal clarity**. A detailed cross-border rulebook and decentralized governance structure were created that clearly outlined the individual roles of central banks and jurisdiction-specific regulatory compliance, facilitating smooth legal coordination.
- **Used a peer-to-peer transaction model**. A decentralized platform was used to enable direct, peer-to-peer cross-border digital currency transactions, significantly reducing reliance on traditional correspondent banking and associated systemic risks. The platform's design supported the use of local currencies in international payments, promoting financial stability by reducing exposure to foreign exchange fluctuations.
- **Employed atomic settlement for foreign exchange risk**. Integrated atomic (payment-versus-payment) settlement features were used to simultaneously settle both legs of cross-border currency exchanges, effectively minimizing foreign exchange settlement risks.

Project Dunbar

- **Used flexible governance structures**. Governance frameworks were adopted that allowed each central bank to retain regulatory autonomy while sharing a common digital currency platform, ensuring compliance with domestic laws and reducing cross-jurisdictional conflicts.
- **Employed a direct access model**. The direct access model enabled commercial banks to directly hold and transact in multiple digital currencies, thereby enhancing liquidity management and significantly reducing credit and intermediary risks.
- **Automated the management of foreign exchange risk**. Smart contracts were used to automate real-time foreign exchange transactions, improving transparency and mitigating settlement risks associated with foreign exchange fluctuations.

Shared security emphasis

Both initiatives emphasized rigorous cybersecurity measures, real-time compliance (know-your-customer/anti-money-laundering checks) and robust operational resilience through decentralized validation structures, collectively reinforcing the security and stability of cross-border digital currency systems. By designing digital currency platforms to minimize reliance on intermediaries, Project mBridge's peer-to-peer transaction structure reduced dependence on correspondent banks, while Project Dunbar's direct digital currency holdings for commercial banks improved liquidity management, thereby enhancing overall financial stability.

Phased and iterative implementation

Project mBridge and Project Dunbar underscored the importance of adopting a phased and iterative approach when developing cross-border digital currency platforms. By progressively testing and refining technical solutions, governance models and operational frameworks, central banks can effectively manage risks, incorporate stakeholder feedback and enhance system resilience.

Project mBridge

- **Used structured pilot phases**. A six-week pilot was conducted in the third quarter of 2022, involving 20 banks across four jurisdictions. The pilot was executed in three consecutive phases, each expanding the number of participating jurisdictions and transaction volumes.
- **Executed real-value transactions**. More than 160 real-value cross-border payment and foreign exchange transactions totalling more than HK $171 million occurred, providing practical insights into operational challenges and system performance.
- **Progressed to a minimum viable product stage**. According to pilot outcomes, Project mBridge advanced to a minimum viable product stage in mid-2024, signalling its readiness for broader real-world applications and inviting further international participation. This phase also allowed for an expansion of participation, and the central bank of Saudi Arabia joined the project.

Project Dunbar

- **Developed concurrent prototypes**. Over a nine-week period, two distinct prototypes using different distributed ledger technologies were developed and tested, which allowed for a comparative analysis of technical approaches.
- **Employed collaborative workstreams**. Central banks, commercial banks and technology partners were engaged in parallel workstreams to address policy, business and technological challenges collaboratively.
- **Created a foundation for future exploration**. Open questions and challenges were identified across policy, business and technology domains, laying the groundwork for ongoing collaboration and development in the multi-digital currency space.

9.4 Monetary policy considerations in a shared framework

In a monetary union, the central bank manages a single monetary policy on behalf of multiple sovereign States. Introducing a digital currency into this context requires careful consideration of the ways the currency may influence liquidity, money supply, interest rates and overall policy transmission, in particular when adoption levels and economic conditions vary among member States.

Potential impact of central bank digital currency on union-wide monetary policy

Digital currencies can alter demand for cash and deposits, affect commercial bank intermediation and introduce new channels for monetary policy transmission. If currencies are not properly designed and monitored, uneven adoption across countries could lead to imbalanced liquidity flows, affecting subregional financial stability.

Maintaining policy coherence across diverse economies

To preserve policy coherence, central banks should:

- Implement uniform core design parameters, such as holding limits, access models and interest-bearing features (if applicable)
- Ensure that digital currency issuance and redemption mechanisms are consistent across all jurisdictions to avoid arbitrage or disintermediation
- Avoid country-specific features that could fragment a unified monetary stance

Recommendations for coordinated liquidity management

- Establish real-time monitoring systems to track digital currency supply and usage by country and sector
- Coordinate with fiscal authorities to align digital currency deployment with broader liquidity objectives
- Use tiered access rules or programmable limits to modulate demand without disrupting the union-wide policy framework

Monitoring tools and early warning indicators

To anticipate and respond to monetary effects, the central bank should implement:

- Digital currency dashboards disaggregated by country and user type
- Liquidity stress tests and simulations to evaluate the impact of a digital currency under various adoption scenarios
- Feedback loops to adjust parameters on the basis of real-time data and economic shifts

By embedding these tools and principles into digital currency design and governance, monetary unions can safeguard policy integrity and macroeconomic stability while embracing digital innovation.

9.5 Regulatory and legal harmonization

The successful implementation of a digital currency in a monetary union depends on a cohesive and modernized regulatory and legal framework that spans all member States. Given the shared nature of currency issuance, legal clarity and harmonization are critical to ensuring that the digital currency is recognized, trusted and operable across jurisdictions.

Amending treaties and central bank statutes

Currently, in most African monetary unions, subregional treaties and central bank laws authorize the issuance of physical currency only. Introducing a digital currency would likely require:

- Revising subregional treaties and legal tender definitions to include digital forms of currency
- Updating the mandates and powers of subregional central banks (e.g. Central Bank of West African States, Bank of Central African States) to explicitly permit digital issuance, distribution and oversight

These changes must be coordinated across all member States through appropriate legislative and intergovernmental mechanisms.

Aligning national legal frameworks

In parallel, member countries must align national laws on:

- The legal tender status of the digital currency in domestic transactions
- The regulation of payments and settlement systems, ensuring consistent rules for digital-currency-related services
- AML/CFT compliance, to prevent regulatory arbitrage and ensure subregional security standards
- Consumer protection and data privacy, ensuring uniform provisions for digital financial services

Harmonization across these domains prevents fragmentation and builds user trust across the subregion.

Establishing a subregional regulatory perimeter

It is recommended that:

- A unified regulatory framework for digital-currency-related entities, including commercial banks, fintech companies and wallet providers, be created.
- A shared supervisory perimeter be defined, led by the subregional central bank in coordination with national regulators, who ensure alignment with domestic laws and provide input, while retaining responsibility within their jurisdictions.
- A subregional licensing or passport regime be developed to regulate digital currency service providers.

Regulatory and legal readiness checklist

To guide implementation, the handbook includes a high-level checklist:

- Have subregional treaties and central bank statutes been updated to authorize digital currency issuance?
- Have all member States adopted laws recognizing a central bank digital currency as legal tender?
- Are AML/CFT rules harmonized and applicable to digital currency transactions?
- Do consumer protection and data privacy laws cover digital currencies in a uniform manner?
- Is a cross-border supervisory coordination mechanism in place?

A harmonized legal foundation ensures that digital currencies function effectively as a subregion-wide monetary instrument, supporting policy objectives and financial integrity across all member States.

9.6 Cross-border use cases and subregional integration

In a monetary union, a digital currency has the potential to go beyond national applications and become a subregionally integrated tool for improving cross-border payments, trade and public service delivery. By leveraging the shared currency and regulatory coordination, digital currencies can serve as a bridge for deeper economic integration.

Enhancing subregional trade and payments

Digital currencies can reduce transaction costs, delays and the number of intermediaries in cross-border retail and commercial payments. This is especially relevant for:

- Small traders who operate across borders within the union
- Micro- and small businesses that have subregional supply chains
- Mobile money and fintech users seeking affordable remittance solutions

Sample use cases

Digital currencies can support such targeted workflows as:

- Cross-border government-to-person disbursements (e.g. digitally delivered subregional aid or social support programmes)
- Utility and bill payments across borders (e.g. electricity or telecommunications payments made by migrants for family who reside in their home country)
- Subregional trade settlements, in which digital payments are executed in real time across national platforms.

Optimizing settlement layers and data flows

For seamless subregional functionality, the digital currency system should:

- Enable real-time settlement finality for cross-border transactions
- Implement shared data standards to allow for reconciliation and reporting across countries
- Support multi-institution settlement layers, ensuring interoperability among commercial banks, fintech companies and mobile money operators within the union

Exploring interoperability beyond the union

Digital currency design should remain forward-compatible with external systems, allowing for:

- Integration with other African subregional payment platforms (e.g. Pan-African Payment and Settlement System)
- Participation in multi-digital currency initiatives for cross-border trade, such as experiments led by the Bank for International Settlements or coordinated by IMF
- Connectivity with diaspora remittance corridors, enabling outbound remittances from the union to countries outside the union, and inbound remittances from countries outside the union to the union

By enabling trusted, fast and low-cost digital payments within and beyond the subregion, digital currencies can become a powerful lever for the integration of subregional finance and global connectivity.

9.7 Implementation pathways and pilot design in a monetary union context

Implementing a digital currency across a monetary union requires a carefully phased approach that reflects the diversity of infrastructure, institutional readiness and policy priorities among member States. A well-designed implementation pathway should allow for gradual scaling, evidence-based refinement and inclusive participation.

Phased roll-outs across countries, sectors and use cases

It is recommended that monetary unions begin with limited-scope pilots in selected countries or sectors that demonstrate both:

- High readiness (infrastructure, regulatory clarity, digital literacy)
- Strategic value (e.g. cross-border remittances, government disbursements)

Roll-outs can then expand in stages:

- **Phase 1**. Controlled testing in a single or dual-country environment
- **Phase 2**. Expansion to broader retail or business-to-business use cases across multiple States
- **Phase 3**. Full subregional integration and open public access.

Each phase should be designed to test core functionalities, interoperability, risk controls and user experience.

Structuring multi-country pilot programmes

Successful pilots depend on strong stakeholder coordination. Central banks should:

- Establish subregional digital currency pilot committees that include representatives from all participating countries
- Define shared technical and policy parameters while allowing for localized features where appropriate
- Coordinate with payment providers, telecommunications firms and regulators to ensure consistent roll-out across borders
- Use sandbox environments to test real-time cross-border flows under monitored conditions

Data should be collected in a harmonized format to assess performance, security and inclusion.

Union-wide and country-specific success metrics

To evaluate pilots effectively, the following is recommended:

- Define common subregional success indicators, such as transaction success rates, interoperability performance and policy alignment
- Enable country-specific metrics to monitor adoption, user satisfaction and operational readiness
- Establish feedback loops from users, regulators and private sector actors to guide refinement before scaling up the pilot

By approaching implementation through a phased, collaborative and metrics-driven process, monetary unions can build momentum and trust in the digital currency while ensuring readiness for full-scale deployment.

9.8 Case studies and international insights

While the majority of digital currency exploration globally has focused on single-country implementations, several subregional and multi-jurisdictional efforts provide valuable insights for African monetary unions pursuing shared digital currencies. The present section highlights early case studies and global parallels that offer transferable lessons on governance, interoperability and cross-border design.

Central Bank of West African States: West African digital currency exploration

The Central Bank of West African States has initiated exploratory work to assess the feasibility of a subregional retail digital currency. The focus areas include improving financial inclusion, streamlining government payments and enhancing intraunion payment efficiency. The Central Bank has emphasized the importance of regulatory harmonization, phased experimentation and technological interoperability across its eight member States.

Eastern Caribbean Central Bank: DCash

The Eastern Caribbean Central Bank launched DCash, the world's first deployed multi-country digital currency pilot. Operating across eight island countries, DCash highlighted the importance of strong subregional coordination, a shared technological platform and clear public communication strategies. The experience also revealed operational challenges, such as infrastructure dependence and offline availability, which are particularly relevant for African unions.

Eurozone: digital euro

The Eurozone project provides relevant insights into:

- Managing a single currency across diverse economies
- Coordinating monetary policy and payment system governance
- Balancing central authority with national legal frameworks

These insights are applicable to African monetary unions as they assess ways to harmonize digital currency policies across sovereign States while ensuring inclusivity and resilience.

Main lessons and best practices

- **Start small, scale smart**. Multi-country digital currency pilots benefit from incremental roll-outs that have strong monitoring and feedback loops.
- **Ensure legal clarity**. It is essential to clearly define digital legal tender across jurisdictions.
- **Coordinate early and often**. Subregional task forces and joint governance mechanisms are crucial to ensure cross-border coherence.
- **Prioritize user trust**. Public communication, transparency and user education are central to success.

By learning from subregional and global examples, African monetary unions can build a digital currency strategy that is pragmatic, resilient and subregionally responsive.

9.9 Strategic considerations for monetary unions

Digital currency exploration within a monetary union presents both a unique opportunity and a set of complex design and governance trade-offs. The present section consolidates essential strategic considerations for subregional central banks and policymakers seeking to develop a unified digital currency across multiple sovereign States.

Critical trade-offs to navigate

- **Centralized versus decentralized issuance models**. A centralized issuance model ensures uniformity and control but may reduce country-level autonomy. A decentralized or federated model offers flexibility and localized management but requires stronger coordination to avoid fragmentation.
- **Harmonized versus modular roll-out**. Full harmonization provides policy coherence and operational consistency, while a modular roll-out allows for adaptation to varying readiness levels. The ideal approach balances common subregional standards with phased, context-specific implementation.
- **Subregional platform versus national integration layers**. A single subregional platform can drive efficiency and interoperability, while layered national integration supports innovation and responsiveness to local conditions. Successful design must ensure seamless connectivity and real-time data-exchange across both.

Guiding principles for the implementation of a subregional central bank digital currency

1. **Coordination**. Joint governance structures and intergovernmental decision-making processes are essential to ensure legitimacy, alignment and stability across the union.

2. **Inclusivity**. Design and roll-out must accommodate the diverse economic, technological and institutional realities of all member States, leaving no country or user group behind.

3. **Adaptability**. A digital currency framework should support continuous learning, policy flexibility and technical scalability in order to respond to evolving subregional needs and innovations.

By applying these principles, monetary unions can position digital currencies as instruments for deepening subregional integration, enhancing monetary sovereignty and advancing inclusive digital transformation.

10 Monitoring and feedback

10.1 Establishing monitoring frameworks

The successful deployment of a central bank digital currency requires a comprehensive monitoring framework to track performance, mitigate risks and ensure that the digital currency remains aligned with national policy objectives. A well-structured monitoring system allows central banks to:

- Measure adoption and impact across critical sectors
- Identify potential risks (e.g. financial stability concerns, cybersecurity threats)

- Ensure continuous policy alignment as economic and financial conditions evolve
- Facilitate adjustments on the basis of data-driven insights

The present section outlines a structured approach for central banks to follow when establishing digital currency monitoring frameworks, including essential metrics, reporting mechanisms and feedback loops.

10.1.1 Defining monitoring objectives

Before designing a monitoring framework, central banks must define what they intend to track. These objectives should align with the country's digital currency policy drivers (see table 33).

Table 33
Policy objectives and monitoring

Policy objective	Monitoring focus
Financial inclusion	Digital currency adoption among unbanked and underserved populations
Monetary policy effectiveness	Impact on interest rates, money supply and inflation
Payment system efficiency	Transaction speeds, costs and user experience
Financial stability	Changes in bank deposits, lending behaviours and systemic risks
AML/CFT compliance	Monitoring suspicious transactions and ensuring regulatory adherence
Cross-border transactions	Volume and efficiency of international digital currency payments

Source: Author's elaboration.

Takeaway

Monitoring must be purpose-driven, tracking data that directly informs policy adjustments and system refinements.

10.1.2 Establishing key performance indicators

To effectively measure the impact of a digital currency, central banks should establish quantifiable key performance indicators (see table 34). These indicators should be regularly reviewed and updated as needed.

Table 34
Quantifiable key performance indicators

Category	Key performance indicators
Adoption and usage	Number of active digital currency wallets, transaction volume, merchant acceptance rate
Payment system efficiency	Average transaction time, processing costs, network uptime
Financial stability	Changes in bank deposits, digital-currency-to-fiat conversion rates
AML/CFT and security	Number of flagged transactions, fraud incidents, cybersecurity threats detected
Economic impact	Digital currency effect on inflation, velocity of money, monetary aggregates

Source: Author's elaboration.

Takeaway

Clear, measurable key performance indicators ensure that central banks can track digital currency performance in real time and make data-driven policy decisions.

10.1.3 Data collection and reporting mechanisms

To operationalize monitoring, central banks need reliable data sources and reporting frameworks.

Data sources for digital currency monitoring include:

- On-chain transaction data: captures real-time digital currency transaction flows
- Financial institutions and fintech reporting: banks and mobile money providers report usage trends
- Consumer and business surveys: gather qualitative insights on adoption barriers and user experiences
- Regulatory law enforcement agencies: provide oversight for AML/CFT compliance

Reporting and dashboarding methods include:

- Automated dashboards that use real-time analytics
- Periodic policy reports (e.g. monthly, quarterly) that assess the economic and financial impact of a digital currency
- Early warning indicators relating to financial stability risks

Takeaway

A centralized reporting system enables policymakers to detect trends early and respond proactively.

10.1.4 Implementing feedback loops for policy adjustments

Digital currency monitoring should not be passive, but instead should actively inform policy decisions. Feedback loops ensure that insights from data collection lead to timely adjustments in digital currency policy and operations.

- Public feedback mechanisms gather user input through surveys and focus groups.
- Regulatory adjustments update laws and regulations in accordance with emerging risks.
- Technical system improvements enhance digital currency infrastructure on the basis of real-world performance.
- Banking and fintech collaboration ensures that the financial sector is able to integrate digital currency smoothly.

Takeaway

A continuous feedback cycle allows digital currency systems to evolve and remain effective in a dynamic economic environment.

10.1.5 Archetype-specific monitoring considerations

The monitoring framework must also reflect the specific challenges and priorities of different archetypes (see table 35).

Table 35
Archetype-specific monitoring considerations

Archetype	Special monitoring focus
Large economies	Impact on wholesale settlement efficiency and financial market stability
High-mobile-money-penetration economies	Interoperability of digital currency with mobile money networks and user adoption trends
Fixed exchange rate economies	Effects on monetary stability and foreign reserves
Highly dollarized economies	Digital-currency-to-foreign-currency conversions and de-dollarization trends
Resource-dependent economies	Integration with fiscal management and transparency in government payments
High-remittance economies	Efficiency and cost reductions in cross-border payments
Small island economies	Tourism-related digital currency transactions and cross-border trade settlements
Fragile-State economies	Offline digital currency usage and financial access for humanitarian aid programmes

Source: Author's elaboration.

Takeaway

Each archetype requires a customized monitoring approach to track its most critical economic and financial risks.

10.1.6 Developing a central bank digital currency monitoring road map

To ensure effective implementation, central banks should establish a structured road map for monitoring (see table 36).

Table 36
Example of a structured road map for monitoring

Phase	Action steps
Phase 1. Pre-launch (pilot stage)	Define key performance indicators, set up data collection infrastructure, establish dashboards
Phase 2. Initial deployment	Track transaction trends, detect early adoption challenges, refine reporting mechanisms
Phase 3. Expansion and scaling	Integrate monitoring with broader financial policy frameworks, introduce advanced analytics tools
Phase 4. Ongoing refinement	Adjust digital currency parameters (e.g. transaction limits, interest rates) on the basis of real-time insights

Source: Author's elaboration.

Takeaway

A phased monitoring strategy allows central banks to track digital currency development and adjust policies dynamically.

A strong digital currency monitoring framework is essential to ensure that digital currency deployment remains aligned with policy objectives and is financially stable and technologically resilient. By setting clear key performance indicators, collecting real-time data and implementing feedback mechanisms, central banks can:

- Detect risks early and adapt policies accordingly
- Enhance financial stability while maximizing digital currency benefits
- Ensure digital currency adoption is efficient, inclusive and effective

10.2 Adaptive pathways

An adaptive pathway ensures that digital currency design, roll-out and policy frameworks are responsive to real-world data, stakeholder input and financial market dynamics. The present section includes guidance on:

- Establishing feedback loops to continuously evaluate digital currency performance
- Refining digital currency features and operational models on the basis of empirical data
- Making policy adjustments to address emerging challenges and opportunities

By embedding adaptability into the digital currency life cycle, central banks can enhance usability, mitigate risks and ensure long-term effectiveness.

10.2.1 Establishing continuous feedback loops

To maintain digital currency relevance and efficiency, central banks must systematically collect, analyse and act upon feedback from users, financial institutions, merchants and regulators.

Sources of digital currency feedback:

- Transaction data and analytics: real-time monitoring of transaction volumes, adoption rates and system performance
- Consumer and business surveys: assessing user experience, barriers to adoption and trust levels
- Financial institution reports: insights from banks, fintech firms and payment providers on system integration and liquidity impact
- Regulatory and compliance reviews: evaluating effectiveness with regard to anti-money-laundering compliance, data privacy and monetary policy execution
- Cross-border interactions: assessing the role of digital currency in regional and international transactions

Takeaway

A well-structured feedback mechanism ensures that digital currency refinements are based on real-world data rather than theoretical assumptions.

10.2.2 Continuous improvement of digital currency systems

Once feedback is collected, central banks should iteratively improve digital currency design and functionality to enhance usability, security and policy effectiveness (see table 37).

Table 37
Refinement strategies for common challenges related to digital currency features

Digital currency feature	Common challenges	Adaptive refinement strategy
Interoperability	Limited connectivity with banks, fintech companies, mobile money providers	Develop application programming interfaces and regulatory standards to ensure seamless integration
Transaction speed and scalability	Network congestion, slow processing times	Upgrade ledger infrastructure and optimize consensus mechanisms
Privacy and security	User concerns over data protection, cyberthreats	Implement privacy-enhancing technologies (e.g. zero-knowledge proofs)
Offline payments	Lack of reliable connectivity in rural or fragile-State environments	Deploy token-based offline solutions and distributed ledger nodes
Cross-border payments	High remittance fees, complex currency conversions	Establish regional digital currency interoperability frameworks

Source: Author's elaboration.

Takeaway

Central bank digital currency systems must be continuously tested and enhanced on the basis of real-world performance and user feedback.

10.2.3 Adaptive policy responses for digital currency systems

As the digital currency system evolves, monetary policy frameworks, regulatory guidelines and risk management strategies must also adapt. Central banks should establish structured pathways for policy adjustments, ensuring that digital currency remains a flexible and resilient tool (see table 38).

Table 38
Adaptive policy considerations

Policy area	When adjustments are needed	Potential adaptive measures
Interest-bearing central bank digital currency	If financial stability is affected by deposit outflows from banks	Adjust digital currency interest rates or introduce holding limits
AML/CFT compliance	If transaction patterns indicate emerging financial crime risks	Strengthen real-time monitoring and adjust know-your-customer/anti-money-laundering regulations
Liquidity and monetary policy	If digital currency circulation disrupts money supply control	Implement tiered access models and adjust digital currency issuance mechanisms
Retail versus wholesale use cases	If digital currency is underutilized or overused in certain segments	Expand or limit use cases on the basis of adoption trends

Source: Author's elaboration.

Takeaway

Central bank digital currency policies should be dynamic, allowing for fine-tuning in accordance with economic conditions and stakeholder needs.

10.2.4 Managing risks and unintended consequences

Adapting digital currency frameworks must also involve proactive risk management (see table 39). A central bank digital currency risk dashboard should be developed to track emerging threats and trigger adaptive responses when necessary.

Table 39
Common risks and adaptive mitigation strategies

Risk type	Potential impact	Adaptive strategy
Cybersecurity threats	System breaches, financial fraud and user data leaks	Enhance security protocols, conduct regular penetration testing and integrate decentralized identity solutions
Loss of public trust	Resistance owing to privacy concerns or lack of usability	Improve public communication, ensure strong privacy guarantees and conduct user education campaigns
Operational failures	System downtimes, transaction failures	Implement redundant infrastructure, maintain failover mechanisms and perform stress testing
Regulatory arbitrage	Gaps in regulation lead to illicit activities	Continuously update compliance frameworks and collaborate with international regulatory bodies

Source: Author's elaboration.

Takeaway

An adaptive risk mitigation strategy ensures digital currency resilience in a rapidly evolving digital financial landscape.

10.2.5 Creating a structured adaptive framework

To formalize an adaptive digital currency implementation strategy, central banks should develop a structured framework that ensures ongoing assessment and refinement (see table 40).

Table 40
Example of a structured adaptive framework

Phase	Primary adaptive actions
Phase 1. Pilot stage	Gather user feedback, conduct transaction monitoring, refine technical infrastructure
Phase 2. Initial roll-out	Evaluate adoption barriers, adjust regulatory frameworks, optimize system performance
Phase 3. Expansion and scaling	Introduce advanced features, refine monetary policy tools, expand cross-border capabilities
Phase 4. Continuous evolution	Monitor market responses, implement iterative improvements, future-proof digital currency design

Source: Author's elaboration.

Takeaway

A structured adaptive framework ensures that the central bank digital currency remains relevant, efficient and aligned with long-term national policy objectives.

10.2.6 Archetype-specific adaptive strategies

Different digital currency archetypes may require customized approaches to adaptability (see table 41).

Table 41
Adaptive considerations for various economic archetypes

Archetype	Adaptive considerations
Large economies	Ensure high-volume scalability, adjust monetary policy settings dynamically
High-mobile-money-penetration economies	Refine digital currency-mobile money interoperability, enhance user experience in accordance with feedback
Fixed exchange rate economies	Fine-tune capital controls and pegged currency stability mechanisms
Highly dollarized economies	Gradually phase out reliance on foreign currency, track adoption trends
Resource-dependent economies	Adjust fiscal transparency tools in accordance with government revenue-collection insights
High-remittance economies	Optimize remittance flows by integrating with new payment corridors
Small island economies	Enhance multi-currency features for tourism-related transactions
Fragile-State economies	Strengthen offline capabilities, refine humanitarian aid distribution

Source: Author's elaboration.

Takeaway

Archetype-specific adaptations ensure that digital currency strategies remain aligned with each country's evolving financial landscape.

A digital currency is not static; it must evolve over time in response to economic shifts, technological advancements and user behaviours. By establishing robust feedback loops, refining technical design, adjusting policies dynamically and managing risks proactively, central banks can ensure that their digital currency remains:

- Flexible: adjusting to user needs, economic changes and market dynamics
- Secure: mitigating cybersecurity threats and financial stability risks
- Efficient: continuously optimizing performance, usability and transaction costs
- Policy-driven: maintaining alignment with national monetary, financial and economic objectives

An adaptive digital currency pathway ensures that central banks can navigate the complexities of digital currency adoption, ultimately enhancing financial inclusion, payment efficiency and economic resilience.

10.3 Success metrics

To ensure the long-term success of a central bank digital currency, central banks must define clear, measurable success metrics that align with their policy objectives, economic priorities and financial system maturity. These metrics should be tracked over time to evaluate the adoption, impact and overall effectiveness of a digital currency.

The present section includes:

- Common success metrics applicable to all digital currency implementations
- Archetype-specific success indicators to reflect the diverse economic and financial realities of different countries
- A structured framework for monitoring digital currency performance against national goals

By establishing well-defined success metrics, central banks can ensure that digital currency deployment remains data-driven, adaptable and aligned with long-term policy priorities.

10.3.1 Common success metrics for central bank digital currency evaluation

While digital currency objectives vary across countries, certain core metrics are universally important for evaluating effectiveness (see table 42).

Table 42
Core success metrics and why they matter

Category	Core success metrics	Why they matter
Adoption and usage	Percentage of population using digital currency, merchant acceptance rate, daily transaction volume	Measures public and business adoption levels
Financial inclusion	Per cent increase in financial access among unbanked populations	Evaluates the impact of a digital currency on economic accessibility
Payment system efficiency	Average transaction processing time, network uptime, cost per transaction	Ensures that digital currency improves digital payments
Monetary policy impact	Change in money velocity, effectiveness of interest rate transmission	Assesses the role of a digital currency in monetary policy execution
Financial stability	Percentage of bank deposits retained, liquidity impact on banking sector	Tracks risks related to disintermediation
Cybersecurity and fraud prevention	Number of security incidents, compliance with AML/CFT frameworks	Ensures system resilience and regulatory compliance
Cross-border interoperability	Digital currency usage in remittance flows, settlement speed for cross-border payments	Evaluates the effectiveness of a digital currency in facilitating global transactions

Source: Author's elaboration.

Takeaway

Common success metrics provide a baseline for evaluating the effectiveness of a digital currency across diverse economic environments.

10.3.2 Success measurement criteria for central bank digital currencies across economies

Each country archetype faces unique economic conditions, requiring tailored success indicators (see table 43).

Table 43
Success metrics by economic archetype

Archetype	Core success metrics
Large economies	Scalability performance, impact on interbank settlements, integration with existing financial infrastructure
High-mobile-money-penetration economies	Digital currency-mobile money interoperability rate, percentage of mobile transactions processed using digital currency
Fixed exchange rate and currency-pegged economies	Stability of foreign exchange reserves after introduction of digital currency, impact on capital flow controls
Highly dollarized economies	Reduction in foreign currency reliance, increase in local digital currency usage
Resource-dependent economies	Percentage of government revenues processed using digital currency, transparency improvements in fiscal transactions
High-remittance economies	Reduction in remittance fees, increase in direct digital currency remittance transactions
Small island economies	Percentage of tourism payments processed using digital currency, cross-border payment efficiency gains
Fragile-State economies	Digital currency availability in offline mode, aid distribution efficiency improved through the use of digital currency

Source: Author's elaboration.

Takeaway

Archetype-specific success metrics ensure that digital currency implementation aligns with unique national priorities.

10.3.3 Monitoring central bank digital currency performance over time

Digital currency success is not static: central banks must track performance continuously and adapt on the basis of real-world outcomes (see table 44).

Table 44
Phased performance evaluation

Phase	Success indicators	Monitoring tools
Pilot phase	Usability of digital currency in test groups, technical system stability, public perception surveys	Controlled pilots, consumer/business surveys
Initial roll-out	Adoption rate, merchant integration, fraud detection rates	Real-time analytics, bank/fintech reporting
Scaling and expansion	Effect of digital currency on monetary policy, financial stability indicators	Macroeconomic impact assessments, regulatory reports
Long-term adoption	GDP impact, remittance efficiency, financial system resilience	Central bank reports, international benchmarks

Source: Author's elaboration.

Takeaway

Central bank digital currency success must be assessed in multiple phases, ensuring gradual refinement and risk management.

10.3.4 Adjusting policy and design on the basis of metrics

As digital currency performance data are collected, central banks should adjust policies and system parameters to optimize success (see table 45).

Table 45
Adjusting policy and design on the basis of metrics

Trend observed	Adjustment to be made
Low adoption rates among unbanked users	Improve public education, enhance offline payment options
High transaction failures	Strengthen infrastructure, optimize payment processing protocols
Increased bank disintermediation	Introduce deposit caps, enhance banking sector incentives
Cybersecurity breaches or fraud cases	Strengthen AML/CFT compliance, improve encryption protocols
Low merchant adoption	Offer tax incentives, simplify digital currency payment integration

Source: Author's elaboration.

Takeaway

Central banks must remain flexible, adjusting digital currency strategies on the basis of performance metrics.

10.3.5 Global benchmarking and best practices

To contextualize digital currency success, central banks should compare their performance against global benchmarks (see table 46).

Table 46
Global digital currency benchmarks and their relevance for African central banks

Global digital currency benchmark	Relevance for African central banks
SandDollar (Bahamas)	Small-scale retail adoption, regulatory clarity on digital currency usage
eNaira (Nigeria)	Mobile wallet integration, challenges related to merchant adoption
e-CNY (China)	Large-scale adoption, role in monetary policy testing
e-Krona (Sweden)	Alternative to cash, impact on banking sector dynamics

Source: Author's elaboration.

Takeaway

Learning from global case studies allows African central banks to refine their digital currency success criteria.

10.3.6 Establishing a success reporting framework

To maintain transparency and accountability, central banks should establish formal reporting mechanisms to track digital currency success.

Recommended reporting practices

- Quarterly digital currency reports, summarizing adoption rates, transaction volumes and economic impact
- Public dashboards displaying real-time digital currency usage statistics for transparency
- Stakeholder consultations engaging banks, fintech firms and consumers to refine digital currency design
- Annual policy reviews assessing the effectiveness of a digital currency and making the necessary adjustments

Takeaway

Structured reporting ensures ongoing assessment and policy alignment for long-term digital currency success.

Defining and tracking success metrics is essential to ensuring that digital currency implementation meets national policy objectives, enhances financial stability and delivers meaningful economic benefits. By adopting a data-driven approach, central banks can:

- Measure adoption and usage trends
- Track financial and economic impact
- Adjust policies and digital currency design as needed
- Ensure long-term sustainability and resilience

A robust success measurement framework will allow central banks to continuously refine digital currency strategies, ensuring that digital currency remains an effective tool for financial inclusion, monetary stability and economic growth.

11 Conclusion and next steps

A comprehensive, structured and context-sensitive framework has been provided in the *Central Bank Digital Currency Handbook for Africa* to guide African central banks and regional monetary authorities in the exploration and potential implementation of central bank digital currencies. Focused on policy objectives, readiness dimensions, design considerations and archetype-specific pathways, the information set out in the handbook supports tailored decision-making aligned with the unique characteristics and strategic priorities of each jurisdiction.

Essential insights

- More than just a technology, digital currencies must be understood as a policy tool designed to advance national and regional objectives, such as financial inclusion, payment system efficiency and monetary sovereignty.
- A modular, step-by-step framework enables central banks to explore digital currency possibilities systematically, beginning with the definition of objectives, readiness assessment, design evaluation, cost-benefit analysis and phased implementation.
- The use of archetypes provides a pragmatic method to tailor guidance on the basis of shared economic attributes, such as high remittances, mobile money adoption or vulnerability to dollarization.
- Regional monetary unions require additional considerations, in particular with regard to governance, legal harmonization and cross-border interoperability, to ensure successful digital currency deployment.

Strategic emphasis

Across all pathways, a consistent message emerges: digital currency initiatives must be anchored in national development goals, responsive to institutional capacity and attuned to user realities. There is no one-size-fits-all model. Success will depend on maintaining flexibility, inclusivity and a strong alignment with country-specific and regional policy drivers.

Next steps

As central banks and regional bodies move from exploration to more concrete planning, the following next steps are recommended:

1. Deepen stakeholder engagement

- Expand structured dialogue with domestic and cross-border stakeholders, including financial institutions, fintech companies, regulators and end users
- Prioritize inclusive consultation processes to build trust and ensure that digital currency design reflects user needs and societal priorities

2. Strengthen regional collaboration

- Leverage existing platforms for regional monetary cooperation, especially in monetary unions, to coordinate policy, legal reforms and the implementation of pilots
- Explore joint participation in multi-digital currency initiatives or continental frameworks, such as the African Continental Free Trade Area and the Pan-African Payment and Settlement System

3. Build institutional capacity

- Invest in technical expertise, legal reform capability and project management for digital currency exploration
- Establish or strengthen digital currency working groups or innovation labs within central banks to lead experimentation and evaluation

4. Promote knowledge-sharing

- Participate in cross-border learning initiatives and case-study exchanges through such forums as those held by African central banks, the Alliance for Financial Inclusion, IMF and the Innovation Hub of the Bank for International Settlements
- Contribute to an evolving regional knowledge base that tracks lessons learned, innovations discovered and challenges faced by African digital currency pilots and studies

Conclusion

Digital currencies offer African countries the opportunity to shape a more inclusive, sovereign and digitally connected financial future. Whether through cautious exploration or targeted deployment, this journey must be guided by policy, not by hype**,** and driven by collaboration, not isolation. By following the frameworks, tools and insights set out in the present handbook, central banks can lead with clarity, purpose and regional cohesion towards a new era of digital public money.

12 Suggested resources

1. Foundational frameworks and principles

- Bank for International Settlements and others, “Central bank digital currencies: foundational principles and core features”, Report No. 1 (Basel, Switzerland, 2020).
- IMF, “Central bank digital currency – initial considerations”, Policy Paper No. 2023/048 (Washington, D.C., 2023).
- Manisha Patel, Safari Kasiyanto and André Reslow, “Positioning central bank digital currency in the payments landscape”, Fintech Notes No. 2024/006 (Washington, D.C., IMF, 2024).

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2. Central bank digital currency design and technical architecture

- Bank for International Settlements and others, “Central bank digital currencies: system design and interoperability”, Report No. 2 (Basel, Switzerland, 2021).
- World Economic Forum, “CBDC technology considerations”, Digital Currency Governance Consortium White Paper Series (Geneva, Switzerland, 2021).

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3. Regulatory and financial stability considerations

- Wouter Bossu and others, “Legal aspects of central bank digital currency: central bank and monetary law considerations”, IMF Working Paper, WP/20/254 (Washington, D.C., IMF, 2020).
- Romain Bouis and others, “Central bank digital currencies and financial stability: balance sheet analysis and policy choices”, IMF Working Paper, WP/YY/226 (Washington D.C., IMF, 2024).
- Marianne Bechara and others, “Private law aspects of token-based central bank digital currencies”, Fintech Notes No. 2025/003 (Washington, D.C., IMF, 2025).

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4. Economic and policy implications

- Bank for International Settlements, “CBCDs: an opportunity for the monetary system” in *Annual Economic Report* (Basel, Switzerland, 2021).
- Tommaso Mancini-Griffoli and others, “Casting light on central bank digital currency”, Staff Discussion Note No. 18/08 (Washington, D.C., IMF, 2018).

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5. Implementation strategies and readiness assessment

- World Economic Forum, "Central bank digital currency policy-maker toolkit", Insight Report (Geneva, Switzerland, 2020).
- Gabriel Soderberg and others, “How should central banks explore central bank digital currencies? A dynamic decision-making framework”, Fintech Notes, 2023/008 (Washington, D.C., IMF, 2023).

- IMF, “Central bank digital currency: progress and further considerations”, Policy Paper No. 2024/052 (Washington, D.C., 2024).

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6. Cross-border central bank digital currencies and interoperability

- Biagio Bossone and Oya Ardic, “Central bank digital currencies for cross-border payments: a review of current experiments and ideas” (Washington, D.C., World Bank, 2021).
- World Economic Forum, “Central bank digital currency global interoperability principles”, White Paper (Geneva, Switzerland, 2023).

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7. Pilot studies and case studies (including Africa)

- Jookyung Ree, “Nigeria’s eNaira, one year after” IMF Working Paper No. WP/23/104 (Washington, D.C., IMF, 2022).
- Digital Euro Association, “Lessons from the first implemented CBDC: the Sand dollar”, 23 June 2023. Available at https://digital-euro-association.de/blog/lessons-from-the-sand-dollar.

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8. Risk management and cybersecurity

- Bank for International Settlements, *Central Bank Digital Currency (CBDC) Information Security and Operational Risks to Central Banks* (Basel, Switzerland, 2023).
- Arvinder Bharath, Anca Paduraru and Tamas Gaidosch, “Cyber resilience of the central bank digital currency ecosystem”, Fintech Notes No. 2024/003 (Washington, D.C., IMF, 2024).

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9. Adoption strategies and public engagement

- Tayo Tunyathon Koonprasert and others, “Central bank digital currency adoption: inclusive strategies for intermediaries and users”, Fintech Notes No. 2024/005 (Washington, D.C., IMF, 2024).
- Bank for International Settlements and others, “Central bank digital currencies: user needs and adoption”, Report No. 3 (Basel, Switzerland, 2021)

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10. Central bank digital currencies and financial inclusion

- Ashley Lannquist and Brandon Tan, “Central bank digital currency’s role in promoting financial inclusion”, Fintech Notes No. 2023/011 (Washington, D.C., IMF, 2023).
- Bank for International Settlements, *CBDCs in Emerging Market Economies*, BIS Papers No. 123 (Basel, Switzerland, 2022).